pyAmpliCol at one loop

The recursive one-loop engine against MadLoop, Recola, OpenLoops and NJet

milestone M-FINAL-b (2026-11-02) · engine `f3d9e361` on `loop` · AMD EPYC 9754, shared host · finite part, binary64, one pinned thread, $B = 128$
pyAmpliCol timed by G1 on 2026-10-31: engine `71e2041e`, native `loop-native-v2-slice-71e2041e` (not re-measured on `f3d9e361`: one final retime at the M-FINAL pin, user Q11)

×1/16 faster · ×1/4 · ×1 run time ÷ fastest provider · ×4 · ×16 slower

Gate pass ×0.25 · fail ×1.4 · open ×1.0 · <5 ms: not gated ×1.4 — time ÷ limit
Stages inputs · numerator · scatter · NinjaRS · masters · glue
missing (why) · no colour: generation, LC · † within estimate · ¶ reference's own (DD) · ‡ user decision
ML MadLoop · Rc Recola · OL OpenLoops

Table 1: **State of the one-loop engine.** How to read it: after Figure 1.

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
pyAmpliCol values: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: `71e2041e`, native per-point inputs, G1 2026-10-31				
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 2$	$gg \to gg$ 4 legs · QCD · 24 families (LC 12) OL·Rc·ML ≤ 6.7e−13 16 pts · median 1.8e−14 · †	**7.63 s** ×54 0.58 GiB peak · plan 0.305 MB **Rc 142 ms** · ML 9.70 s	**1.52 ms** ×0.75 $B = 128$ **ML 2.02 ms** · OL 3.20 ms · Rc 10.0 ms no gate (2→2) numerator 56 %	**0.657 ms** LC/FC 0.43 $B = 128$ · FC 1.52 ms
$2 \to 3$	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL ≤ 6.7e−13 · Rc 3.3e−10 16 pts · median 8.7e−14 · ¶	**8.06 s** ×20 0.58 GiB peak · plan 0.282 MB **Rc 406 ms** · ML 9.87 s	**1.96 ms** ×1.3 $B = 128$ **OL 1.52 ms** · ML 3.17 ms · Rc 3.18 ms <5 ms: not gated ×1.00 numerator 40 %	**1.25 ms** LC/FC 0.63 $B = 128$ · FC 1.96 ms
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML ≤ 8.4e−12 · Rc 7.2e−11 16 pts · median 1.7e−13 · ¶	**8.03 s** ×17 0.58 GiB peak · plan 0.388 MB **Rc 462 ms** · ML 10.9 s	**2.79 ms** ×0.95 $B = 128$ **OL 2.92 ms** · Rc 5.55 ms · ML 8.59 ms <5 ms: not gated ×0.73 numerator 46 %	**1.59 ms** LC/FC 0.57 $B = 128$ · FC 2.79 ms
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families (LC 7) Rc·ML ≤ 1.4e−13 16 pts · median 9.1e−15 · †	**11.9 s** ×17 0.62 GiB peak · plan 0.310 MB **Rc 701 ms** · ML 14.0 s	**0.702 ms** ×0.74 $B = 128$ **ML 0.948 ms** · Rc 1.08 ms · OL no library <5 ms: not gated ×0.74 numerator 39 %	**0.484 ms** LC/FC 0.69 $B = 128$ · FC 0.702 ms
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) ML·Rc ≤ 1.5e−12 · OL 8.4e−11 16 pts · median 6.7e−14 · ¶	**8.42 s** ×13 0.59 GiB peak · plan 0.656 MB **Rc 634 ms** · ML 17.6 s	**5.47 ms** ×0.44 $B = 128$ **OL 12.4 ms** · Rc 25.3 ms · ML 28.2 ms pass ×0.34 numerator 57 %	**2.77 ms** LC/FC 0.51 $B = 128$ · FC 5.47 ms
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) ML·Rc·OL ≤ 1.0e−6 16 pts · median 4.9e−13 · †	**9.93 s** ×1.5 0.63 GiB peak · plan 1.85 MB **Rc 6.63 s** · ML 47.0 s	**20.4 ms** ×0.21 $B = 128$ **OL 97.0 ms** · ML 247 ms · Rc 351 ms pass ×0.16 numerator 58 %	**8.48 ms** LC/FC 0.42 $B = 128$ · FC 20.4 ms
$2 \to 4$	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families (LC 33) Rc·ML ≤ 9.4e−11 16 pts · median 6.3e−14 · †	**12.9 s** ×16 0.63 GiB peak · plan 0.624 MB **Rc 833 ms** · ML 24.8 s	**5.01 ms** ×0.49 $B = 128$ **Rc 10.2 ms** · ML 27.3 ms · OL no library pass ×0.38 numerator 58 %	**1.74 ms** LC/FC 0.35 $B = 128$ · FC 5.01 ms
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) OL·Rc·ML ≤ 1.4e−8 16 pts · median 1.9e−11 · †	**13.9 s** ×0.23 0.72 GiB peak · plan 3.71 MB **Rc 61.4 s** · ML 218 s	**98.5 ms** ×0.21 $B = 128$ **OL 472 ms** · Rc 1.97 s · ML 5.45 s pass ×0.16 numerator 47 %	**32.4 ms** LC/FC 0.33 $B = 128$ · FC 98.5 ms
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) Rc 9.3e−12 · OL 1.6e−10 16 pts · median 1.5e−12 · †	**13.9 s** ×0.43 0.73 GiB peak · plan 4.54 MB **Rc 32.8 s**	**77.4 ms** ×0.21 $B = 128$ **OL 374 ms** · Rc 908 ms · ML not exported pass ×0.16 numerator 63 %	**30.3 ms** LC/FC 0.39 $B = 128$ · FC 77.4 ms
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) Rc·OL·ML ≤ 2.2e−7 8 pts · median 3.0e−11 · †	**33.1 s** ×0.016 1.27 GiB peak · plan 16.1 MB **Rc 35.4 min** · ML 144.6 min	**371 ms** ×0.13 $B = 128$ **OL 2.84 s** · Rc 17.6 s · ML 38.7 s pass ×0.10 numerator 56 %	**155 ms** LC/FC 0.42 $B = 128$ · FC 371 ms

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol values: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: 71e2041e, native per-point inputs, G1 2026-10-31			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 5$	$u\bar{u} \to e^-e^+d\bar{d}g$ 7 legs · QCD · 242 families (LC 114) ML·OL·Rc ≤ 5.2e−11 16 pts · median 7.7e−13 · †	**15.8 s** ×2.3 0.66 GiB peak · plan 1.13 MB **Rc 6.73 s** · ML 58.6 s	**16.1 ms** ×0.45 **Rc 35.3 ms** · OL 51.0 ms · ML 132 ms $B = 128$ pass ×0.35 numerator 53 %	**6.33 ms** LC/FC **0.39** $B = 128$ · FC 16.1 ms
	$u\bar{u} \to e^-e^+ggg$ 7 legs · QCD · 463 families (LC 164) ML·Rc·OL ≤ 1.1e−10 16 pts · median 1.5e−12 · †	**16.1 s** ×1.2 0.68 GiB peak · plan 2.45 MB **Rc 13.4 s** · ML 146 s	**59.7 ms** ×0.24 **Rc 251 ms** · OL 346 ms · ML 1.42 s $B = 128$ pass ×0.18 numerator 70 %	**10.7 ms** LC/FC **0.18** $B = 128$ · FC 59.7 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · QCD · 579 families (LC 164) Rc·OL·ML ≤ 8.7e−9 16 pts · median 2.7e−11 · †	**17.3 s** ×0.62 0.75 GiB peak · plan 2.13 MB **Rc 27.7 s** · ML 165 s	**62.5 ms** ×0.15 **OL 404 ms** · Rc 620 ms · ML 3.04 s $B = 128$ pass ×0.12 numerator 63 %	**11.6 ms** LC/FC **0.19** $B = 128$ · FC 62.5 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · EW · 1 599 families OL·ML ≤ 2.6e−9 16 pts · median 6.8e−10	**318 s** ×14 8.93 GiB peak · plan 4.47 MB **Rc 23.5 s** · ML 11.9 min	**153 ms** ×0.15 **Rc 987 ms** · OL 1.40 s · ML 7.07 s $B = 128$ EW: indicative, not gated numerator 52 % model: the workspace's py3 G_mu copy	***not supported*** refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4; typed colour-declaration-coverage)
	$u\bar{u} \to gggg$ 7 legs · QCD · 5 258 families (LC 2 712) ML 1.9e−12 · Rc 2.9e−10 3 pts · median 7.1e−11 · ¶	**134 s** ×0.013 3.03 GiB peak · plan 69.2 MB **Rc 170.3 min** · ML 12.2 h	**1.72 s** ≥×0.0094 **Rc ≤182 s** · ML 569 s · OL no library $B = 128$ pass ×0.0030 numerator 66 %	**581 ms** LC/FC **0.34** $B = 128$ · FC 1.72 s
	$gg \to gggg$ 7 legs · QCD · 4 707 families (LC 4 707) NJet DD 4.4e−8 10 pts · median 3.2e−9 · † · top-free (particle_veto = [t]), Nf = 5 ‡ pts 4, 8 above 2.0e−8: accepted (user 2026-10-25)	**384 s** 9.54 GiB peak · plan 160 MB ML diagrams 17.4 h, stopped in export · Rc unfinished	**7.26 s** $B = 128$ no gate (no matched provider) numerator 51 %	**4.03 s** LC/FC **0.56** $B = 128$ · FC 7.26 s
	PRELIMINARY $2 \to 6$ (M-FINAL item 1, below Figure 1): pyAmpliCol only, no provider exists at $2 \to 6$, so no ratio, no gate and no colour · engine 71e2041e (`work/v2-bench-2to6`: 15d4ef5a + harness-only 8a062a52, d5ed94c2), native `loop-native-v2-slice-71e2041e` · pre-optimisation; runs stopped by the user on 2026-11-01, checks incomplete			
$2 \to 6$	$gg \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 1.0e−11 · E ≤ 1.7e−3 (2 pts: rescue) · **FC** rotations: pt 16 unstable (rescue) pending: Catani, Ward, DD (LC); poles, Catani, Ward, DD (FC)	**3.7 h** FC 235.2 GiB peak · plan 3 060 MB · store 63.8 min artifact 2.565 GB · load 19.4 min (159.9 GiB) **LC** 3.8 h · 217.2 GiB · artifact 2.14 GB	**478 s** one rotation block: 4 points x 4 copies · FFT $B = 16$ no provider: no gate inputs 56 % peak 239.2 GiB · **ran partly in swap: time likely inflated**	**119 s** 2 blocks of 16 $B = 16$ numerator 73 % peak 327.1 GiB
	$u\bar{u} \to gggggg$ 8 legs · QCD · 23 437 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 4.6e−12 · E ≤ 8.3e−7 pending: Catani, Ward, DD (LC)	**56.1 min** 41.8 GiB peak · plan 209 MB · store 580 s artifact 0.178 GB · load 74.9 s (12.6 GiB)	***not run*** not generated before the user's stop (its dry run passes)	**19.9 s** 2 blocks of 128 $B = 128$ numerator 63 % peak 90.9 GiB
	$u\bar{u} \to u\bar{u}gggg$ 8 legs · QCD · 13 318 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 6.7e−14 · Catani ε^{-1} 9.8 %: LC definitions differ · Ward ≤ 2.5e−12 · binary64 = DD to 2.3e−11 · E ≤ 2.6e−6	**23.6 min** 17.3 GiB peak · plan 53.7 MB · store 45.3 s artifact 0.023 GB · load 13.1 s (1.55 GiB)	***not run*** its store was killed by the host-memory floor 15 s after its start; not re-run before the user's stop	**4.72 s** 2 blocks of 128 $B = 128$ numerator 62 % peak 23.3 GiB

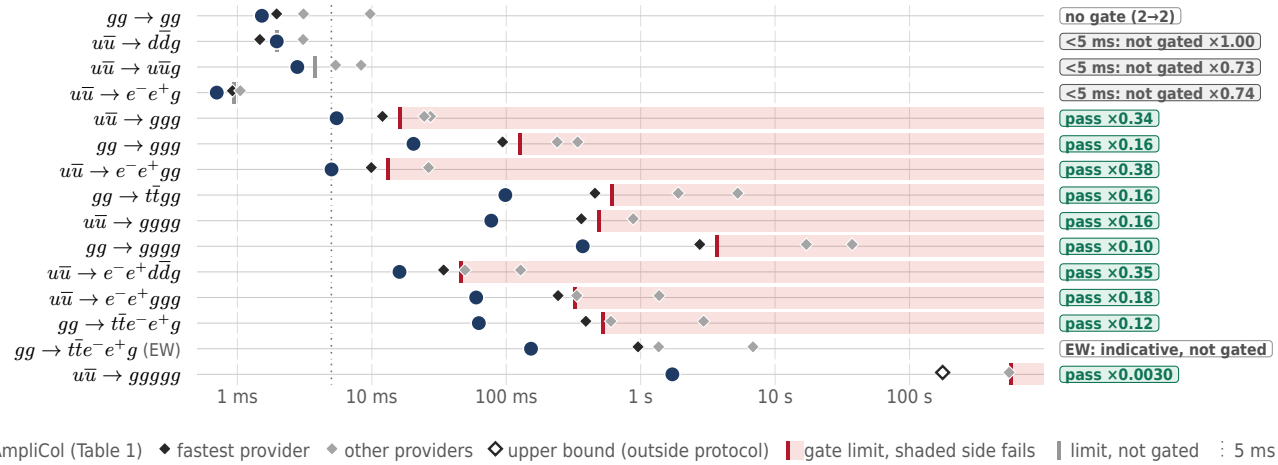


Figure 1: **Batched full colour against the gate** (finite part, $B = 128$; the rows of Table 1 with a matched provider).

Reading the summary

Table 1 has one row per test process, by multiplicity. A cell gives pyAmpliCol's value in bold and, at its right, the multiplier *pyAmpliCol ÷ the fastest provider* on this host (in bold among the provider values). A provider time marked ≤ ($u\bar{u} \to ggggg$) comes from a run outside the protocol that does the production work and more (Recola's reference run evaluates the IR poles too), so it bounds that provider's production time from above, and the multiplier against it is a lower bound (≥); it is never a gate limit. Only run times are coloured: green faster, white at parity, red slower ($\log_2$ scale, clamped at ×1/16 and ×16). Generation cells carry their multiplier uncoloured (generation time is not the main metric, user 2026-10-24); leading colour has no provider ratio. Grey cells say why a value is missing. Run times are per phase-space point in batches, finite part, stability checks off; providers in production mode (Section 1). Every value is the **most recent measured configuration** of its cell (by git ancestry of the revisions; a newer check that only repeats part of an older comparison does not replace it): $2 \to 2$ to $2 \to 5$: the engine on `loop` with native per-point inputs, gate G1 at `71e2041e` (2026-10-31). The engine before G-int (Python per-point inputs, G1 at `1c971289`) is history in Section 6. The column header of Table 1 names these configurations once.

Gate chips (Table 1 and Figure 1): the time over the binding limit (MadLoop's time, or 1.30 × the faster of Recola and OpenLoops), **pass** or **fail** when the whole uncertainty band is below or above 1, else **open**. The user's gate rule of 2026-10-24: a process whose time per point at $B = 128$ is below 5 ms has no blocking gate, its verdict is "below 5 ms: not gated (user 2026-10-24)" and its chip ("<5 ms: not gated") keeps the measured ratio ($u\bar{u} \to d\bar{d}g$ 1.96 ms, 1.00 of its limit, $u\bar{u} \to u\bar{u}g$ 2.79 ms, 0.73 of its limit and $u\bar{u} \to e^-e^+g$ 0.702 ms, 0.74 of its limit). $gg \to gg$ ($2 \to 2$) has no gate; $gg \to t\bar{t}e^-e^+g$ (EW, `data/msm-g1.json`) is timed and reported **indicatively**, chip "EW: indicative, not gated": the EW speed gate binds in M-FINAL (m-sm/PLAN.md [U5] and the milestone plan); its leading colour is refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4; typed colour-declaration-coverage). The gate is taken at $B = 128$ only (user 2026-10-28, `gate_decisions.gate_batch`): $B = 1$ and the $2 \to 5$ rows' $B = 16$ are measured and **reported** with their ratios (Table 11, Table 12), without a verdict or a verdict colour.

Stage shares (the bar of the full-colour cell; the largest is named; Table 13): inputs = trees, Born and counterterm inputs (on `loop` the executor's own timer of its native trees, since lc-perf); glue = point parameters, input conversion, assembly and the rest of the call. Times use the protocol's schedule B.

References and deviations. ML MadLoop 3.7.0, Rc Recola 2.2.5 (COLLIER), OL OpenLoops 2.1.5 (prebuilt libraries: no generation time), NJet DD NJet 3.1.2 in double-double. Deviation $|a - b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$ of the binary64 finite part at the timed points (median: over the points, the largest over the references); † the largest lies within the binary64 stability estimate of the same configuration at that point; ¶ the largest is the reference's own: pyAmpliCol's double-double value at that point lies as far from it, and its binary64 value lies within E of the double-double one. ‡ $gg \to ggggg$: the reference check's points 4 and 8 (4.4×10^{-8} and 2.1×10^{-8} from NJet DD) exceed its tolerance 2×10^{-8} in binary64: **accepted (user 2026-10-25): O(1e-8) binary64, within the estimate, DD = NJet** (estimates E 1.7×10^{-6} and 3.3×10^{-7}; pyAmpliCol's double-double values equal the reference's); the check keeps its tolerance and, asserting double-double at these points since the user's E1 ruling (2026-10-29), passes (`data/milestone.json validation_decisions`).

Configurations (main-configuration rule of `scripts/build_summary.py`; `data/summary.json` names the configuration and source of every value). $2 \to 2$ to $2 \to 5$: the engine on `loop`, gate G1 at `71e2041e`, build `loop-native-v2-slice-71e2041e`, native per-point inputs (Section 6). Generation excludes model loading; unbatched development timings (e.g. a seven-gluon point with its rotation check) are not in this table (Section 8). G1 generates every row cold since milestone m1026a (`11542fd5`), in its own process with a fresh cache directory, so the counterterm caches are rebuilt in every run (Section 6, generation protocol).

Notes under a value (`data/milestone.json` `cell_notes`) qualify it and leave it as measured, in amber a known limitation of the value, in grey its run conditions: $gg \to t\bar{t}e^-e^+g$, full colour ("model: the workspace's py3 G_mu copy"): timed on reference/models/loop_qcd_qed_sm_Gmu_py3 (the harness's processes.json default, as MS-VALIDATE's earlier run), not on the shipped patched UFO (UFOmodeldb 6ca4b120 + the user's issue-1 patch) that the M-SM campaign validated; an indicative timing, no verdict.

The $2 \to 6$ rows (M-FINAL item 1; `data/2to6.json` through `summary.json` `preliminary`; their records and caveats below) are **preliminary** and outside the main-configuration rule: pyAmpliCol only (no provider exists at $2 \to 6$), so no multiplier, no gate and no colour in any cell; engine `71e2041e` (`work/v2-bench-2to6`, harness-only commits on `15d4ef5a`), not this milestone's pin; generation = generate + plan export (store, artifact and load beside it; "FC" marks the full-colour generation where both were stored); the timed evaluation is binary64, finite part, checks off, one pinned thread. The process cell lists the internal checks done (ε^{-2}: double pole against the universal value; E: the recommended stability estimate; rotations; Ward; binary64 against double-double; Catani) and those pending.

Figure 1 places each row's batched full-colour value of Table 1 (dots; the most recent measured configuration: $2 \to 2$ to $2 \to 5$: `71e2041e`, native per-point inputs, G1 2026-10-31) against the providers (diamonds, the darkest the fastest) and the gate limit (red bar, the shaded side fails; grey bar: the limit of a process below 5 ms, not gated; dotted line: 5 ms), for the rows with a matched provider, at $B = 128$. The ratios and bands at every batch size: Table 12.

$2 \to 6$: preliminary records (M-FINAL item 1)

The $2 \to 6$ rows of Table 1 come from `data/2to6.json`, which `scripts/build_2to6.py` derives byte for byte from the existing records of the $2 \to 6$ benchmark (harness runs, watchdog and guard logs, dry runs and the stored artifacts' `generation.json`; 58 sources): harvested from existing records only (no new 8-leg run: STOP-8LEG); pre-optimisation engine; checks incomplete; runs stopped by the user. Engine: `work/v2-bench-2to6` = `15d4ef5a` plus the harness-only commits `d5ed94c2` and `8a062a52` (no engine source change), native `loop-native-v2-slice-71e2041e` (digest `f8d77c97`…), i.e. the engine of M-SM (`71e2041e`), not this milestone's pin. Protocol: binary64, finite part only (poles off), no stability check, one thread on one pinned core, warm-up excluded, paired blocks, setup heap frozen out of the cyclic collector particle veto [6] (top-free massless five-flavour QCD, user Q-A), 16 lanes. No external reference exists at $2 \to 6$: acceptance is internal (double pole against the universal value, Catani, Ward, double-double, rotations) and the speed is reported without a gate (cells uncoloured).

Table 2: **The $2 \to 6$ records** (`data/2to6.json` `rows`): generation stages (wall time · peak), the stored artifact and its load, the timed evaluation (finite part, poles and checks off; warm-up excluded, including lazy compilation) and the checks done or not obtained. Peaks are the harness's per-stage high-water marks (GiB = 2^{30} bytes).

Record	Generate	Export · plan	Store	Artifact · load	Evaluation	Checks
$gg \to gggggg$ LC · 47 348 families	77.2 min · 91.8 GiB	151.0 min · 217.2 GiB · 1 870 MB	47.6 min · 228.6 GiB	2.14 GB · 14.2 min, 139.6 GiB	**119 s/pt** at $B = 16$ (32 pts); warm-up 50.3 min peak 327.1 GiB	double pole, stability; not obtained: catani single pole, ward, double double
$gg \to gggggg$ FC · 47 348 families	46.3 min · 62.6 GiB	175.9 min · 235.2 GiB · 3 060 MB	63.8 min · 246.5 GiB	2.565 GB · 19.4 min, 159.9 GiB	**478 s/pt** at $B = 16$ (16 pts, 4 × 4 rotations); warm-up 163.9 min peak 239.2 GiB	rotation spread; not obtained: double pole, catani single pole, ward, double double
$u\bar{u} \to gggggg$ LC · 23 437 families	27.6 min · 15.4 GiB	28.5 min · 41.8 GiB · 209 MB	580 s · 31.6 GiB	0.178 GB · 74.9 s, 12.6 GiB	**19.9 s/pt** at $B = 128$ (256 pts); warm-up 435 s peak 90.9 GiB	double pole, stability; not obtained: catani single pole, ward, double double
$u\bar{u} \to u\bar{u}gggg$ LC · 13 318 families	15.6 min · 8.56 GiB	485 s · 17.3 GiB · 53.7 MB	45.3 s · 15.7 GiB	0.023 GB · 13.1 s, 1.55 GB	**4.72 s/pt** at $B = 128$ (256 pts); warm-up 98.1 s peak 23.3 GiB	double pole, catani single pole, ward, double double, stability

Caveats (`data/2to6.json` `global_caveats`, `row_caveats`):

- **Pre-optimisation engine.** The optimisation rounds the user asked for on 2026-11-01 (~23:00: "stop your 8-point runs now and do a couple round of optimizations first", RAM first) target exactly these generation, load and evaluation peaks; they run on the slice after this milestone's pin (round 1, mf-8g-2: bounded evaluation memory, integrated at `e071d022`; round 2 in progress) and are not part of this push. No 8-leg run of any kind is made until the coordinator lifts the stop: the numbers are a baseline, not the M-FINAL result.
- **Host.** One pinned thread on a shared, loaded host (other projects held ~600–950 GB); several steps were killed by host-memory floors (a "host memory" end is not a defect); timings indicative. $gg \to 6g$ full colour ran partly in zram swap: its trees time (265.8 s/pt against 20.0 at LC) is likely inflated, its block high-water mark understates the footprint, and its time is one rotation block of 16 evaluations (4 stream points × 4 copies, FFT colour contraction), not the paired two-block protocol of the LC rows.
- **Stability.** The full-colour rotation block finds stream point 16 binary64-unstable (spread 2.1: no correct digit), the other points below 10^{-5}: a **rescue case** for the stability stack (Section 9, Python API; not applied in this run). At LC the recommended estimate exceeds 10^{-5} at points 0, 1 (rescue candidates); no double-double cross-check exists at $gg \to 6g$.
- **Checks done and pending.** Double pole = universal value on every LC row (≤ 1.0e−11); on $u\bar{u} \to u\bar{u}gggg$ Ward (≤ 2.5e−12) and double-double (binary64 error 2.3e−11 within its estimate 3.3e−10) pass. Not obtained: Catani at $gg \to 6g$ and $u\bar{u} \to 6g$, Ward and double-double at both, every full-colour pole check (host-memory kills, then the user's stop).
- **Catani LC-definition offset.** On $u\bar{u} \to u\bar{u}gggg$ at LC the double pole passes (≤ 6.7e−14) but the single pole is 9.8 % off, while the two engines' LC Borns differ too: the tree engine's LC (colour-flow truncation) is not the loop engine's formal-N_c LC for quark lines, so the comparison is not like for like. The same route agrees at full colour on $gg \to ggg$ (2.8e−13); the decisive full-colour check is pending (its store was killed by host memory 15 s after its start).
- **Dry runs.** All 7 `Generator.plan` dry runs pass, the top-included $gg \to 6g$ full colour among them.

Table 3: **ESTIMATES [inf] ±50 %, not measured (8g0-report §5 'now -> after MF-8G-1/-2')**: the mf-8g-0 study's expected $2 \to 6$ costs after the optimisation packages, for orientation only; never a measurement or a Table 1 value.

Stage	$gg \to 6g$ LC, now (8g0-report)	FC, now	estimate after MF-8G-1/-2
generate	4630 s / 92 GiB	2779 s / 63 GiB	1500-2500 s / ≤ 60 GiB
export; store	9063 s / 217; 2858 s / 229 GiB	10553 s / 235; 3827 s / 247 GiB	900-1800 s / ≤ 75 GiB; 200-400 s, no new peak
load	854 s / 140 GiB	≈ 1100 s / 160 GiB	1-3 min / ≤ 20 GiB
eval peak B = 128	≈ 450-490 GiB (base 225-250 + state 213 + rows 15-25)	≈ 410-445 GiB	≤ 64 GB (§7)
eval time	119 s/pt	478 s/pt (swap)	dressing first, then trees

1 Status at this milestone: M-FINAL-b

On loop (engine f3d9e361; merges since 15d4ef5a, engine 71e2041e)

821ae808	MF-CORR	Recola correlation oracles
2234ee33	MF-RESCUE	Stability rescue stack
e1a1fea2	MF-SMALL	Small fixes
e2f0c78e	MF-GM-HC	HC_NLO_X0 genericity (+ 6605f6b4)
dbc92efb	MF-UP	SymJIT 2.28.3
25d39f6b	MF-CLEAN	Legacy agreement check removed
9ad0c77f	MF-MAP	Permutation aliases and group members
eddb4b49	MF-DEC-1	Tree-level 1 -> n decays
7cdd32e9	MF-8G-0	2 -> 6 feasibility study
b36af289	MF-RUST	EW export parses in Rust
28b24a2a	MF-DEC-2	One-loop 1 -> n decays
617b59cd	MF-SMALL2	Determinism and spin-source guard
f3d9e361	mf-map-fix	Mapping test spin classes (test only)

Before (engine 71e2041e): v2 waves 0-1 (gates S1, G0), native executor, colour, stability API, schedule M, observables, NJet reference, public v2 engine with plan v4 artifacts, counterterms as plain tree insertions (addendum J), runtime pole selection, Rust A0-A3, the G1 harness, G-int (native per-point inputs and generation stages), the summary-first report; at milestone m1025a ct-perf, ew-poles (EW guards, QED Catani operator), lc-perf, artifact-portability and ew-lines; at m1025c native-dag A4, ew-gauge EWI-4a, precision-ap, ew-ct EWI-0/EWI-3, measure-g1-7leg (G1 at 2->5), reference-relocate, model-determinism, native-lowering D phase 1 part 1 and onshell-consistency; at m1026a ew-gauge EWI-4b, native-seeds B phase 1, native-cells C phase 1, genericity-fixes, ew-poles EWI-6b, fermion-lines and generation-scale-2; at m1028a ew-ct EWI-3R, ew-epsilon EWI-7b, fail-closed, sdk-parity, ew-leptons EWI-5, identical-flavour-lc and gate-b128; at M-QCD onshell-threshold, fail-closed-2, glibc-floor, MQ-WARD, MQ-FFT, MQ-REF, MQ-MODEL, MQ-STAB, MQ-VALIDATE, MQ-NC, MQ-5F, MQ-EWJOIN, MQ-SELECT, MQ-LEGACY-REMOVE and g1-accuracy; at M-SM ew-native, ew-validate, MS-REF, MS-PHOTON, MS-CMS-N, MS-MODEL, MS-CMS, MS-GEN, MS-POLES, MS-UFODB, MS-UFO-PATCH, MS-VALIDATE, ms-policy-fix and MS-MASTERS. **Native build** `loop-native-v2-slice-b36af289` (digest `58b40383`...); G1 ran on `loop-native-v2-slice-71e2041e` (engine `71e2041e`).

M-FINAL-b (item 2 complete and the item-4 packages, `m-final/PLAN.md` §3.1 wave 5; campaigns at `617b59cd`, the pin adds a test-only commit)

- **Decays $1 \to n$** (Section 2, Table 28): 8 rows XX01–XX08 ($1 \to 2$ and $1 \to 3$, QCD and EW, `loop_sm` and both shipped EW UFOs), 8 pass, 0 fail. Largest: DD from MadLoop's quadruple precision 9.6×10^{-14} (XX08), binary64 from DD 5.2×10^{-11} (XX08), Catani and QED poles in DD 5.7×10^{-27} (XX08); $Z \to u\bar{u}$ = the massless vector form factor, Ward on XX07, LC/NLC = the N_c scan; public route and native inputs. The decaying particle at its real mass with zero own width.
- **HC_NLO_X0 genericity** (Table 5): HC01, HC02, HC03, HC04, HC06 = MadLoop 3.7.0 to $\le 2.4\times10^{-13}$; HC05 ($gg \to$ x0 gg) blocked, typed `unlowered-counterterm` (five-point $ffff$ R_2 contact).
- **Correlations against Recola** (Table 6): 16 rows on 4 processes pass, correlated Borns $\le 6.5\times10^{-14}$, one-loop Laurent coefficients $\le 4.1\times10^{-10}$ of the row's scale.
- **Product** (Table 24): the stability rescue stack (Python native evaluate API only; `test_loop_v2_rescue.py` 16/16); permutation aliases and permuted group members served from the stored target (`test_loop_v2_mapping.py` 17/17 at `f3d9e361`); SymJIT 2.28.3; virtual C ABI version 2 (legacy agreement check removed); EW export parses in Rust.

Gate G1 (Section 6; finite part, $2 \to 3$ and beyond)

- **Not re-timed here** (user Q11: one final retime at the M-FINAL pin): Table 1 and Figure 1 keep M-SM's G1 at `71e2041e` (`loop-native-v2-slice-71e2041e`). Regression spot check of the pin's engine against `71e2041e` (same core, paired; the four largest gated rows, the smallest and the EW row): plans identical, time per point ×0.96–1.08: no regression.
- **Most recent configuration** (Table 1 and Figure 1): native per-point inputs on `loop` (G1 at `71e2041e`) at $B = 128$: 10 of 10 gated processes pass; $u\bar{u} \to d\bar{d}g$, $u\bar{u} \to u\bar{u}g$ and $u\bar{u} \to e^-e^+g$ run below 5 ms per point: **not gated** (user 2026-10-24; 1.00, 0.73, 0.74 of the limit). $B = 1$ (16 lanes, the runs of `28a16255`, not re-timed) is **reported**, not gated (user 2026-10-28): ×1.0–5.3 of the limit (Table 12).
- **$2 \to 5$** (full and leading colour at $B = 128$; $B = 16$ reported): 4 of 4 gated rows pass ($u\bar{u} \to e^-e^+d\bar{d}g$ 16.1 ms, 0.35 of its limit; $u\bar{u} \to e^-e^+ggg$ 59.7 ms, 0.18 of its limit; $gg \to t\bar{t}e^-e^+g$ 62.5 ms, 0.12 of its limit; $u\bar{u} \to ggggg$ 1.72 s, 0.0030 of its limit, ≥×0.0094 against Recola's upper bound); $gg \to ggggg$ 7.26 s at full and 4.03 s at leading colour per point, no matched provider and no gate.
- **Before G-int** (Python per-point inputs, G1 at `1c971289`; history, Section 6): failed every gated process it generated, at ×11–20 of the binding limit at $B = 128$.
- **EW row** indicative (`data/msm-g1.json`, not gated: [U5], binding in M-FINAL): $gg \to t\bar{t}e^-e^+g$ (EW) 153 ms per point at $B = 128$, ×0.15 of Recola's time (Section 4).

M-SM, kept (QCD + QED/EW on the shipped `loop_qcd_qed_sm(_Gmu)`; `m-sm/PLAN.md` §5; re-run at this pin)

- **Validation campaign** at `617b59cd` (Section 4, Table 29): 71 rows in 10 classes, 69 pass, 2 partial, 0 fail (LL02, SC02: F7, accepted); no deviation above 10^{-5}, no wrong pole or sector; against M-SM's campaign no status change and no deviation change above 10^{-5} (40 values in 8 rows moved by $\le 7.4 \times 10^{-11}$). Largest: DD from the reference 4.9×10^{-9} (SC02), binary64 from DD 1.9×10^{-10} (TV06), Catani DD 7.2×10^{-26} (CM07) at rule 4′s standard tolerance (relaxed by the user's ruling: TT05 $1.1\times10^{-29} \le 1\times10^{-14}$, TT06 $4.2\times10^{-30} \le 1\times10^{-13}$, SC03 $1.1\times10^{-6} \le 1\times10^{-5}$, SC05 $4.1\times10^{-14} \le 1\times10^{-12}$).
- **CMS** (Table 8): 9 rows = MadLoop CMS; the acceptance row CM07 ($gg \to t\bar{t}e^-e^+g$) DD 6.9×10^{-13}, binary64 from DD 5.6×10^{-11}, native inputs.

M-QCD, kept (QCD on the stock `loop_sm`; `m-qcd/PLAN.md` §5; re-run at this pin)

- **Campaign re-run** at `617b59cd` (2026-11-02, Section 5, Table 30): 78 rows, 76 pass, 1 partial, 1 fail (F7, F11, accepted), the statuses and check counts of M-QCD's campaign (`e18a024b`); against M-SM's re-run 42 of its 1212 values moved, all binary64 (QCD-only rows too; the largest, information only: A14′s finite part against the providers $1.5 \times 10^{-8} \to 5.6 \times 10^{-8}$), every double-double value bitwise. Largest: DD from the reference 1.2×10^{-11} (A18), binary64 from DD 2.3×10^{-10} (C05), Catani DD 1.2×10^{-22} (A09).
- **At `e18a024b`** (not repeated: no status changed): public route on the shipped `loop_sm`, one row per class, DD from MadLoop $\le 3.6\times10^{-13}$; FFT seed contraction = direct to rounding, ×0.46–5.9 of direct (Table 10). **Colour:** full colour by default; LC and NLC pass rule 7 on 13 rows. **Ward** up to seven gluons (passed (MQ-WARD)).

- **Public route** on both shipped EW UFOs: `test_loop_v2_ew_public_models.py` 25/25 at the pin; wheel self-test 7/7 in two fresh venvs, its EW virtual from the shipped prepared artifact = MadLoop to 5.4e-15.
- **EW products:** per-leg photon tagging (`loop.external_photons`, default `"final"`), poles `uv | ir | split` by the model's regulators, native complex pole masses, EW generation stages in Rust, binary64 of a stored artifact without Symbolica (Table 24).

$2 \to 6$ (M-FINAL item 1, preliminary Table 1, records and caveats)

- **From existing records only** (`data/2to6.json`; engine `71e2041e`, pre-optimisation): $gg \to 6g$ 119 s per point at LC and 478 s at full colour (one rotation block, partly in swap), $u\bar{u} \to 6g$ 19.9 s and $u\bar{u} \to u\bar{u}gggg$ 4.72 s at LC; generation up to 3.7 h and 235.2 GiB (generate + plan export).
- **Internal checks** (no reference exists): double pole = universal value on every LC row; Ward and double-double pass on $u\bar{u} \to u\bar{u}gggg$, whose LC Catani differs by the LC definitions (full colour pending); the full-colour point 16 of $gg \to 6g$ is binary64-unstable, a rescue case; the rest is pending.
- **User stop** (2026-11-01 ~23:00): no 8-leg run until the coordinator lifts it; optimisation rounds first (RAM, generation and runtime).

Next: M-FINAL-a (`m-final/PLAN.md` §3.1)

- The optimisation rounds on the slice after this pin (round 1, bounded evaluation memory: mf-8g-2, integrated; round 2: load-time and evaluation memory, evaluation runtime), then the $2 \to 6$ runs again: storage below 150 GB, the pending checks, full colour.
- The SDK v4 extras (resolved, colour/spin-correlated, double-double and arbitrary precision in every SDK language and UMAMI; integrated after this pin, mf-sdk), the final G1 retime with the new processes and the **binding EW speed gate**, then the M-FINAL gate and push.

Missing or incomplete (typed refusal keys, listed and shown by the dry run)

- **Refused (typed):** loop-induced processes, including loop-induced decays ($h \to gg$) and the loop-squared members of multiparticle requests (`loop-induced`; a multiparticle request generates its members with a Born and records the others as skipped); decay chains and `$` (`madloop-syntax`); Majorana fermions (`majorana-fermion`) and multi-fermion contact vertices (`multi-fermion-contact`); the five-point $ffff$ R_2 contact of HC05 (`unlowered-counterterm`: its colour is a multi-trace state); numerical process-group analysis (`process-group-analysis`; exact aliases and members are served); photon-only and partial EW vetoes on the NLOCT UFOs (`veto-unlabelled-counterterms`: their counterterm couplings are not split by loop content); the complex-mass scheme for a width whose mass is an internal parameter (`cms-internal-mass`: `loop_sm`'s W), an external unstable species under CMS (`cms-external-width`), a Yukawa input under CMS (`cms-yukawa-input`); a width outside CMS (`electroweak-width`, `coloured-width`; a decaying particle has zero own width); tagged photons without an $\alpha(0)$ input (`external-photon-scheme`); Dirac–Weyl currents the tree lanes cannot close (`dirac-weyl-closure`); a spin-correlated leg without a vector source (refused at generation).
- **M-FINAL-a, in progress:** the $2 \to 6$ validation (Catani, Ward, double-double, full-colour poles; after the optimisation rounds, records); the **binding EW speed gate** with the final G1 retime (the EW row is indicative here); the SDK v4 extras (after this pin).
- **Not in M-FINAL** (rulings): loop-induced processes (user Q8); helicity and colour-flow selection at NLC and full colour (LC has it; refused for mapped processes); the publication-grade timing campaign and the provider protocol-mode columns (user Q11).
- **Not supported, by design:** EW LC/NLC (`colour-declaration-coverage`, permanent: user 2026-10-30; no plan to lift it).
- **Accepted at the pin:** F7 — LL02 and SC02 partial (rule 4 not formed: the massive-muon G_μ copy writes the field renormalisation of massless leptons; rule 1 = MadLoop); SC03's τ-mass relaxation of rule 4 to 10^{-5} (the EW counterterms omit the τ mass, the user's retained model limitation; MadLoop has the same offset). M-QCD's A09@collinear (F7: typed `Unstable` / `CutFailed{DegenerateGram}`, never a value) and A24 (F11: rule 4 not formed at seven gluons, poles = NJet).
- **Recorded test outcomes** (not gate failures): `test_loop_v2_companion.py::test_companion_seeds_on_the_slice_certificates_ggg` misses a bitwise claim by ~10^{-16} relative since SymJIT 2.28.3 (below the lax-precision line; routed to mf-8g-1b); the five NinjaRS light-like spinor xfails stay (NinjaRS #10 open).
- **Compatibility notes:** the shipped EW UFOs are UFOmodeldb `6ca4b120` plus the user's issue-1 patch (PROVENANCE records the local delta); MadLoop never inserts R2_G0tt_b on the unpatched models (mg5amcnlo#475): pyAmpliCol keeps the UFO-complete R_2 insertion (user Q7).
- **Fast binary64 lanes:** NinjaRS light-like spinors near the $-z$ axis (~ 10^{-5}) and model units ~10^4 from GeV (silent percent-level errors with checks off): the recommended checks flag them, and the rescue stack escalates such points in the Python API.
- **No longer missing:** $1 \to n$ decays at tree level and one loop, QCD and EW (M-FINAL-b); permutation aliases, process-group members and correlations composed with them; independent Recola oracles of correlated Borns and virtuals; the stability rescue stack (Python API); the EW corrections, CMS and external-photon artifacts of M-SM.

Performance gates and timing protocol: definitions

- **Gate.** Faster than MadLoop and at most 1.30× each matched Recola and OpenLoops time, on the **finite part**, from $2 \to 3$ upward, **taken at $B = 128$ only** (user 2026-10-28, MASTER_PLAN §5.8: $B = 1$ and the $2 \to 5$ rows' $B = 16$ are measured and reported with their ratios, without a verdict; a configuration without a $B = 128$ timing has none); $2 \to 2$ is reported without a gate. A process whose time per point at $B = 128$ is below 5 ms has no blocking gate: its ratio is shown, its verdict is "below 5 ms: not gated (user 2026-10-24)" (user decision; `data/milestone.json` `gate_decisions.threshold_rule`); optimisation effort goes to higher multiplicities. The electroweak row is timed and reported indicatively at M-SM, without a verdict ([U5]; `data/msm-g1.json`): its gate binds in M-FINAL. Aspiration: fastest of all tools from $2 \to 3$.
- **Protocol.** Providers in production mode with poles off where they allow it (OpenLoops `truepoles 0`, Recola without IR poles, MadLoop production build); pyAmpliCol finite-only (`poles = "none"`) with stability checks off, one pinned thread, binary64, 16 lanes, the same frozen points (pyAmpliCol receives their exact rationals, the providers their binary64 decimal text); since milestone m1025a pyAmpliCol's setup heap is frozen out of Python's cyclic collector before the warm-up (a loaded runtime holds no generation objects; in the committed harness since `de8d3eb8`, which times G1 at `71e2041e`; Section 6). Statistic: median per point; the uncertainty band combines pyAmpliCol's block spread and the provider's repeat spread; a gate whose band contains the limit is **open**.
- **Timing boundary.** pyAmpliCol: the public v2 runtime call (`NativeExecutor.evaluate`) for a batch of B distinct points, everything included (per-point inputs, loop stage, assembly); the public `Runtime.evaluate` of a loaded artifact adds its input parsing (Section 6). Providers have no batch API: one time per point serves every B. Generation: pyAmpliCol `LoopSlice.generate` plus the plan export, in G1 on one pinned core, cold (fresh cache directory) since `11542fd5` (m1026a); a development record's generation in Table 1 ran on the shared host under load (threads per increment: Table 21), without a multiplier in a G1 row; MadLoop graph generation, export and compilation; Recola process generation; OpenLoops libraries are prebuilt (no generation time).
- **Labels.** Table 1 and Figure 1 show the most recent measured configuration of each cell (main-configuration rule of `scripts/build_summary.py`; `data/summary.json` names the configuration and source of every value); the column header of Table 1 states it once, with branch and commit, and "Reading the summary" says when it is a development measurement not yet on `loop`. The detailed G1 tables (Section 6) show the configuration on `loop` and, labelled, its history: the runs of milestones M-QCD (`e18a024b`), m1028a (`28a16255`), m1026a (`11542fd5`) and m1025a (`1d27426b`) and of G-int (`4e9b5a9e`), the Python per-point inputs before G-int and the development runs of `39824f9b`. Missing values are marked, never estimated or extrapolated; test durations are never used as event timings. The legacy per-diagram engine's timings of 18 October are history (data kept in `data/measurements.json`, not shown).

2 Decays $1 \to n$ (M-FINAL item 2)

Table 4: **Decay capability matrix** at 617b59cd (native loop-native-v2-slice-b36af289, campaign of 2026-11-01 data/dec-matrix.json from data/dec.json; per-row data: Table 28): every row against MadLoop's quadruple-precision reference of the same decay (MF-DEC-1 fixtures; the decaying particle at its real mass with zero own width, its absorptive self-energy dropped, as MadLoop does).

Row	Process	Corr.	Model · scheme	Sectors	DD vs ML	b64/DD	Poles (DD)	LC/NLC	Public · inputs	Status
XX01 1→2	$t \to bw^+$	QCD	loop_sm no_b_mass · QCD, 5F	QCD2 QED2	3.9e−16	1.4e−14	1.2e−31 Catani	N_c scan ✓	public ✓	pass
XX02 1→2	$t \to bw^+$	QCD+QED	$G_\mu \cdot G_\mu$	QCD2 QED2, QED4	2.9e−15	2.0e−13	2.3e−30 Catani + QED	—	native inputs ✓	pass
XX03 1→2	$z \to e^-e^+$	QED	qcd_qed_sm · $\alpha(m_Z)$	QED4	2.9e−15	2.0e−15	9.3e−31 Catani + QED	—	—	pass
XX04 1→2	$z \to u\bar{u}$	QCD+QED	qcd_qed_sm · $\alpha(m_Z)$	QCD2 QED2, QED4	6.1e−15	3.8e−13	1.9e−30 Catani + QED	—	—	pass form factor
XX05 1→2	$w^+ \to e^+$ ve	QED	qcd_qed_sm · $\alpha(m_Z)$	QED4	1.8e−15	5.4e−14	7.5e−31 Catani + QED	—	—	pass
XX06 1→2	$h \to b\bar{b}$	QCD	loop_sm · QCD, 4F	QCD2 QED2	4.1e−16	3.0e−15	1.9e−32 Catani	—	—	pass
XX07 1→3	$t \to bw^+g$	QCD	loop_sm no_b_mass · QCD, 5F	QCD4 QED2	5.6e−15	1.4e−13	4.5e−31 Catani	N_c scan ✓	public ✓	pass Ward
XX08 1→3	$z \to u\bar{u}g$	QCD+QED	qcd_qed_sm · $\alpha(m_Z)$	QCD4 QED2, QCD2 QED4	9.6e−14	5.2e−11	5.7e−27 Catani + QED	—	—	pass

Reading the matrix. The decay campaign re-ran every row at the pin's Python tree (617b59cd, the native loop-native-v2-slice-b36af289): 8 pass, 0 partial, 0 fail, as at its integration (no status change; the largest metric change 7.9×10^{-31}). Rows XX01–XX08 cover $1 \to 2$ and $1 \to 3$, QCD (loop_sm, five and four flavours) and EW (QED and joint QCD+QED corrections on both shipped EW UFOs, G_μ and $\alpha(m_Z)$); the deviations are the largest over every Laurent coefficient, sector, point and gating variant: DD from MadLoop's quadruple precision (rule 1), binary64 from DD where binary64 is asserted (rule 2), the poles against Catani, plus the QED operator for EW corrections, in DD (rule 4). XX02 runs on the native per-point inputs; XX01@public and XX07@public generate through Generator on the shipped loop_sm and evaluate the stored artifact with the recommended checks. **Further checks:** XX04: $Z \to u\bar{u}$ (QCD) equals the analytic massless vector form factor (1 test, pass) XX07: the Ward identity on the external gluon of $t \to bw^+g$ (8 checks; virtual 5.0×10^{-31} and Born 1.3×10^{-31} of the physical scale in DD, binary64 within its estimate) XX01 and XX07: LC and NLC are the kept powers of the N_c scan (3 tests, pass). Loop-induced decays are refused typed (loop-induced, e.g. $h \to gg$), chains and $ too (madloop-syntax).

3 Model genericity and correlation oracles (M-FINAL item 4)

Table 5: **HC_NLO_X0 rows against MadLoop 3.7.0** (data/hc.json at 617b59cd, loop-native-v2-slice-b36af289 tests/unit/test_loop_v2_hc.py 15/15 passed): the Higgs characterisation UFO at NLO (v1.3) with its restriction and parameter cards (κ settings), the largest relative deviation deviation |ours - MadLoop| / max(|ours|, |MadLoop|, |Born| alpha_s/(2 pi)) per Laurent coefficient (acceptance 10^{-5}), the poles against Catani, the rotation estimate and double-double where run.

Row	Process	Restriction · veto	Pts · values	Born	Finite	ε^{-1}	ε^{-2}	Catani	Rotation	Status · DD
HC01	$gg \to$ x0	heft · / t	1 · 6	3.5e−16	1.3e−15	2.0e−15	8.4e−16	1.9e−15	1.9e−14	pass
HC02	$gg \to$ x0 g	heft · / t	2 · 12	1.7e−15	1.8e−14	3.1e−15	2.0e−15	5.2e−15	1.4e−13	pass DD 1.5e−14
HC03	$u\bar{u} \to$ x0 g	heft · / t	2 · 8	1.4e−15	2.4e−13	1.4e−14	2.6e−15	3.4e−14	6.5e−13	pass
HC04	$u\bar{u} \to$ x0 z	no_b_mass	2 · 8	1.5e−14	8.4e−14	8.2e−15	8.1e−15	2.4e−14	1.2e−13	pass
HC06	$gg \to$ x0 $t\bar{t}$	no_b_mass	2 · 8	2.2e−15	2.9e−14	2.2e−14	1.6e−14	9.6e−15 (ε^{-2} only)	6.2e−13	pass
HC05	$gg \to$ x0 gg	heft · / t	blocked typed unlowered-counterterm: five-point f*f*f*f R2 contact (G G G G X0) needs the multi-trace colour state (DESIGN §6.1); recorded limitation (optional row)							

The HC rows ran through the generic fixes of MF-GM-HC (e2f0c78e): the UFO namespace head, the Feynman-gauge refusal only where massive vectors run in the loops (ew_policy.require_loop_gauge), positional UFO counterterm keys with zero extra positions (other key lengths are the listed source refusal ufo-counterterm-keys) and a four-point contact with a colour-singlet leg split through an adjoint auxiliary. HC05 ($gg \to$ x0 gg) stays refused at generation: the five-point R_2 contact [G G G G X0] carries the colour $ffff$ that the tree colour flow holds only as the multi-trace colour state of DESIGN §6.1, so the reachable vertex is refused, never dropped (addendum J); a recorded limitation (optional row).

Table 6: **Colour, spin and spin-colour correlations against Recola** (data/corr.json at 617b59cd, tests/unit/test_loop_v2_corr_recola.py 12/12 passed; references tests/fixtures/loop/required/corr/<process>.json (Recola2 2.2.5, binary64)): per process and kind the entries and points, the correlated Born of the tree engine and of the one-loop runtime relative to Recola's (acceptance 10^{-11}), and the one-loop Laurent coefficients relative to the row's largest Recola coefficient (acceptance 10^{-8}).

Process	Kind	Entries · pts	Tree Born	Loop Born	Finite	ε^{-1}	ε^{-2}	Status
$u\bar{u} \to gg$	plain Born	2 · 2	1.7e−15	9.2e−16	9.4e−14	2.7e−14	1.3e−15	pass
$u\bar{u} \to gg$	colour	24 · 2	1.4e−15	7.7e−16	1.7e−13	8.1e−14	2.6e−14	pass
$u\bar{u} \to gg$	spin	4 · 2	1.8e−15	7.2e−16	1.0e−13	2.5e−14	2.1e−15	pass
$u\bar{u} \to gg$	spin-colour	12 · 2	1.5e−15	6.9e−16	2.5e−13	6.8e−14	4.1e−15	pass
$gg \to t\bar{t}$	plain Born	2 · 2	1.1e−15	6.7e−16	1.2e−15	1.5e−16	2.7e−17	pass

Process	Kind	Entries · pts	Tree Born	Loop Born	Finite	ε^{-1}	ε^{-2}	Status
$gg \to t\bar{t}$	colour	24 · 2	2.0e−15	9.6e−16	2.8e−14	2.6e−15	6.5e−15	pass
$gg \to t\bar{t}$	spin	4 · 2	2.0e−15	1.2e−15	2.1e−15	1.1e−15	1.7e−15	pass
$gg \to t\bar{t}$	spin-colour	12 · 2	1.7e−15	6.3e−16	1.3e−14	2.4e−15	1.4e−15	pass
$u\bar{u} \to d\bar{d}g$	plain Born	2 · 2	4.8e−15	2.0e−14	1.2e−10	4.7e−11	7.4e−12	pass
$u\bar{u} \to d\bar{d}g$	colour	40 · 2	1.3e−14	6.5e−14	4.1e−10	1.8e−10	3.4e−11	pass
$u\bar{u} \to d\bar{d}g$	spin	2 · 2	5.1e−15	1.9e−14	9.7e−11	3.9e−11	6.5e−12	pass
$u\bar{u} \to d\bar{d}g$	spin-colour	8 · 2	1.2e−14	2.8e−14	3.5e−10	1.7e−10	3.4e−11	pass
$gg \to ggg$	plain Born	2 · 2	5.0e−15	1.7e−14	1.9e−12	7.7e−13	4.0e−13	pass
$gg \to ggg$	colour	40 · 2	9.5e−15	2.7e−14	1.0e−10	4.0e−11	6.1e−12	pass
$gg \to ggg$	spin	10 · 2	7.4e−15	2.3e−14	3.1e−12	1.5e−12	5.2e−13	pass
$gg \to ggg$	spin-colour	40 · 2	1.3e−14	2.6e−14	1.1e−10	3.9e−11	7.0e−12	pass

4 Electroweak capability matrix (M-SM)

Table 7: **Electroweak capability matrix** at 617b59cd, per class of m-sm/PLAN.md §1.1 (campaign of 2026-11-01/02 at the M-FINAL-b pin, M-SM's of 2026-10-31 re-run, on the shipped patched EW UFOs, data/msm.json; legend below the table, rows: Table 29).

Class	Rows	Public route	DD vs ref.	binary64	Catani (DD)	Native inputs	CMS	Status
W-lep lepton-only [virt=QED], lines through the loop, Bhabha, W pairs, tops · up to $2 \to 3$ · qcd_qed_sm, G_μ, G_μ, m_μ	**7** pass 6 partial 1	LL01, self-test (LL03) ✓ bitwise the tools view; 1 against MadLoop QP	**4.6e−13** LL08	**2.3e−12** LL08 · ref. 2.3e−12†	**3.5e−28** LL08 · LL02 not formed	— public route: native	CM01 ✓ same processes	accepted 6 of 7 pass; F7 LL02, SC02 (accepted)
W-DY neutral and charged Drell-Yan, lepton-hadron, + jets, alpha(MZ) twin · up to $2 \to 4$ · qcd_qed_sm, G_μ	**7** pass 7	DY01 ✓ bitwise the tools view; 1 against MadLoop QP	**9.1e−13** DY05	**4.7e−12** DY02 · ref. 4.7e−12	**1.1e−28** DY07	— public route: native	CM02, CM03, CM05 ✓ same processes	pass 7 of 7 pass; F4 DY05, VJ03, VV06 (open)
W-V V/photon + jet, VV, VVV, VH, VBF H · up to $2 \to 3$ · G_μ	**13** pass 13	VJ02, VV02, VJ03 ✓ bitwise the tools view; 2 against MadLoop QP	**5.3e−11** VV07	**9.2e−11** VV07 · ref. 4.2e−9†	**1.1e−26** VV07	— public route: native	—	pass 13 of 13 pass; F4 DY05, VJ03, VV06 (open)
W-top t t~ (joint QCD+QED), + jet, four tops, massive b, t t~ V/H/ll(+g) · up to $2 \to 5$ · G_μ	**11** pass 11	TT01 ✓ bitwise the tools view; 1 against MadLoop QP	**1.4e−11** TT03	**1.9e−10** TV06 · ref. 9.6e−5†	**6.4e−26** TV07 · TT05 ≤ 1×10^{-14} · TT06 ≤ 1×10^{-13}	TT04, TV07 ✓ public route: native	CM04, CM07 ✓ same processes	pass
W-singletop s- and t-channel, tW; massless and massive b · up to $2 \to 2$ · G_μ	**4** pass 4	— no public case	**3.0e−12** ST04 · MadLoop + Recola	**2.3e−11** ST04 · ref. 2.3e−11†	**4.8e−27** ST04	— campaign rows	—	pass
W-jets QCD processes with EW orders, every split order · up to $2 \to 3$ · G_μ	**4** pass 4	JJ01 ✓ bitwise the tools view	**6.9e−13** JJ04	**7.6e−12** JJ04 · ref. 8.3e−12	**2.0e−28** JJ04	— public route: native	—	pass
W-photon final and initial photons, tagged per leg · up to $2 \to 2$ · G_μ	**4** pass 4	VJ03 ✓ bitwise the tools view	**9.7e−15** PI04	**2.9e−13** PI03 · ref. 2.9e−13	**1.0e−29** PI04	— public route: native	CM09 ✓ same processes	pass
W-CMS internal widths > 0, coloured too; tagged photons; loop_sm · up to $2 \to 5$ · G_μ, loop_sm	**9** pass 9	— no public case	**6.9e−13** CM07	**1.8e−10** CM10 · ref. 4.8e−9†	**7.2e−26** CM07 · CM08 no sector (uncorrected-exchange) · CM10 no sector (uncorrected-exchange)	CM07 ✓ campaign rows	9 rows ✓ against MadLoop CMS (Table 8)	pass
W-multi multiparticle labels, no-Born members skipped (Q2) · up to $2 \to 2$ · G_μ	**2** pass 2	— no public case	**4.2e−14** MP01	**1.5e−12** MP01 · ref. 1.5e−12†	**1.0e−29** MP02	— campaign rows	—	pass
W-scheme alpha(MZ), massive mu, CKM, massive b, massive initial states, eps sign · up to $2 \to 4$ · G_μ, G_μ, m_μ, loop_sm	**10** pass 9 partial 1	— no public case	**4.9e−9** SC02	**1.1e−10** SC06 · ref. 3.4e−4†	**1.1e−6** SC03 · SC02 not formed · SC03 ≤ 1×10^{-5} · SC05 ≤ 1×10^{-12}	SC06, SC07, SC08, SC09 ✓ campaign rows	CM03 ✓ same processes	accepted 9 of 10 pass; F7 LL02, SC02 (accepted)

Rows: campaign rows and their status (passing every rule of their tier; partial: a rule not formed); class status **accepted**: every other row is a finding the coordinator accepted. Models: G_μ = loop_qcd_qed_sm_Gmu, qcd_qed_sm = loop_qcd_qed_sm (both shipped: UFOmodeldb 6ca4b120 + the user's issue-1 patch), G_μ, m_μ its massive-muon copy. **Public route**: the campaign's cases of test_loop_v2_ew_public_models.py (Generator on the shipped UFO with its shipped profile, Runtime.load, evaluate; Born and virtual bitwise the tools view at two points, and = MadLoop's quadruple precision where the case names a fixture) at the row of their process, and the wheel self-test's EW virtual from the shipped prepared artifact. **DD vs ref.**: largest relative deviation of the double-double Laurent coefficients (Born, finite, poles, every sector) from the row's reference (rule 1; MadLoop in quadruple precision, "double": its double-precision control exports). **binary64**: from DD where binary64 is asserted (rule 2, estimate $E \leq 10^{-9}$); "ref." from the reference at every point (information; † the class has points with $E > 10^{-9}$, where DD is asserted). **Catani**: DD poles against the Catani operator (rule 4, every point and predicted sector; "≤" a tolerance relaxed by the user's ruling of 2026-10-29 ~10:20; "no sector": no predicted sector, typed reason in Table 8). **Native inputs**: the rows the campaign ran on the native per-point inputs (production route); the public route is native. **CMS**: the complex-mass rows of the class's processes.

Reading the matrix. The campaign ran every row of its tier at the pin 617b59cd (native loop-native-v2-slice-b36af289) as whole test files, one process each: 23 nightly file jobs, four restriction-smoke files, the default tier and the campaign tier (TV07, its QCD twin and CM07 through the harness CLI), the manifest's points in binary64 and double-double (rules 1–4; rule 5 and the extras on the rows that name them; m-sm/PLAN.md §1.4 under the user's ruling of 2026-10-29 ~10:20: a deviation below ~10^{-5} is not a finding). Its counts: 69 pass, 2 partial, 0 fail, 0 pending, 0 not run; the 13 smokes and multiparticle expansions all pass. The two partial rows are the accepted F7: rule 4 is not formed on the massive-muon G_μ copy, which writes the field renormalisation of massless leptons (rule 1 = MadLoop). Notes of the campaign (data/msm.json coverage_notes): rule 4 is not formed on LL02 and SC02 (F7): a model property, the rows report partial; the Ward extra is a diagnostic under the complex-mass scheme (CM05, CM07, CM09; PLAN 1.4/1.5) and not run on VV05 and ST03 in the real-mass scheme (F3); rule 7 (nc) is not run: EW LC/NLC is never supported (USER Q4, 2026-10-30 ~08:10; typed colour-declaration-coverage); a row whose evaluation raises records the error as its one failed check (the F1 and F2 rows did until the fixes: 52eaff60, and 19619e62 / ms-ufo-patch's decided_number); on the shipped patched models (MS-UFO-PATCH) MadLoop's exports call every R2 coupling the engine inserts (the R2 rule finds no omission; on the unpatched models R2_G0tt_b on LL10, TT04, SC09, TT05).

Table 8: **Complex-mass scheme against MadLoop CMS** (data/msm-matrix.json cms, from data/msm.json and the manifest): CM01–CM05 and CM07–CM10 at 617b59cd (CM06 was cut: its policy is an MS-CMS refusal test). Widths fixed at generation (GeV; "card widths": the restriction's), the per-point inputs, the points and checks (variants included), the largest deviation of double-double from MadLoop CMS (rule 1; quadruple precision, CM01 against its double-precision exports), binary64 from DD (rule 2), the poles against Catani in DD (rule 4) and the Ward identity in DD, a diagnostic only under CMS (m-sm/PLAN.md §1.4, §1.5). CM07 is the acceptance row ([U4]).

Row	Process	Model	Widths	Inputs	Pts · checks	DD	b64/DD	Catani	Ward (DD)	Variants · notes · status
CM01	$e^-e^+ \to m^-m^+$	G_μ	card widths	python	4 · 278	1.2e−13	1.0e−13	2.0e−30	—	zero, small, source pass
CM02	$u\bar{u} \to e^-e^+$	G_μ	WT 1.37758 WZ 2.49877 WW 2.09291 WH 0.00407468	python	4 · 134	8.8e−15	1.5e−13	2.2e−29	—	pass
CM03	$u\bar{d} \to e^+\,\mathrm{ve}$	G_μ	WT 1.37758 WZ 2.49877 WW 2.09291 WH 0.00407468	python	4 · 134	2.2e−14	3.9e−12	3.1e−30	—	pass
CM04	$gg \to t\bar{t}$	G_μ	WT 0 WZ 2.49877 WW 2.09291 WH 0.00407468	python	4 · 230	1.3e−15	1.5e−14	9.1e−31	—	pass
CM05	$u\bar{u} \to e^-e^+g$	G_μ	WT 1.37758 WZ 2.49877 WW 2.09291 WH 0.00407468	python	4 · 142	1.1e−13	5.1e−13	4.8e−30	3.7e−31	pass
CM07 U4	$gg \to t\bar{t}e^-e^+g$	G_μ	WT 0 WZ 2.49877 WW 2.09291 WH 0.00407468	native	3 · 86	6.9e−13	5.6e−11	7.2e−26	1.2e−3	pass
CM08	$u\bar{d} \to e^+\,\mathrm{ve}\,b\bar{b}$	G_μ	WT 1.37758 WZ 2.49877 WW 2.09291 WH 0.00407468	python	4 · 66	1.3e−14	1.8e−13	—	—	QCD2 QED8 unpredicted (uncorrected-exchange) pass
CM09	$u\bar{d} \to e^+\,\mathrm{ve}\,a$	G_μ a0	WT 1.37758 WZ 2.49877 WW 2.09291 WH 0.00407468	python	4 · 142	2.7e−14	1.6e−12	2.4e−29	2.9e−30	pass
CM10	$u\bar{d} \to w^+b\bar{b}$	loop_sm no_b_mass	WT 1.4915 WZ 0 WW 0 WH 0	python	4 · 66	1.6e−13	1.8e−10	—	—	QCD2 QED6 unpredicted (uncorrected-exchange) pass

G1, electroweak row indicative (data/msm-g1.json, the gate tool on the EW records; [U5]: its speed gate binds in M-FINAL, so no verdict here). $gg \to t\bar{t}e^-e^+g$ (EW) on loop_qcd_qed_sm_Gmu_py3 (the workspace's Python-3 reference copy of loop_qcd_qed_sm_Gmu, the harness default; not the shipped patched UFO, which the campaign used), native per-point inputs, full colour, binary64 finite part, checks off, one pinned core (159), at $B = 128$: **153 ms per point** (±0.7 %) against Recola 987 ms, OpenLoops 1.40 s and MadLoop 7.07 s (provider records reused, not re-run): ×0.155 of Recola's time, 0.119 of 1.30 × Recola (Table 1 row, Figure 1). Generation 234 s + 84.1 s, peak 8.93 GiB. Accuracy: against its fixture (3 points) 2.0×10^{-10} (DD asserted at point 2, where $E > 10^{-9}$; binary64 6.8×10^{-10}); against the providers at 16 stream points in binary64: ML 2.6×10^{-9}, OL 1.8×10^{-9}. Earlier run (MS-VALIDATE, cb012428): $B = 1$ 1.35 s, $B = 128$ 150 ms.

5 QCD capability matrix (M-QCD)

Table 9: **QCD capability matrix** at 617b59cd, per class of m-qcd/PLAN.md §1.1 (campaign of 2026-11-02, data/mqcd.json; legend below the table, rows: Table 30).

Class	Rows	Public route	DD vs ref.	binary64	Catani (DD)	LC/NLC	FFT	Status
Q-massless gluons and massless quark lines, every initial state · up to $2 \to 5$	**22** pass 20 partial 1 fail 1	A12 ✓ DD 3.8e−14 · 3 pts	**1.2e−11** A18	**1.1e−10** A18 · ref. 7.3e−7†	**1.2e−22** A09 · A24 not formed	rule 7 ✓ 8 rows · smoke pass	**×0.46-5.92** seed · 9 rows; total ×0.93-1.29	accepted 20 of 22 pass; F7 A09@collinear (accepted); F11 A24 (accepted)
Q-top t t~ + jets, t t~ with quark lines, four tops · up to $2 \to 4$	**6** pass 6	M01@P0, M01@P2 ✓ DD 2.0e−15 · 8 pts	**3.4e−13** B03	**1.2e−10** B01 · ref. 1.7e−7†	**3.0e−26** B01	rule 7 ✓ 1 row · smoke pass	**×1.13** seed · 1 row; total ×0.96	pass
Q-heavy massive b (4FS) and c, b-Yukawa Higgs · up to $2 \to 4$	**10** pass 10	C01 ✓ DD 9.3e−16 · 4 pts	**5.0e−14** C02	**2.3e−10** C05 · ref. 2.3e−10	**1.1e−28** C09	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-EWBorn QCD corrections to Borns with QED >= 1 · up to $2 \to 5$	**25** pass 25	D05 ✓ DD 2.3e−14 · 4 pts	**1.8e−12** D28	**1.4e−11** D28 · ref. 3.1e−11†	**7.3e−28** D29 · D19 no sector	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-ttX t t~ + H/Z, t t~ + leptons, e+ e- > t t~ (h) · up to $2 \to 5$	**4** pass 4	E01 ✓ DD 3.6e−13 · 4 pts	**3.6e−13** E01	**7.4e−12** E01 · ref. 5.7e−9†	**5.7e−27** E06	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-singletop s- and t-channel, tW, 4FS t-channel · up to $2 \to 3$	**5** pass 5	F03 ✓ DD 1.7e−15 · 4 pts	**8.0e−15** C07	**1.6e−14** C07 · ref. 1.6e−14	**3.6e−30** F04 · C07 no sector	rule 7 ✓ 1 row · smoke pass	auto no cost row	pass
Q-mixed several Born orders, every squared split order · up to $2 \to 2$	**2** pass 2	G02 ✓ DD 2.0e−15 · 4 pts	**4.6e−15** G01	**3.1e−14** G02 · ref. 2.9e−14	**3.7e−30** G02	smoke pass no rule-7 row	auto no cost row	pass
Q-CKM non-diagonal CKM on W lines · up to $2 \to 2$	**3** pass 3	H01 ✓ DD 1.4e−15 · 4 pts	**1.4e−15** H01	**7.2e−14** H01 · ref. 7.3e−14	**4.7e−29** H01	smoke pass no rule-7 row	auto no cost row	pass
Q-multi multiparticle labels (p p > t t~) · up to $2 \to 2$	**1** pass 1	M01@P0, M01@P2 ✓ DD 2.0e−15 · 8 pts	**2.0e−15** M01	**4.1e−14** M01 · ref. 4.3e−14	**4.0e−30** M01	smoke pass no rule-7 row	auto no cost row	pass

Rows: campaign rows and their status (passing every rule of their tier; partial: a rule not formed); class status **accepted**: every other row is an open finding the coordinator accepted for the gate. **Public route** (at e18a024b): the class representative generated by Generator on the shipped loop_sm, Runtime.load and evaluate, against MadLoop (rules 1–3; Q-top and Q-multi by the campaign's public variants of $pp \to t\bar{t}$). **DD vs ref.**: largest relative deviation of the double-double Laurent coefficients (Born, finite, poles, every sector) from the binding reference (rule 1; MadLoop in quadruple precision unless named). **binary64**: from DD where binary64 is asserted (rule 2, estimate $E \le 10^{-9}$); "ref." from the reference at every point (information; † the class has points with $E > 10^{-9}$, where DD is asserted). **Catani**: DD poles against the Catani operator (rule 4, every point and predicted sector). **LC/NLC**: rule 7 (formal-N_c projection = N_c scan, and MQ-NC's independent oracles) and the smoke on every shipped restriction. **FFT**: seed contraction FFT ÷ direct of the class's cost rows (Table 10); auto runs the certified FFT wherever the plan has a certified pair.

Reading the matrix. The campaign ran every row of its tier at the pin as whole test files, one process each, the manifest's points in binary64 and double-double (rules 1–4; rules 5–7 and the extras on the rows that name them; m-qcd/PLAN.md §1.4 under the user's ruling of 2026-10-29 ~10:20: a deviation below ~10^{-5} is not a finding). Its counts: 76 pass, 1 partial, 1 fail, 0 pending, 0 not run. The two non-passing rows are accepted limitations, not wrong values: A24 ($gg \to ggggg$, F11) forms no rule 4 because the seven-gluon colour-correlated trees exceed addendum H's memory budget, and its poles equal NJet's double-double values (rule 1); A09 ($u\bar{u} \to ggg$, F7) fails only through its collinear variant, where the executor returns typed failures and no value (binary64 Unstable at $\theta = 10^{-3}$–10^{-5}, DD CutFailed{DegenerateGram} at 10^{-4}, 10^{-5}); A09 itself and its soft variant pass. D19 and C07 have no predicted pole sector (MQ-EWJOIN's separated-exchange rule: their QCD loop selection omits IR-divergent loops, as MadLoop's does) and are checked against MadLoop per split order. **Re-run at 617b59cd** (2026-11-02): no status or check count changed; against M-SM's re-run 42 of 1212 values moved, all binary64 (QCD-only rows too; likely the summation order of SymJIT 2.28.3, not proven; every double-double value bitwise; Section 1); the public-route and FFT columns are M-QCD's runs at e18a024b (public route on a scratch manifest copy).

Table 10: **FFT seed contraction against the direct one** (MQ-FFT, color.contraction): time for the batch of 128 G1-stream points (binary64, poles none, checks off, one pinned thread; g g > u u~ d d~ on the six-gluon stream; one paired round, best of two below $2 \to 5$; indicative), FFT / direct and their ratio (green faster, red slower; Table 1′s scale), for the whole evaluation and for the executor's seed-contraction stage. The values agree to rounding at stable points; at the cost runs' worst binary64 points (21 point/carrier checks, binary64-unstable: E up to $O(10^2)$) FFT – direct stays within 2.8 % of the acceptance envelope $10(E_{\mathrm{FFT}} + E_{\mathrm{direct}}) + 16uS$ in binary64 and within 1.7 % in DD (relative ≤ 1.1×10⁻¹⁹); test_loop_v2_seed_fft.py (default, nightly, seven legs) passed at e18a024b.

Process	Class	group	pairs	evaluation, s	ratio	seed contraction, ms	ratio
$gg \to gg$	Q-massless	S_3	2	0.266 / 0.205	**×1.29**	2.00 / 0.339	**×5.92**
$gg \to ggg$	Q-massless	S_4	2	2.58 / 2.76	**×0.93**	10.5 / 7.03	**×1.50**
$gg \to gggg$	Q-massless	S_5	2	49.3 / 52.1	**×0.95**	180 / 255	**×0.70**
$gg \to ggggg$	Q-massless	S_6	2	1275 / 1246	**×1.02**	8 160 / 17 700	**×0.46**
$u\bar{u} \to ggg$	Q-massless	S_3	19	0.800 / 0.825	**×0.97**	11.4 / 6.73	**×1.69**
$u\bar{u} \to gggg$	Q-massless	S_4	40	10.6 / 10.2	**×1.04**	357 / 312	**×1.15**
$u\bar{u} \to ggggg$	Q-massless	S_5	74	240 / 247	**×0.97**	24 300 / 34 700	**×0.70**
$gg \to t\bar{t}gg$	Q-top	S_4	40	13.2 / 13.7	**×0.96**	224 / 198	**×1.13**
$gg \to u\bar{u}d\bar{d}$	Q-massless	S_2	2340	4.95 / 4.55	**×1.09**	260 / 89.8	**×2.89**
$u\bar{u} \to d\bar{d}g$	Q-massless · no certified pair: direct	—	—	0.519 / 0.420	**×1.24**	2.88 / 2.88	**×1.00**

6 Gate G1: end-to-end timings of the v2 engine

G1 times the complete Born–virtual evaluation, **finite part only**, through the call of the public v2 runtime (`NativeExecutor.evaluate`, which `V2VirtualRuntime` calls) with stability checks off, on one pinned core and one thread, binary64, 16 SIMD lanes, in batches of B distinct points of a frozen stream (REQUIREMENTS_ADDENDUM D, plan §5.8); pyAmpliCol receives the stream's exact rational points, the providers their binary64 decimal text. The configurations:

- **native inputs, on `loop`** — the engine at `71e2041e` (milestone M-SM: the engine of M-QCD, `e18a024b`, with M-SM's 14 packages, ew-native to MS-MASTERS; not re-timed at M-FINAL-b, whose engine `f3d9e361` adds the packages of Section 1; build `loop-native-v2-slice-71e2041e`, digest `f8d77c97`): the executor forms the trees, the Born and the counterterm inputs per point (binary64 SymJIT plane applications, tree closures in lanes of points), skips the counterterm interactions that only feed poles in finite-only mode, then runs the loop stage. $gg \to ggggg$ is timed from its stored process. The public path ($u\bar{u} \to d\bar{d}g$): the public generator's artifact has the harness's plan digest, `Runtime.load` in a fresh interpreter makes no generation call, and the loaded executor returns the timed run's binary64 values bitwise at all 128 stream points (`public_path_check` in `data/g1.json`; its plan digest is also M-QCD's). **Values against M-QCD** (`e18a024b`, `values_note`): bitwise in 22 of 30 configurations; the other 8 are the full- and leading-colour configurations of the four rows with an e^-e^+ pair (electroweak couplings), changed at rounding level (largest 1.2×10^{-9}, $u\bar{u} \to e^-e^+d\bar{d}g$ full colour). The binary64-unstable stream points of M-QCD's analysis (at 6 of $gg \to ggggg$'s 128 points, 17, 89, 93, 95, 109 and 112, binary64 is wrong by $O(1)$ or more; the recommended checks flag every one, E = 8 to 5×10^3) lie in configurations whose values are bitwise M-QCD's, whose stability rows stay inputs; of the 65 stability rows re-run here, 60 are bitwise M-QCD's and the others within 5.8×10^{-12}. The timing runs evaluate every point with checks off. Earlier campaigns (M-QCD `e18a024b`, m1028a `28a16255`, m1026a `11542fd5`, m1025a `1d27426b`, G-int `4e9b5a9e`, the development campaign of `39824f9b`) are `earlier_runs`, history;
- **Python inputs** history — the engine on `loop` before G-int, G1 at `1c971289` (LC at `d591a2f7`): trees, Born amplitudes and counterterm scalars formed per point in Python, then the loop stage natively.

Timing protocol at `71e2041e` (`timing_protocol` in `data/g1.json`): the committed harness, binary64, finite part only, no stability check, one thread on one pinned core, setup heap frozen out of Python's cyclic collector before the warm-up, the frozen streams. The native digest changed (`1be09f92` ⟶ `f8d77c97`), so every configuration was re-timed at $B = 128$ (three blocks; at $2 \to 5$ after a 16-point warm-up), full and leading colour, in one round (timings indicative, user 2026-10-28: no repeat for host contention; no verdict was open, so no verdict repeat). $B = 1$ and the $2 \to 5$ rows' $B = 16$ were **not re-timed**: the tables show m1028a's runs (`28a16255`, each configuration's last earlier run) marked °. Host: AMD EPYC 9754, shared host, cores 128-223 without SMT siblings: milestone m1031-msm's G1 on 2026-10-31 (07:31-09:19 UTC), at most one job per CCX of 8 cores, each on the least busy core of a CCX whose least busy core was below 50 % busy (g g > g g g g g FC on the reserved CCX 216-223; cores per record: host_cpus); host load average 186-344 (other users' processes and the milestone's whole-suite gate and campaign ran at the same time); timings indicative (user 2026-10-28), one round, no repeats for contention.

Generation (`generation_protocol`): every row generated cold in its own process with fresh caches, as in m1026a; against M-QCD, model + generate + export changed by −19 % to +5 % (median −3 %), plan sizes and structure equal everywhere, and the seven-gluon plans are M-QCD's ($gg \to ggggg$: full colour 171 + 213 s, 9.5 GiB; LC 178 + 213 s, 10.0 GiB). The provider records are reused unchanged (`provider_records`). With $u\bar{u} \to e^-e^+gg$ at 5.01 ms per point (M-QCD: 4.93 ms) one more process is above the 5 ms floor and gated. The electroweak row is timed by the same protocol in `data/msm-g1.json`, indicative and not gated (Section 4); the tables of this section are `data/g1.json`'s.

Table 11: **Gate G1, full colour**, ms per point (gate statistic: median), finite part, one thread. Rows $B = 1$ and $B = 128$ (the Python-input $2 \to 4$ rows only at $B = 1$: their Python inputs took seconds per point). Multiplier and colour: time ÷ the fastest provider of the row (legend of Table 1). Grey rows: generation — pyAmpliCol generation plus plan export with its peak memory; MadLoop graph generation, export and compilation · the share of its loop-numerator coefficients in its time (Table 13); Recola process generation; OpenLoops: the prebuilt library and its load plus first call. MadLoop $u\bar{u} \to gggg$ was not exported: the $gg \to gggg$ export took 2.4 h here (HANDOFF §0.1), and OpenLoops binds the gate. Gate: upper line Python inputs (history), lower line native inputs (`loop`); the number is the time over the binding limit (MadLoop's time or 1.30× the faster of Recola and OpenLoops); **open**: the band contains the limit; **not gated**: the configuration runs below 5 ms per point at $B = 128$ (user rule of 2026-10-24; ratio as measured); **reported**: the verdict is taken at $B = 128$ only, so the $B = 1$ ratios are measured and shown without a verdict (user 2026-10-28; bands: Table 12); ° native $B = 1$: the run of `28a16255`, not re-timed at `71e2041e` (uncoloured, history; its ratio over this campaign's limit).

Process	B	Python inputs[hist]	Native inputs, loop	MadLoop	Recola	OpenLoops	Gate
$gg \to gg$ $2 \to 2$, no gate	1	**140** ×69	15.0° ×7.4	**2.02** fastest	**10.0** ×5.0	**3.20** ×1.6	no gate
	128	**132** ×66	**1.52** ×0.75	**2.02** fastest	**10.0** ×5.0	**3.20** ×1.6	no gate
	gen.	10.4 s · 0.3 GiB	7.63 s · 0.6 GiB	9.70 s · 64 %	0.142 s	ppjj · 6.80 ms	
$u\bar{u} \to d\bar{d}g$ $2 \to 3$	1	**40.6** ×27	10.5° ×7.0	**3.17** ×2.1	**3.18** ×2.1	**1.52** fastest	reported 20.61 reported 5.35°
	128	**32.4** ×21	**1.96** ×1.3	**3.17** ×2.1	**3.18** ×2.1	**1.52** fastest	**fail** ×16 **not gated** 1.00
	gen.	9.94 s · 0.3 GiB	8.06 s · 0.6 GiB	9.87 s · 73 %	0.406 s	ppjjg · 4.73 ms	
$u\bar{u} \to u\bar{u}g$ $2 \to 3$	1	**68.9** ×24	18.5° ×6.3	**8.59** ×2.9	**5.55** ×1.9	**2.92** fastest	reported 18.12 reported 4.87°
	128	**54.5** ×19	**2.79** ×0.95	**8.59** ×2.9	**5.55** ×1.9	**2.92** fastest	**fail** ×14 **not gated** 0.73
	gen.	10.4 s · 0.3 GiB	8.03 s · 0.6 GiB	10.9 s · 75 %	0.462 s	ppjjg · 8.43 ms	
$u\bar{u} \to e^-e^+g$ $2 \to 3$	1	*not generable*	4.37° ×4.6	**0.948** fastest	**1.08** ×1.1	*no library*	— reported 4.61°
	128	*not generable*	**0.702** ×0.74	**0.948** fastest	**1.08** ×1.1	*no library*	— **not gated** 0.74
	gen.	CT coverage refused	11.9 s · 0.6 GiB	14.0 s · 79 %	0.701 s	—	
$u\bar{u} \to ggg$ $2 \to 3$	1	**215** ×17	48.1° ×3.9	**28.2** ×2.3	**25.3** ×2.0	**12.4** fastest	reported 13.38 reported 2.98°
	128	**179** ×14	**5.47** ×0.44	**28.2** ×2.3	**25.3** ×2.0	**12.4** fastest	**fail** ×11 **pass** 0.34
	gen.	12.3 s · 0.4 GiB	8.42 s · 0.6 GiB	17.6 s · 76 %	0.634 s	ppjjg · 23.7 ms	
$gg \to ggg$ $2 \to 3$	1	**2 760** ×28	205° ×2.1	**247** ×2.5	**351** ×3.6	**97.0** fastest	reported 21.86 reported 1.62°
	128	**2 560** ×26	**20.4** ×0.21	**247** ×2.5	**351** ×3.6	**97.0** fastest	**fail** ×20 **pass** 0.16
	gen.	27.9 s · 0.6 GiB	9.93 s · 0.6 GiB	47.0 s · 78 %	6.63 s	ppjjg · 159 ms	
$u\bar{u} \to e^-e^+gg$ $2 \to 4$	1	*not generable*	44.8° ×4.4	**27.3** ×2.7	**10.2** fastest	*no library*	— reported 3.37°
	128	*not generable*	**5.01** ×0.49	**27.3** ×2.7	**10.2** fastest	*no library*	— **pass** 0.38
	gen.	CT coverage refused	12.9 s · 0.6 GiB	24.8 s · 83 %	0.833 s	—	
$gg \to gggg$ $2 \to 4$	1	**101 000** ×36	3 850° ×1.4	**38 700** ×14	**17 600** ×6.2	**2 840** fastest	reported 27.42 reported 1.04°
	128	*not timed*	**371** ×0.13	**38 700** ×14	**17 600** ×6.2	**2 840** fastest	— **pass** 0.10
	gen.	335 s · 4.1 GiB	33.1 s · 1.3 GiB	8 680 s · 51 %	2 130 s	ppjjjj · 4 950 ms	
$gg \to t\bar{t}gg$ $2 \to 4$	1	**7 310** ×15	817° ×1.7	**5 450** ×12	**1 970** ×4.2	**472** fastest	reported 11.92 reported 1.33°
	128	*not timed*	**98.5** ×0.21	**5 450** ×12	**1 970** ×4.2	**472** fastest	— **pass** 0.16
	gen.	62.5 s · 0.9 GiB	13.9 s · 0.7 GiB	218 s · 64 %	61.4 s	pptt̄jj · 821 ms	
$u\bar{u} \to gggg$ $2 \to 4$	1	**5 760** ×15	801° ×2.1	*not timed*	**908** ×2.4	**374** fastest	reported 11.85 reported 1.65°
	128	*not timed*	**77.4** ×0.21	*not timed*	**908** ×2.4	**374** fastest	— **pass** 0.16
	gen.	56.1 s · 0.7 GiB	13.9 s · 0.7 GiB	—	32.8 s	ppjjjj · 606 ms	
$u\bar{u} \to e^-e^+d\bar{d}g$ $2 \to 5$	1	*not timed*	*not timed*	**132** ×3.7	**35.3** fastest	**51.0** ×1.4	— —
	128	*not timed*	**16.1** ×0.45	**132** ×3.7	**35.3** fastest	**51.0** ×1.4	— **pass** 0.35
	gen.	—	15.8 s · 0.7 GiB	58.6 s	6.73 s	ppllj̄jj · 108 ms	
$u\bar{u} \to e^-e^+ggg$ $2 \to 5$	1	*not timed*	*not timed*	**1 420** ×5.7	**251** fastest	**346** ×1.4	— —
	128	*not timed*	**59.7** ×0.24	**1 420** ×5.7	**251** fastest	**346** ×1.4	— **pass** 0.18
	gen.	—	16.1 s · 0.7 GiB	146 s	13.4 s	ppllj̄jj · 727 ms	
$gg \to t\bar{t}e^-e^+g$ $2 \to 5$	1	*not timed*	*not timed*	**3 040** ×7.5	**620** ×1.5	**404** fastest	— —
	128	*not timed*	**62.5** ×0.15	**3 040** ×7.5	**620** ×1.5	**404** fastest	— **pass** 0.12
	gen.	—	17.3 s · 0.8 GiB	165 s	27.7 s	pplltt̄j · 1 060 ms	
$u\bar{u} \to ggggg$ $2 \to 5$	1	*not timed*	*not timed*	**569 000** ≥×3.1	**≤ 182 000** (outside protocol, poles on: upper bound)	*no library*	— —
	128	*not timed*	**1 720** ≥×0.0094	**569 000** ≥×3.1	**≤ 182 000** (outside protocol, poles on: upper bound)	*no library*	— **pass** 0.0030
	gen.	—	134 s · 3.0 GiB	44 000 s	—	—	
$gg \to ggggg$ $2 \to 5$, no gate (no matched provider)	1	*not timed*	*not timed*	*not exported*	*not generated*	*no library*	no gate
	128	*not timed*	**7 260**	*not exported*	*not generated*	*no library*	no gate
	gen.	—	384 s · 9.5 GiB	—	—	—	

The providers run in production mode on the same stream: OpenLoops with truepoles 0, Recola without IR poles (one amplitude call per point), MadLoop at its production stability setting on exports built with MG5′s production flags. OpenLoops has no installed library with QCD corrections to $\ell^+\ell^-$ + 1 or 2 jets (pplj_ew carries electroweak corrections only; existing provider builds only), so the V+jets gates bind on MadLoop and Recola. Providers have no batch API: one time per point serves every B. **Gate statistic:** the median per point on both sides — over pyAmpliCol's timed blocks of one B, and over the provider's timed points (the stream's first 16; 8 for MadLoop $gg \to gggg$, Recola $gg \to gggg$, OpenLoops $gg \to gggg$, Recola $u\bar{u} \to gggg$, MadLoop $u\bar{u} \to ggggg$ and Recola $u\bar{u} \to ggggg$). The provider times are the records of the measure-g1 campaign (23–24 October, on cores 129–345), not re-run since, and for the $2 \to 5$ rows the production-mode runs of package measure-g1-7leg (25 October, each provider with its production Collier cache: on for Recola's COLLIER 1.2.8 and OpenLoops, off for MadLoop's COLLIER 1.2.9 with its seven-leg cache defect; $u\bar{u} \to ggggg$ imported from the reference campaign's three points); pyAmpliCol ran on 25 cores between 128 and 216 (per run: host_cpus in data/g1.json) on 2026-10-31, at most one job per eight-core group sharing an L3 cache, each on the least busy core of a group whose least busy core was below 50 % busy, while the milestone's whole-suite gate and other users' processes loaded the host (host in data/g1.json: cores per record: host_cpus)). The three $B = 128$ blocks of a process spread by at most 3.7 % (half the range over the median; $u\bar{u} \to d\bar{d}g$ 2.034, 1.948, 1.961 ms). The gate's uncertainty is pyAmpliCol's block spread (half the range of three blocks, the median's standard error from five) plus the provider's half-range over its repeated runs (8 %, the campaign's largest, for the single runs of Recola $gg \to gggg$, Recola $u\bar{u} \to gggg$, OpenLoops $u\bar{u} \to e^-e^+d\bar{d}g$, MadLoop $u\bar{u} \to e^-e^+d\bar{d}g$, OpenLoops $u\bar{u} \to e^-e^+ggg$, Recola $u\bar{u} \to e^-e^+ggg$, MadLoop $u\bar{u} \to e^-e^+ggg$, OpenLoops $gg \to t\bar{t}e^-e^+g$, Recola $gg \to t\bar{t}e^-e^+g$, MadLoop $gg \to t\bar{t}e^-e^+g$ and MadLoop $u\bar{u} \to ggggg$). A gate passes or fails only if the whole band does; otherwise it is **open** and the configuration is repeated once (plan §5.8 item 10). Every timed call returned all its points (failed points: 0); rescue is off by protocol (no checks, no escalation), so the rescue rate is 0.

Table 12: **The gate at every batch size** (native inputs on loop, full colour): time over the binding limit and its uncertainty band. **The speed gate is taken at $B = 128$ only** (user 2026-10-28, MASTER_PLAN §5.8; data/milestone.json gate_decisions.gate_batch): only that column has a verdict and a colour (green pass, red fail, amber open, grey not gated: below 5 ms per point at $B = 128$, user rule of 2026-10-24); every other batch size is **reported**: measured, its ratio and band shown in a white cell, no verdict. $B = 1$ is timed with the default 16 lanes (a one-point call pays a 16-lane chunk); the one-lane column is the development campaign's (39824f9b, history: the public path always runs 16 lanes in binary64), a configuration without a $B = 128$ timing and therefore without any verdict (gate missing: no B = 128 timing (timed at B = 1); G1 is taken at B = 128 only (user 2026-10-28)). This campaign timed $B = 128$ only (no verdict was open, so no repeat: repeats in data/g1.json); ° grey: the earlier run of 28a16255, 39824f9b, 4e9b5a9e at that batch size, its median over the same binding limit with its time (no band; an earlier engine under the committed protocol; history).

Process	$B = 1$, 16 lanes		$B = 1$, 1 lane (39824f9b)		$B = 8$		$B = 16$		$B = 32$		$B = 128$ (gate)	
$u\bar{u} \to d\bar{d}g$	*5.35°*	10.5 ms	2.13	2.10–2.16	*1.46°*	2.87 ms	—		*1.15°*	2.26 ms	**1.00**	0.96–1.03
$u\bar{u} \to u\bar{u}g$	*4.87°*	18.5 ms	1.63	1.59–1.67	*1.16°*	4.40 ms	—		*0.87°*	3.31 ms	**0.73**	0.72–0.75
$u\bar{u} \to e^-e^+g$	*4.61°*	4.37 ms	1.61	1.60–1.62	*1.07°*	1.01 ms	—		*0.85°*	0.805 ms	**0.74**	0.74–0.75
$u\bar{u} \to ggg$	*2.98°*	48.1 ms	0.84	0.81–0.86	*0.68°*	11.0 ms	—		*0.74°*	11.9 ms	**0.34**	0.33–0.35
$gg \to ggg$	*1.62°*	205 ms	0.45	0.45–0.45	*0.42°*	52.7 ms	—		*0.34°*	42.9 ms	**0.16**	0.16–0.16
$u\bar{u} \to e^-e^+gg$	*3.37°*	44.8 ms	0.78	0.78–0.78	*0.62°*	8.20 ms	—		*0.42°*	5.54 ms	**0.38**	0.37–0.38
$gg \to gggg$	*1.04°*	3 850 ms	0.31	0.30–0.31	*0.30°*	1 120 ms	—		*0.26°*	977 ms	**0.10**	0.10–0.10
$gg \to t\bar{t}gg$	*1.33°*	817 ms	0.37	0.36–0.37	*0.35°*	215 ms	—		*0.28°*	172 ms	**0.16**	0.16–0.16
$u\bar{u} \to gggg$	*1.65°*	801 ms	0.43	0.43–0.44	*0.47°*	230 ms	—		*0.37°*	178 ms	**0.16**	0.16–0.16
$u\bar{u} \to e^-e^+d\bar{d}g$	—		—		—		*0.34°*	15.4 ms	—		**0.35**	0.32–0.38
$u\bar{u} \to e^-e^+ggg$	—		—		—		*0.18°*	57.2 ms	—		**0.18**	0.16–0.20
$gg \to t\bar{t}e^-e^+g$	—		—		—		*0.12°*	61.2 ms	—		**0.12**	0.11–0.13
$u\bar{u} \to ggggg$	—		—		—		*0.0030°*	1 730 ms	—		**0.0030**	0.0028–0.0033

6.1 Where the time goes

Table 13: **Stage split**, ms per point, full colour, at the largest batch timed (Python-input $2 \to 4$: $B = 1$). Trees/Born/CT: the per-point inputs, timed in Python before G-int (history: trees, counterterm program, Born and counterterm colour projections, scalars); on loop the executor forms them and, since lc-perf (88657260), times them itself (its trees stage: programs, closures, interferences). Numerator: seeds, fused factors and the dressing of the open-loop objects (the pure numerator evaluation). Scatter: closures into NinjaRS lanes. NinjaRS: the reduction without the masters; masters: the master integrals (canonical arguments, cache, oneloopmaster). Glue: point parameters, carrier conversion, assembly and, where the inputs are timed, the rest of the Python call (all rows of this table). MadLoop (instrumented copy; its points less one, 3 for $gg \to gggg$): Born/counterterm HELAS calls, loop-numerator coefficients, reduction plus masters, residual (stability test, colour sums). Bar: shares, colours as the column heads.

Process		Trees, Born, CT	Numerator	Scatter	NinjaRS	Masters	Glue	Total	Shares
$gg \to gg$	native, loop	0.180	0.854	0.0718	0.120	0.0418	0.255	**1.52**	
	Python^{hist}	128	0.851	0.0919	0.119	0.0381	2.96	**132**	

Process		Trees, Born, CT	Numerator	Scatter	NinjaRS	Masters	Glue	Total	Shares
	MadLoop	0.0315	1.27	—	0.585		0.107	**1.99**	
$u\bar{u} \to d\bar{d}g$	native, `loop`	0.0837	0.786	0.407	0.154	0.129	0.405	**1.96**	
	Python[hist]	28.1	0.749	0.462	0.137	0.111	2.66	**32.4**	
	MadLoop	0.0479	2.35	—	0.683		0.133	**3.22**	
$u\bar{u} \to u\bar{u}g$	native, `loop`	0.124	1.27	0.421	0.253	0.192	0.514	**2.79**	
	Python[hist]	48.5	1.23	0.479	0.232	0.171	3.99	**54.5**	
	MadLoop	0.137	6.44	—	1.15		0.814	**8.54**	
$u\bar{u} \to e^-e^+g$	native, `loop`	0.0452	0.271	0.123	0.0291	0.0200	0.212	**0.702**	
	MadLoop	0.0184	0.754	—	0.133		0.0460	**0.951**	
$u\bar{u} \to ggg$	native, `loop`	0.460	3.10	0.456	0.520	0.245	0.676	**5.47**	
	Python[hist]	168	2.92	0.476	0.459	0.225	5.75	**179**	
	MadLoop	0.346	18.5	—	2.54		2.87	**24.3**	
$gg \to ggg$	native, `loop`	4.33	11.9	0.508	2.08	0.502	1.12	**20.4**	
	Python[hist]	2 530	11.7	0.519	2.03	0.483	10.5	**2 560**	
	MadLoop	2.42	184	—	11.3		39.5	**237**	
$u\bar{u} \to e^-e^+gg$	native, `loop`	0.178	2.90	0.627	0.464	0.251	0.593	**5.01**	
	MadLoop	0.289	22.7	—	2.38		2.11	**27.5**	
$gg \to gggg$	native, `loop`	100	209	2.37	42.7	5.79	10.5	**371**	
	Python[hist]	97 300	3 750	4.08	38.6	5.81	137	**101 000**	
	MadLoop	88.9	18 600	—	209		17 700	**36 500**	
$gg \to t\bar{t}gg$	native, `loop`	11.5	46.6	1.03	23.9	6.50	8.70	**98.5**	
	Python[hist]	6 490	751	3.55	20.7	6.58	29.3	**7 310**	
	MadLoop	24.2	3 290	—	105		1 740	**5 160**	
$u\bar{u} \to gggg$	native, `loop`	9.56	48.7	1.79	9.39	2.92	5.27	**77.4**	
	Python[hist]	4 910	806	2.01	9.64	2.99	24.1	**5 760**	
$u\bar{u} \to e^-e^+d\bar{d}g$	native, `loop`	0.542	8.48	1.87	1.97	1.30	1.76	**16.1**	
$u\bar{u} \to e^-e^+ggg$	native, `loop`	1.70	41.6	2.10	8.13	2.69	3.53	**59.7**	
$gg \to t\bar{t}e^-e^+g$	native, `loop`	2.82	39.3	0.827	10.9	4.11	4.81	**62.5**	
$u\bar{u} \to ggggg$	native, `loop`	223	1 130	9.46	253	45.1	54.6	**1 720**	
$gg \to ggggg$	native, `loop`	2 990	3 700	11.5	461	41.6	31.8	**7 260**	

Verdict of gate G1 (processes from $2 \to 3$; $2 \to 2$ without a gate).

- **Native inputs on `loop`** (G1 at `71e2041e`) at $B = 128$: **pass** for all 10 gated processes, $u\bar{u} \to ggg$, $gg \to ggg$, $u\bar{u} \to e^-e^+gg$, $gg \to gggg$, $gg \to t\bar{t}gg$, $u\bar{u} \to gggg$, $u\bar{u} \to e^-e^+d\bar{d}g$, $u\bar{u} \to e^-e^+ggg$, $gg \to t\bar{t}e^-e^+g$ and $u\bar{u} \to ggggg$. **Not gated** (below 5 ms per point, user decision of 2026-10-24; ratios as measured): $u\bar{u} \to d\bar{d}g$ 1.96 ms (1.00 of its limit), $u\bar{u} \to u\bar{u}g$ 2.79 ms (0.73 of its limit) and $u\bar{u} \to e^-e^+g$ 0.702 ms (0.74 of its limit). $B = 1$ (16 lanes) and the $2 \to 5$ rows' $B = 16$ are **reported** without a verdict (user 2026-10-28): $B = 1$ at ×1.0–5.3 of the limit (a one-point call pays a 16-lane chunk, Table 12); with one lane per call, in the development runs of `39824f9b` (history), ×0.31–2.1.
- **Python inputs** history (the engine before G-int, G1 at `1c971289`): failed every process it generated, at $B = 128$ at ×11–20 of the binding limit, the per-point Python inputs taking 87 %–99 % of its time (Table 13); at $B = 1$ at ×12–27, where $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$ were timed only. It did not generate $u\bar{u} \to e^-e^+g$ and $u\bar{u} \to e^-e^+gg$ (Furry zeros, WP3.5; Z fermion signs, WP3.2).
- **Fastest of all tools** (aspiration), native inputs at $B = 128$, within the uncertainty band: faster than every provider for $u\bar{u} \to u\bar{u}g$, $u\bar{u} \to e^-e^+g$, $u\bar{u} \to ggg$, $gg \to ggg$, $u\bar{u} \to e^-e^+gg$, $gg \to gggg$, $gg \to t\bar{t}gg$, $u\bar{u} \to gggg$, $u\bar{u} \to e^-e^+d\bar{d}g$, $u\bar{u} \to e^-e^+ggg$ and $gg \to t\bar{t}e^-e^+g$; faster than every provider with a production-mode time for $u\bar{u} \to ggggg$ (MadLoop; Recola has only an upper bound of its production time, ≲ 182 s); at parity with the fastest for none; slower than the fastest for $u\bar{u} \to d\bar{d}g$ (OpenLoops). NJet, not a matched provider, takes 4.59 s (NJet 3.1.2) for its two-evaluation full-colour $gg \to gggg$ test (Table 22).
- Seven-leg rows ($u\bar{u} \to e^-e^+d\bar{d}g$, $u\bar{u} \to e^-e^+ggg$, $gg \to t\bar{t}e^-e^+g$, $u\bar{u} \to ggggg$, $gg \to ggggg$) are generated by the WP3.1 generator, on `loop` since G-int, and timed by G1 since package `measure-g1-7leg` (full and leading colour at $B = 16$ and 128; re-timed at `71e2041e`); $gg \to ggggg$ has no matched provider timing (addendum F, H): reported without a gate.

Table 14: **Levers**, measured on the gated rows of Table 13 ($2 \to 3$ and beyond; shares at $B = 128$). Gains are measured where a column says so; otherwise they are the stage's share, an upper bound.

Lever	Evidence	Expected gain	Owner
Native per-point inputs (WP3.1 I3–I6: trees, Born, counterterm scalars in the executor)	Python inputs 87 %–99 % of the Python-input time at $B = 128$ (history)	measured, Python ÷ native inputs at equal B: — at $B = 1$, ×17–125 at $B = 128$ ($2 \to 3$)	done: G-int, on `loop` since `4e9b5a9e`
Binary64 point inputs (the providers' input; the protocol passes the stream's exact rationals)	paired run on the loaded artifact of G-int (`4e9b5a9e`, history), $B = 128$: $u\bar{u} \to d\bar{d}g$ 1.68 against 2.29 ms, $u\bar{u} \to u\bar{u}g$ 2.94 against 3.39 ms	measured: ×1.37 and ×1.15 (information; not a G1 configuration)	none: moot under the 5 ms gate rule (user, 2026-10-24); the protocol keeps exact inputs
Compiled native trees (kernels as the tree engine's SymJIT planes, lanes batched)	native trees (with the Born and counterterm inputs; the executor's timer) 3 %–27 % of the time at $B = 128$; $gg \to gggg$ 27 %	up to the share; ×1.4 for the largest	in part on `loop`: SymJIT plane applications (`7db32901`), lane-chunked tree closures (`88657260`); next: the native tree program (A5, `native-dag`)
Lanes by batch in binary64 (`plan_export.lanes_for`: 16 lanes also for one point)	$B = 1$: one 16-lane chunk per call ($B = 1$ is reported, not gated: user 2026-10-28)	measured: — at $B = 1$ (16 lanes on `loop` ÷ 1 lane of `39824f9b`, different builds)	executor policy (WP3.1 or wave 4)
Finite-only result path (sum the virtual natively, no per-family lane export, Python assembly only for checks and observables)	glue 3 %–30 % of the native-input time; skipping the pole-only counterterm interactions natively (generation-scale-2's I5 record) saved 2–5 % of the total at $u\bar{u} \to gggg$ and $gg \to gggg$	up to the share (largest at $2 \to 3$ with quarks)	in part on `loop` (generation-scale-2); next `plan_export`, executor (wave 4)
Lane-major closures (no scatter, no second copy into the reducer's batch)	scatter 1 %–21 %	up to the share	executor, reducer adapter (wave 4)
AVX2 / AVX-512 kernels (runtime selection)	numerator 39 %–70 %	×1.0–1.4 on the dressing (gate G0)	wave 4 (WP4.6)
NinjaRS and masters: parallel reduction, transposed reduction, stage-2 sharing, a cheaper master cache key	NinjaRS + masters 7 %–31 %; masters 12 %–46 % of the reduction, oneloopmaster alone 1 %–13 %: the cache bookkeeping costs more than the integrals	one thread: up to the share; threads: DESIGN §5.6	NinjaRS requests (DESIGN §11.5)

Below 5 ms per point. $u\bar{u} \to d\bar{d}g$, $u\bar{u} \to u\bar{u}g$ and $u\bar{u} \to e^-e^+g$ take 1.96, 2.79 and 0.702 ms per point at $B = 128$; by the user's decision of 2026-10-24 a process below 5 ms has no blocking gate and its slowdown against the providers need not be pursued: optimisation effort goes to higher multiplicities. Their ratios stay in the tables ($u\bar{u} \to d\bar{d}g$: 1.96 ms against a limit of 1.97 ms, OpenLoops 1.52 ms × 1.30, 1.00 of it; measured inconclusive, band 0.96–1.03). The question whether G1 should pass the points as exact rationals (the protocol) or as the binary64 text the providers read is moot under this rule, and the protocol stays. For information only (not G1 configurations, no verdict formed from them): on `71e2041e` the public `Runtime.evaluate` of the loaded artifact (poles off, core 221) takes 11.71 ms per point at $B = 1$ and 2.01 ms at $B = 128$ for $u\bar{u} \to d\bar{d}g$, against the campaign's 1.96 ms of the harness's executor at $B = 128$ (core 147) and 1.98 ms in its smoke run, on other cores and at other times (`public_path_check` in `data/g1.json`, the campaign's `STATE.md`; M-QCD's run, 2.20 ms on a busy host, was 1.83 ms at m1028a); at G-int (`4e9b5a9e`, committed protocol) a paired run (one core, ten alternating rounds) gave 1.68 ms on the binary64 text against 2.29 ms on the exact points (×0.73; $u\bar{u} \to u\bar{u}g$: 2.94 against 3.39 ms, ×0.87).

6.2 Leading colour

Table 15: **Leading colour** (LC), ms per point, same protocol as Table 11; in each cell the full-colour (FC) time of the same configuration and batch, and LC/FC. No provider ratio: the providers time full colour. LC uses the committed QCD colour continuation (`examples/loop/source_colour`); the V+jets rows are timed at LC with native inputs only. This campaign timed LC at $B = 128$; ° grey: the earlier run of `39824f9b` at $B = 1$ with its FC time (an earlier engine under the committed protocol; history).

Process	Python inputs[hist]				Native inputs, loop			
	$B = 1$		largest batch		$B = 1$		largest batch	
$gg \to gg$	*not timed*		*not timed*		*not timed*		**0.657**	FC 1.52 · 0.43
$u\bar{u} \to d\bar{d}g$	**40.5**	FC 40.6 · 1.00	**37.4**	FC 32.4 · 1.15	*5.25°*	FC 10.9 · 0.48	**1.25**	FC 1.96 · 0.63
$u\bar{u} \to u\bar{u}g$	*not timed*		*not timed*		*9.00°*	FC 19.1 · 0.47	**1.59**	FC 2.79 · 0.57
$u\bar{u} \to e^-e^+g$	*not timed*		*not timed*		*not timed*		**0.484**	FC 0.702 · 0.69
$u\bar{u} \to ggg$	**184**	FC 215 · 0.86	**171**	FC 179 · 0.96	*22.8°*	FC 50.6 · 0.45	**2.77**	FC 5.47 · 0.51
$gg \to ggg$	**2 350**	FC 2 760 · 0.85	**2 300** ($B = 32$)	FC 2 550 · 0.90	*91.1°*	FC 211 · 0.43	**8.48**	FC 20.4 · 0.42
$u\bar{u} \to e^-e^+gg$	*not timed*		*not timed*		*not timed*		**1.74**	FC 5.01 · 0.35
$gg \to gggg$	*not timed*		*not timed*		*2 070°*	FC 4 100 · 0.50	**155**	FC 371 · 0.42
$gg \to t\bar{t}gg$	*not timed*		*not timed*		*418°*	FC 840 · 0.50	**32.4**	FC 98.5 · 0.33
$u\bar{u} \to gggg$	*not timed*		*not timed*		*380°*	FC 860 · 0.44	**30.3**	FC 77.4 · 0.39
$u\bar{u} \to e^-e^+d\bar{d}g$	*not timed*		*not timed*		*not timed*		**6.33**	FC 16.1 · 0.39
$u\bar{u} \to e^-e^+ggg$	*not timed*		*not timed*		*not timed*		**10.7**	FC 59.7 · 0.18
$gg \to t\bar{t}e^-e^+g$	*not timed*		*not timed*		*not timed*		**11.6**	FC 62.5 · 0.19
$u\bar{u} \to ggggg$	*not timed*		*not timed*		*not timed*		**581**	FC 1 720 · 0.34
$gg \to ggggg$	*not timed*		*not timed*		*not timed*		**4 030**	FC 7 260 · 0.56

LC is cheaper in the loop stage (fewer families and skeletons) and, since lc-perf (`88657260`: LC-aware pruning of the closure blocks, native tree closures in lanes of points), also end to end: with native inputs at $B = 128$ LC/FC is 0.18–0.69 ($u\bar{u} \to e^-e^+ggg$ gains most, $u\bar{u} \to e^-e^+g$ least), so LC is faster than full colour on every row; at G-int it was 0.55–1.28, slower than full colour for $u\bar{u} \to ggg$, $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$ (committed protocol; Section 6). The native trees, with the Born and counterterm inputs, take a larger share at LC ($u\bar{u} \to ggg$: 11 % at LC, 8 % at full colour; $gg \to gggg$: 30 % at LC, 27 % at full colour), the lever of compiled trees (Table 14). The $2 \to 5$ rows follow: LC/FC 0.18–0.56 at $B = 128$ ($u\bar{u} \to e^-e^+ggg$ 0.18, $gg \to t\bar{t}e^-e^+g$ 0.19, $u\bar{u} \to ggggg$ 0.34, $u\bar{u} \to e^-e^+d\bar{d}g$ 0.39, $gg \to ggggg$ 0.56). At seven gluons the stored LC plan of $gg \to ggggg$ keeps all 4 707 full-colour loop families, where the lower gluon multiplicities halve them (154 ⟶ 77 and 1 100 ⟶ 550 at five and six gluons, with the top). This is not a defect (DESIGN §7.9): a family is keyed by its propagators' momenta and masses, so a closed massless-quark loop shares its family with the gluon and ghost rings, and the prune drops a family only when none of its skeletons is live; with the top the families halve, under this generation's `particle_veto = [t]` they cannot, while the tries shrink alike. The LC plan generates in 3.0 min against 2.9 min at full colour (`LoopSlice.generate` alone; on `1d27426b` 40.6 against 37.9 min).

Identical-flavour quark lines: since identical-flavour-lc (`5cb5ea83`, milestone m1028a) the leading and next-to-leading colour of processes with identical-flavour quark lines (from $u\bar{u} \to u\bar{u}g$ up, any number of lines) are exact formal-N_c projections: the loop-request blocks and the counterterm program split by U(1)-subtraction N_c grade in the Python and Rust lanes, with order-free block grades (second genericity audit, G2-M1; validated against the legacy engine's LC and NLC, now frozen captures, and since MQ-NC by rule 7 against N_c scans and independent oracles, Section 5). G1′s LC of $u\bar{u} \to u\bar{u}g$ is a validated value (1.59 ms per point at $B = 128$).

6.3 Throughput, model recalibration and schedule M

Table 16: **Kernel throughput and the design recipe** (native inputs, full colour, largest batch). Work: the executor's modelled complex multiply-adds of the chosen kernel forms per point (one lane); throughput: work × lanes over the dressing time (fused-factor builds excluded). Native loop: seeds, factors, dressing, scatter and reduction; outside: everything else of the call (trees, glue). Recipe: DESIGN §1.3, finite only, where the design has a row; measured ÷ recipe for the total and for the NinjaRS stage (masters included) against its model R.

Process	Work, MCMAC/pt	Dressing, ms	G CMAC/s	Native loop, ms	Outside, ms	Total, ms	Total ÷ recipe	NinjaRS ÷ R
$gg \to gg$	0.932	0.694	1.34	1.09	0.433	**1.52**	1.0–3.0	1.03–1.17
$u\bar{u} \to d\bar{d}g$	0.556	0.453	1.23	1.47	0.487	**1.96**	no row	—
$u\bar{u} \to u\bar{u}g$	1.09	0.881	1.24	2.14	0.650	**2.79**	no row	—
$u\bar{u} \to e^-e^+g$	0.168	0.131	1.28	0.444	0.258	**0.702**	no row	—
$u\bar{u} \to ggg$	3.18	2.51	1.27	4.33	1.14	**5.47**	no row	—
$gg \to ggg$	14.3	10.9	1.31	15.0	5.45	**20.4**	no row	—
$u\bar{u} \to e^-e^+gg$	2.89	2.28	1.27	4.24	0.771	**5.01**	no row	—
$gg \to gggg$	268	202	1.33	260	111	**371**	0.9–2.7	0.95–1.11
$gg \to t\bar{t}gg$	56.0	43.5	1.29	78.0	20.4	**98.5**	0.6–1.8	1.13–1.58
$u\bar{u} \to gggg$	56.7	43.8	1.29	62.8	14.6	**77.4**	no row	—
$u\bar{u} \to e^-e^+d\bar{d}g$	8.80	7.08	1.24	13.6	2.45	**16.1**	1.0–3.0	1.00–1.79

Process	Work, MCMAC/pt	Dressing, ms	G CMAC/s	Native loop, ms	Outside, ms	Total, ms	Total ÷ recipe	NinjaRS ÷ R
$u\bar{u} \to e^-e^+ggg$	47.8	39.3	1.22	54.5	5.19	**59.7**	0.9–2.4	0.84–1.16
$gg \to t\bar{t}e^-e^+g$	49.3	38.0	1.29	55.2	7.36	**62.5**	0.8–2.3	1.07–1.47
$u\bar{u} \to ggggg$	1 170	936	1.25	1 440	285	**1 720**	0.7–2.1	1.02–1.38
$gg \to ggggg$	4 060	3 540	1.15	4 220	3 040	**7 260**	0.9–2.7	0.50–0.59

Recalibration. The kernels run 1.15–1.34 G complex MAC/s per core on the real plans, the rate gate G0 measured on its synthetic programs (1.29–1.37); the dressing term of the recipe holds. The NinjaRS stage, masters included, lands on the design's model R (measured ÷ R 0.50–1.79, Table 16). What the recipe misses is the **outer factor**: it assumed trees, seeds and counterterms add 5–10 % (DESIGN §1.3), but with the current native inputs the total is 1.15–2.2 times dressing plus reduction at full colour (up to 3 at LC). Table 17 re-estimates every design row with these measured terms.

Table 17: **Design estimates recalibrated by G1** (`tools/loop_v2_costmodel.py --g1`), ms per point, finite only, one thread. Recipe: DESIGN Table 1a's finite-only band (outer factor 1.05–1.10). G1: the recipe's dressing term, the NinjaRS model R × 0.502–1.793 (the band this campaign measures, Table 16) and the measured outer factor of the row's class representative at $B = 128$ with native inputs (same colour where measured). Measured: G1's own time where the row was timed. Gate: G1's binding limit at the row's largest batch where G1 timed the providers (1.3× the fastest production provider, or MadLoop's time where it binds), else 1.3× the best production provider of DESIGN Table 1b (protocol of 20 October), with the headroom gate ÷ upper estimate. The outer factor shrinks with compiled trees (Table 14); until then these replace the Table 1a targets.

Design row	Col.	Recipe	Outer	Class	G1 estimate	Measured	Gate	Headroom
$gg \to gg$	FC	0.502–1.52	1.51	$gg \to gg$	**0.619–2.28**	1.52	—	—
$gg \to gg$	LC	0.204–0.542	2.08	$gg \to gg$	**0.333–1.15**	0.657	—	—
$gg \to gggg$	FC	139–409	1.45	$gg \to gggg$	**161–598**	371	3 690	6.18
$gg \to gggg$	LC	55.7–146	1.54	$gg \to gggg$	**64.9–235**	155	—	—
$gg \to ggggg$	FC	2 710–7 920	1.75	$gg \to ggggg$	**3 780–14 000**	7 260	—	—
$gg \to ggggg$	LC	1 070–2 830	1.36	$gg \to ggggg$	**1 100–4 020**	4 030	—	—
$u\bar{u} \to ggggg$	FC	827–2 630	1.38	$u\bar{u} \to ggggg$	**941–3 620**	1 720	569 000	157.12
$u\bar{u} \to ggggg$	LC	206–526	1.41	$u\bar{u} \to ggggg$	**212–813**	581	—	—
$u\bar{u} \to e^-e^+ggg$	FC	24.5–68.1	1.15	$u\bar{u} \to e^-e^+ggg$	**21.5–82.7**	59.7	326	3.94
$u\bar{u} \to e^-e^+ggg$	LC	3.34–8.31	1.49	$u\bar{u} \to e^-e^+ggg$	**3.66–13.7**	10.7	—	—
$u\bar{u} \to e^-e^+d\bar{d}g$	FC	5.34–16.5	1.37	$u\bar{u} \to e^-e^+d\bar{d}g$	**5.74–24.2**	16.1	45.9	1.89
$u\bar{u} \to e^-e^+d\bar{d}g$	LC	1.19–3.14	1.81	$u\bar{u} \to e^-e^+d\bar{d}g$	**1.59–6.48**	6.33	—	—
$gg \to t\bar{t}gg$	FC	54.8–156	1.31	$gg \to t\bar{t}gg$	**55.8–214**	98.5	613	2.87
$gg \to t\bar{t}gg$	LC	14.6–40.1	1.55	$gg \to t\bar{t}gg$	**17.6–65.4**	32.4	—	—
$gg \to t\bar{t}e^-e^+g$	FC	27.4–76.1	1.16	$gg \to t\bar{t}e^-e^+g$	**24.3–93.0**	62.5	525	5.64
$gg \to t\bar{t}e^-e^+g$	LC	4.06–11.0	1.58	$gg \to t\bar{t}e^-e^+g$	**4.94–18.4**	11.6	—	—

Table 18: **Schedule M against B** (native inputs, one thread), pure-gluon classes, the only classes where M merges seed entries (DESIGN §5.6). The schedule-dependent stages are seeds, factors, dressing, scatter and the NinjaRS reduction ("the reduction included", §1.6 item 5); the trees, Born and counterterm inputs and the glue are the same work in both schedules. M: the development campaign's runs (`39824f9b`, not repeated); B: the configuration on `loop` (`71e2041e`), a later engine (native tree closures and plane applications, frozen setup heap) on the same processes, so the ratios pair runs of different builds and campaigns (the gate tool pairs them; history for M): end to end at $B = 128$ they set the M time with the older, slower trees against this engine's B time and do not measure M; LC was re-timed at $B = 128$ only. M is adopted where M/B ≤ 0.85 on the schedule-dependent stages at every common batch size; ± is the timing uncertainty of the ratio.

Process	Col.	Stages M/B, $B = 1$	Stages M/B, $B = 128$	End to end, $B = 1$	End to end, $B = 128$	Adopt M
$gg \to ggg$	FC	—	**0.99** ± 0.03	—	2.48	no
$gg \to ggg$	LC	—	**0.90** ± 0.05	—	5.50	no
$gg \to gggg$	FC	—	**0.91** ± 0.06	—	2.44	no
$gg \to gggg$	LC	—	**0.77** ± 0.02	—	6.57	**yes**

B against M (decision, provisional until the trees are compiled). On the schedule-dependent stages M passes the 15 % rule for $gg \to gggg$ at LC (0.77; DESIGN §5.6 measured 0.75–0.80 on the dressing side), and not for $gg \to ggg$ at FC (0.99), $gg \to ggg$ at LC (0.90) and $gg \to gggg$ at FC (0.91). The same rows qualify as in the development campaign's paired runs, on which the decision was made (`data/g1.json` of `c3a7be09`). End to end the gain is diluted by the tree inputs, which both schedules share and which compiled trees will shrink (Table 14). With a quark line M's entries do not merge at all (M/B work 1.00, DESIGN §5.6; equal by test). **Decision (G1):** schedule M for pure-gluon classes at leading colour from $n = 6$; schedule B for every other class at $n \leq 6$, including pure-gluon full colour; M stays the planned default from $n \geq 8$. The seven-leg comparison of M with B is open (the schedule-M run of `measure-g1-7leg` stopped; G1 and Table 1 time schedule B).

Table 19: **Values at the timed points**: pyAmpliCol's binary64 finite part (native inputs on `loop`) against each provider's production-mode value on the common stream points (the providers read the binary64 decimal text of the exact points), largest deviation, metric $|a-b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$; Recola's finite part converted with the universal double pole (it returns no poles in production). Worst

point: the point of the row's largest deviation, evaluated again with one rotation and the reduction-direction check in binary64 and in double-double (DD) by the stability runs of `e18a024b`, not repeated (their binary64 values equal the timed ones to rounding level, more at binary64-unstable points): E is the binary64 estimate, DD the DD value's deviation from the closest provider (named). Campaign: the first three points against the committed references (`tests/fixtures/loop/required`; $gg \to ggggg$: NJet's ten points), fixture metric and tolerance, asserted in binary64 where its estimate $E \leq 10^{-9}$ and in DD elsewhere (user 2026-10-29; earlier records: binary64 throughout), native inputs (`loop`) and Python inputs (history); ‡ points above the tolerance that the user accepted (below the table).

Process	Pts	vs MadLoop	vs Recola	vs OpenLoops	Worst point: E	DD: closest provider	Campaign, native	Campaign, Python[nist]
$gg \to gg$	16	6.7×10^{-13}	3.1×10^{-13}	3.0×10^{-13}	pt 4: 5.1×10^{-12}	9.4×10^{-14} OpenLoops	—	—
$u\bar{u} \to d\bar{d}g$	16	6.7×10^{-13}	3.3×10^{-10}	6.7×10^{-13}	pt 1: 7.8×10^{-12}	3.2×10^{-13} MadLoop	3.6×10^{-10}	3.6×10^{-10}
$u\bar{u} \to u\bar{u}g$	16	8.4×10^{-12}	7.2×10^{-11}	6.1×10^{-12}	pt 1: 7.9×10^{-12}	2.4×10^{-13} MadLoop	8.6×10^{-11}	8.6×10^{-11}
$u\bar{u} \to e^-e^+g$	16	1.4×10^{-13}	1.0×10^{-13}	no library	pt 0: 2.5×10^{-12}	7.5×10^{-15} Recola	—	—
$u\bar{u} \to ggg$	16	1.4×10^{-12}	1.5×10^{-12}	8.4×10^{-11}	pt 2: 1.4×10^{-12}	1.3×10^{-14} MadLoop	—	—
$gg \to ggg$	16	1.0×10^{-6}	1.0×10^{-6}	1.0×10^{-6}	pt 10: 2.0×10^{-5}	6.0×10^{-12} Recola	—	—
$u\bar{u} \to e^-e^+gg$	16	9.4×10^{-11}	4.7×10^{-11}	no library	pt 8: 3.4×10^{-9}	4.3×10^{-13} Recola	—	—
$gg \to gggg$	8	2.2×10^{-7}	2.8×10^{-8}	2.8×10^{-8}	pt 1: 2.1×10^{-6}	4.4×10^{-12} Recola	7.1×10^{-11}	8.7×10^{-8} **above 2.0×10^{-8}**
$gg \to t\bar{t}gg$	16	1.4×10^{-8}	7.9×10^{-9}	7.9×10^{-9}	pt 13: 1.0×10^{-7}	5.5×10^{-15} OpenLoops	1.8×10^{-11}	1.1×10^{-11}
$u\bar{u} \to gggg$	16	—	9.3×10^{-12}	1.6×10^{-10}	pt 9: 5.6×10^{-10}	1.9×10^{-13} OpenLoops	—	—
$u\bar{u} \to e^-e^+d\bar{d}g$	16	3.7×10^{-11}	5.2×10^{-11}	5.2×10^{-11}	pt 15: 1.4×10^{-9}	5.5×10^{-14} OpenLoops	8.9×10^{-13}	—
$u\bar{u} \to e^-e^+ggg$	16	8.9×10^{-11}	1.1×10^{-10}	1.1×10^{-10}	pt 8: 1.6×10^{-9}	7.7×10^{-14} OpenLoops	4.1×10^{-13}	—
$gg \to t\bar{t}e^-e^+g$	16	8.7×10^{-9}	4.4×10^{-9}	5.6×10^{-9}	pt 2: 5.7×10^{-8}	5.8×10^{-13} OpenLoops	1.0×10^{-8}	—
$u\bar{u} \to ggggg$	3	1.9×10^{-12}	2.9×10^{-10}	no library	pt 0: 3.2×10^{-11}	1.4×10^{-12} MadLoop	2.9×10^{-10}	—
$gg \to ggggg$	0	not generated	not generated	no library	—	—	7.1×10^{-9}	—

Reading the values. The Born agrees with every provider to 1.7×10^{-13} or better. For the finite part, the double-double (DD) value at each worst point (Table 19) separates the sources: it agrees with the closest provider to 6.0×10^{-12} or better everywhere. Where pyAmpliCol's binary64 estimate E exceeds the deviation — $gg \to ggg$, $gg \to gggg$ and $gg \to t\bar{t}gg$ (above 10^{-9}) and $u\bar{u} \to gggg$ (1.9×10^{-13} in DD against 1.6×10^{-10} in binary64 at point 9) — the deviation is pyAmpliCol's own binary64 loss, which E flags. The other deviations above 10^{-11} stay in DD and are the providers' own accuracy: Recola's at $u\bar{u} \to d\bar{d}g$ (3.3×10^{-10}), $u\bar{u} \to u\bar{u}g$ and $u\bar{u} \to e^-e^+gg$ (COLLIER's default target 10^{-8}, HANDOFF §7), OpenLoops' at $u\bar{u} \to ggg$ (8.4×10^{-11}) and MadLoop's at $u\bar{u} \to e^-e^+gg$ (1.1×10^{-10}); MadLoop's production values also stay off the DD values at the ill-conditioned gluon and top points (up to 2.4×10^{-7}). At the campaign points the check asserts binary64 where its estimate $E \leq 10^{-9}$ and double-double elsewhere (user E1 ruling 2026-10-29 ~03:45, package g1-accuracy): $gg \to gggg$ point 1 (E = 2.1×10^{-6}) is asserted in DD, the check passing at 7.1×10^{-11} (tolerance 2×10^{-8} binary64 there 1.5×10^{-8}, recorded); its DD value agrees with Recola to 4.4×10^{-12}. ‡ $gg \to ggggg$ against NJet: points 4 and 8 exceed the tolerance 2×10^{-8} in binary64 (4.4×10^{-8} and 2.1×10^{-8} from NJet's double-double values, at most 7.1×10^{-9} over all its references), **accepted (user 2026-10-25): O(1e-8) binary64, within the estimate, DD = NJet**: estimates E 1.7×10^{-6} and 3.3×10^{-7}, and pyAmpliCol's double-double values equal NJet's; the check keeps its tolerance and, asserting DD at these points ($E > 10^{-9}$, E1 ruling), passes.

7 Required processes: references and development validation

The plan requires the $2 \to 5$ processes of Table 1 (plan §5.7), compared with MadLoop and Recola, and with OpenLoops where a public library exists. A reference provider is run only if its generation completes in about 45 minutes, or in a time comparable to pyAmpliCol's own generation of the process (HANDOFF §0.1, addendum F). Otherwise pyAmpliCol is validated by Ward and gauge identities, Catani poles, LC/NLC/FC and DD/AP consistency, stability checks, lower multiplicities and existing references.

Table 20: **Provider generation on this host and the 45-minute rule** (HANDOFF §0.1): the rows of Table 1 where a provider generation takes more than 45 minutes or did not finish. MadLoop: graph generation + source export + compilation; Recola: process generation; OpenLoops: prebuilt public libraries (no generation time). Records: `data/g1.json` (G1 rows) and `data/measured.json` (`provider_generation`, the reference runs of the $2 \to 5$ rows), through `data/summary.json`. Every other row of Table 1 generates within 45 minutes with every provider (longest: MadLoop 11.9 min for $gg \to t\bar{t}e^-e^+g$ (EW)).

Process	MadLoop	Recola	OpenLoops	45-minute rule
$gg \to gggg$	2.4 h (285 + 8 354 + 37 s)	35.4 min	prebuilt ppjjjj	**MadLoop over 45 min; results exist and are used**
$u\bar{u} \to ggggg$	12.2 h (5 793 + 38 040 + 122 s)	2.8 h	no public library	**MadLoop, Recola over 45 min; results exist and are used**
$gg \to ggggg$	diagrams 17.4 h; unfinished (last state: exporting)	unfinished (last state: running, 2026-10-20)	no public library	**stopped 22 Oct; not to be restarted (HANDOFF section 0.1)**

Table 21: **Native validation of the required set** DEV (`work/v2-generation-scale` before its integration into `loop` at `4e9b5a9e`, DESIGN WP3.1: ninth increment (`6f5a64cc`; threads not recorded) for $u\bar{u} \to e^-e^+d\bar{d}g$, $gg \to gggg$, $gg \to ggggg$; eighth increment (`0ea83af2`; threads not recorded) for $u\bar{u} \to e^-e^+ggg$, $gg \to t\bar{t}e^-e^+g$ (QCD), $gg \to t\bar{t}gg$; seventh increment (`fef92189`; threads not recorded) for $u\bar{u} \to ggggg$): generation (`LoopSlice.generate`, then the plan export; peaks of each stage; shared host under load), then binary64 virtuals with native per-point inputs and the production counterterm route at the fixture points, against the references committed on that branch (`tests/fixtures/loop/required`; fixture tolerance 2×10^{-8}). Largest deviation over the references and points, metric of Table 1; the per-reference values are in `data/measured.json`. For $gg \to ggggg$ the references are NJet (double-double and two binary64 runs) and the Catani operator for the poles, top-free with $N_f = 5$; its estimate column is the rotation plus reduction-direction check per point.

Process	References	Born	Finite	$1/\varepsilon$	$1/\varepsilon^2$	Estimate E	Generation + export
$u\bar{u} \to e^-e^+d\bar{d}g$	ML, Rc, OL · 3 pts	1.9×10^{-14}	8.7×10^{-13}	2.3×10^{-13}	1.0×10^{-13}	1.2×10^{-12}, 2.8×10^{-12}, 1.1×10^{-12}	37.7 s 0.4 GiB
$u\bar{u} \to e^-e^+ggg$	ML, Rc, OL · 3 pts	2.3×10^{-14}	8.5×10^{-13}	4.6×10^{-13}	2.0×10^{-13}	—	36.7 s 0.5 GiB
$gg \to t\bar{t}e^-e^+g$ (QCD)	ML, Rc, OL · 3 pts	1.4×10^{-14}	5.7×10^{-9}	3.7×10^{-12}	5.3×10^{-12}	—	43.7 s 0.6 GiB
$gg \to t\bar{t}gg$	ML, Rc, OL · 3 pts	2.1×10^{-14}	1.1×10^{-11}	1.0×10^{-10}	1.6×10^{-11}	—	38.5 s 0.7 GiB
$gg \to gggg$	Rc, OL · 2 pts	4.7×10^{-14}	4.5×10^{-11}	1.2×10^{-11}	5.2×10^{-12}	1.5×10^{-10}, 1.4×10^{-10}	154 s 2.2 GiB
$u\bar{u} \to ggggg$	ML, Rc · 3 pts	1.7×10^{-13}	2.9×10^{-10}	6.1×10^{-10}	4.4×10^{-11}	—	517 s 6.1 GiB
$gg \to ggggg$	NJet DD, NJet R1, NJet R2, Catani · 1 pt	8.1×10^{-13}	3.8×10^{-9}	1.5×10^{-9}	1.1×10^{-12}	2.0×10^{-8}	51.5 min 28.7 GiB

The $gg \to t\bar{t}e^-e^+g$ (QCD) finite part deviates by 5.7×10^{-9} at one of its three points and by at most 3.7×10^{-12} at the others: the point's known binary64 conditioning (earlier record, DESIGN WP3.1 fourth increment: $\le 3.6\times10^{-12}$ at two points; third point 5.7×10^{-9} in binary64 (estimate 1.7×10^{-9}), MadLoop to 1.9×10^{-14} in DoubleFloat). The electroweak target is validated by the M-SM campaign (Section 4, Table 29), row TV07 (loop_qcd_qed_sm_Gmu, native inputs, 3 points, 124 checks): double-double = MadLoop QP, Recola and OpenLoops to 7.4×10^{-13}, binary64 from double-double 5.1×10^{-11}, poles = Catani to 6.4×10^{-26} (DD), gluon Ward identity at a stable top 3.2×10^{-28} (DD); its QCD twin TV07@qcd 1.4×10^{-12} (91 checks) G1 times it indicatively (`data/msm-g1.json`, Section 4; Table 1), and its generation and validation are the integrated configuration's, not a development record (so it is not in the table above). The QCD classes are M-QCD's (Section 5). The $u\bar{u} \to ggggg$ row comes from DESIGN.md WP3.1 seventh increment (fef92189), the newest record of that process: the later re-runs on the merged heads (eighth and ninth increments) do not include it. It uses the same production counterterm route; its check is in Table 23.

8 Seven gluons: NJet and pyAmpliCol

8.1 NJet: Table 5 of arXiv:1209.0100, re-measured on this host

NJet is the independent reference for $gg \to ggggg$, where MadLoop and Recola cannot generate under the 45-minute rule and OpenLoops has no library. It is massless five-flavour QCD, and it is not one of the gate's matched providers.

Table 22: **NJet, s per point**, double precision, two evaluations each (NJet's scaling test). Paper: Xeon E3-1240 at 3.3 GHz, NJet 1.0. This host: one pinned core, idle at launch, of the shared host; 10 frozen points per channel, mean (spread ≤ 2 %); the multiplier is time ÷ the paper's. De-symmetrised sums equal the standard sums only after phase-space integration; they are timing figures, not point values.

Channel	Table 5	NJet 3.1.2		NJet 2.1.1		Table 5	NJet 3.1.2		NJet 2.1.1	
	standard full-colour sum					de-symmetrised sum				
$gg \to gg$	0.03	**0.00295**	×0.1	**0.00291**	×0.1	0.03	*no class*		*no class*	
$gg \to ggg$	0.22	*defective channel*		**0.141**	×0.6	0.07	*defective channel*		**0.0452**	×0.6
$gg \to gggg$	6.19	**4.59**	×0.7	**4.58**	×0.7	0.57	**0.39**	×0.7	**0.38**	×0.7
$gg \to ggggg$	171.31	**142**	×0.8	**246.7**	×1.4	3.07	**2.41**	×0.8	**4.19**	×1.4

At seven gluons one NJet 3.1.2 evaluation without the scaling test takes 71.2 s (standard) and 1.22 s (de-symmetrised; means over 10 points), at 69 MB; double-double costs 1 990 s per point. NJet 3.1.2's numerical five-gluon channel is 9/2 too large (NJet 2.1.1 is used there), and at the unrotated seven-gluon points NJet keeps only 4–6 digits in double precision (NaN at 3 of 10), so the reference values are taken at exactly rotated points.

8.2 pyAmpliCol at seven gluons DEV

On the WP3.1 branch before its integration (G-int; Table 21) the full $gg \to ggggg$ virtual, with the counterterms of the recursion and native per-point inputs, top-free with $N_f = 5$, agrees with NJet's double-double values to 3.8×10^{-9} in the finite part (per point: 3.8×10^{-9} over all references, point 0, ninth increment) and to 1.1×10^{-12} in the double pole; each point's rotation plus reduction-direction estimate bounds its difference. Generation (ninth increment) took 44.2 min at 14.7 GiB and the plan export 7.3 min at 28.7 GiB (peak RSS of each stage; 160 MB of plan JSON; the whole validation run, point evaluations included, peaked at 38.6 GiB), inside addendum H's budget (hours, below 150 GB). DESIGN WP3.1 attributes the cost: 325 s before the counterterm recursion, 1,122 s with its counterterm program, 2,766 s once poles-uvir's always-generated companion merged (eighth increment), and 2,652 s after the ct-simplify merge (ninth increment, this record), 1,843 s of it the companion's seed table, which built generic seeds. Since then native-seeds' in-crate colour reductions and generation-scale-2's lean companion (sharing the slice's certificates) brought the companion's seeds to 4.7 s (DESIGN WP3.1), and G1's stored process on `loop` (`71e2041e`, full colour) generates in 2.9 min plus 3.5 min of export at a 9.5 GiB peak (one pinned core, cold caches; Table 11). A point with its rotation and reduction-direction checks took 82.3 s, a single-point development evaluation (threads not recorded): not a batched one-thread time and not comparable with NJet's 71.2 s per evaluation on one pinned core. The batched one-thread timing is G1's (Table 1, Section 6), and so is the check of the timed configuration against NJet's ten points (Table 23).

8.3 Validation without a matched provider

Table 23: **Checks of the seven-leg QCD results.** First and third rows: G1's checks of the timed configurations (engine `71e2041e` on `loop`, native per-point inputs, full colour; binary64 finite part and Born, and the double pole of G1's physical rows against the universal one); second row DEV on `work/v2-generation-scale`; Ward row WP2.5 on `loop`; deviation metric of Table 1. NJet values: fixture `gg_ggggg_njet.json` (NJet 3.1.2 double-double at exactly rotated points, scaling-test estimates ≤ 1.3×10^{-24}; points 0–2 are G1's stream points 0–2 at NJet's scale).

Check	Result	Status
$gg \to ggggg$ full virtual against NJet (top-free, $N_f = 5$), 10 points (G1)	finite ≤ 4.4×10^{-8} against NJet's double-double values (median 3.2×10^{-9}; NJet's binary64 values ≤ 7.1×10^{-9}), Born ≤ 9.3×10^{-13}; every point within its estimate E; asserted in DD at points 0, 1, 4, 5, 6, 7, 8, 9 ($E > 10^{-9}$, E1 ruling); above the tolerance: point 4 4.4×10^{-8} ($E = 1.7\times10^{-6}$) and point 8 2.1×10^{-8} ($E = 3.3\times10^{-7}$), where the double-double values equal NJet's; double pole against the universal one ≤ 5.1×10^{-12} (points 0, 1, 2)	passed (tolerance 2×10^{-8})
Top-free full virtual against NJet at lower multiplicity	$gg \to gg$: ≤ 2.1×10^{-14} at 10 points, full virtual; $gg \to ggg$: ≤ 2.3×10^{-13} at the stable points; the flagged point 3×10^{-14} in DoubleFloat; $gg \to gggg$: 1.0×10^{-11} at point 0; point 1 1.6×10^{-7} in binary64 (estimate 2.7×10^{-6}), 4.7×10^{-15} in DoubleFloat (`test_loop_v2_topfree.py`)	passed
$u\bar{u} \to ggggg$ full virtual against MadLoop and Recola, 3 points (G1, the timed stream points)	finite: MadLoop ≤ 1.9×10^{-12}, Recola ≤ 2.9×10^{-10} (Recola's own: pyAmpliCol's double-double value lies 2.9×10^{-10} from it and 1.4×10^{-12} from MadLoop); Born ≤ 1.7×10^{-13}; double pole against the universal one ≤ 6.8×10^{-13}; the providers agree to 6.2×10^{-10} (reference merge)	passed (tolerance 2×10^{-8})
Ward identity, one gluon's $\varepsilon \to p$, at seven gluons	MQ-WARD (`d3081568`): the generic gauge-Ward check on the native per-point input path, top-free $gg \to ggggg$, legs 1, 4 and 7: virtual ≤ 2.9×10^{-10} (binary64) and 2.8×10^{-26} (DD) of the physical scale, Born ≤ 9.8×10^{-16} / 5.3×10^{-32} (nightly `test_loop_v2_ward.py`; `data/milestone.json` `status_notes`)	**passed (MQ-WARD) at seven gluons (addendum H)**

9 Capabilities at this milestone

Table 24: **Capability matrix** of the v2 engine on `loop` at f3d9e361, with branch work marked.

Feature	Status	Boundary
Recursive engine	supported	Trees, families and loop numerators from recursions; no loop diagram is enumerated (FeynKit graphs are test oracles only). Any number of coloured fermion lines, identical or distinct flavours, QCD and EW (fermion-lines, db1ed5cb: the fold of mixed colour-connection parities before the evaluation reuse, Python and Rust lanes; $u\bar{u} \to d\bar{d}s\bar{s}$, $u\bar{u} \to u\bar{u}d\bar{d}$, $u\bar{u} \to u\bar{u}u\bar{u}$ and $u\bar{u} \to u\bar{u}e^+e^-$ equal MadLoop in quadruple precision and Recola). Since ew-leptons (EWI-5, b82d8280) any number of lepton and colour-singlet fermion lines, through the loop too, with one reading-sign rule and one fermion classifier taken from the tree engine (Bhabha against MadLoop; genericity audit 2 G2-M5, G2-M6); since ew-epsilon (6da37352) one root-closure reading sign on every fermionless vertex (G2-M7). Listed, typed refusals replace every silent-wrong-value path of the second genericity audit (fail-closed, fb54246c; visible in the dry run). Counterterms: see below.
Default engine	supported	v2 is the only one-loop engine (MQ-LEGACY-REMOVE, 539dffdd; user 2026-10-28 ~16:10 Q3): the per-diagram legacy engine, its plan v3 artifacts and its precision-rescue options are removed (the built-in rescue stack of M-FINAL-b replaces them in the Python API, see Stability); `loop.engine`, `--loop-engine` and a non-default `[evaluation.stability]` are refused with the removal reason; tests that used the legacy engine as an oracle keep an independent one (frozen captures or providers).
Shipped models, public route	supported	The wheel ships MG5 3.7.0's `loop_sm` and its restrictions with a `[loop.model]` profile (MQ-MODEL, 321efdb4): every QCD class generates through `Generator` / TOML / CLI without a `tools/` helper or hand-written declaration, a multiparticle label one entry per subprocess, bitwise the tools route (Section 5); MadLoop syntax mapped (`[virt=QCD]` → `loop_coupling_orders`, `QED=n` → `max_coupling_orders`, `/ x` → `particle_veto`), decay chains and `$` refused typed (decays are served since M-FINAL-b); `pyamplicol self-test` evaluates a `loop_sm` virtual against MadLoop. **Since M-SM** the wheel also ships `loop_qcd_qed_sm` and `loop_qcd_qed_sm_Gmu`, UFOmodeldb 6ca4b120 plus the user's issue-1 patch (R_2 of `R2_G0tt` split by loop origin, exactly unitary CKM, the G_μ bottom-mass charge counterterm; PROVENANCE and `MANIFEST.sha256` record the local delta; MS-MODEL 2ae8b88b, MS-UFODB 0e5930c2, MS-UFO-PATCH 3eb4db9f), with digest-keyed profiles and the structural Goldstone-flag repair: every EW class generates through the public route, bitwise the tools view (Section 4); a multiparticle request generates its members with a Born and records the loop-induced members as skipped (`loop-induced`, Q2); the self-test also evaluates an EW virtual from the shipped prepared artifact (= MadLoop to 5.4e-15).
Full, NLC and LC colour	supported	**Full colour is the default** of loop requests (user 2026-10-28 Q1); LC and NLC are opt-in through the shipped `loop_sm` N_c declaration (MQ-NC, d8d9349f: every counterterm vertex, resolved by structural digest; formal-N_c projections equal to N_c scans and checked against independent oracles, rule 7; an uncovered vertex refused, `colour-declaration-coverage`). **LC helicity and colour-flow selection** (MQ-SELECT, e4964fc3): `helicities=` / `color_flows=` at generation (recursion pruned to the selection) and at evaluation, tree-engine IDs; a complete set of selections sums to the inclusive LC value; NLC and full colour keep their API (selectors refused there). Exact seed tables in N_c, built in Rust since native-seeds (7e3841f3), with native colour cells, seed records and dipole orbits since native-cells (5ebcebea; the Python stages are the oracles); LC/NLC by generated classes and truncated cells. LC is faster than full colour end to end on every G1 row at $B = 128$, the $2 \to 5$ rows included (LC/FC 0.18–0.69, Table 15); the top-free seven-gluon LC plan keeps every full-colour family by construction (Section 6.2). Since identical-flavour-lc (5cb5ea83) LC and NLC of identical-flavour quark lines (from $u\bar{u} \to u\bar{u}g$ up, any number of lines) are exact formal-N_c projections: blocks and counterterm program split by U(1)-subtraction N_c grade in both lanes, order-free grades (G2-M1; the limitation of m1026a is removed, Section 6.2); a block that still mixes grades is refused (`colour-accuracy-nc-exact`, fail-closed). **EW LC/NLC is not supported, by design** (user 2026-10-30 Q4: the typed `colour-declaration-coverage` refusal is permanent); electroweak requests run at full colour.
FFT colour contraction	supported	Certified symmetric-group FFT of the seed contraction in the native executor at full colour and NLC, binary64, double-double and MPFR (MQ-FFT, da6980b0): generic over the plan's certified pairs, policy `auto | direct | fft` (the algorithm run is reported; a forced FFT without a certified pair is refused, `forced-seed-fft-unserved`), public `color.contraction` route; FFT = direct to rounding up to $gg \to ggggg$; the seed stage ×0.46–0.70 of direct at S_5, S_6, slower below, the whole evaluation unchanged within noise (Table 10).
Colour and spin correlations	supported	Directed colour histories at LC/NLC/FC, spin sources, resolved helicities, flavour aliases, Ward sources (WP2.5). Since generation-scale-2 the directed metrics are native and stored with the process (correlated and resolved calls of a stored process regenerate nothing; one seed column per self-adjoint dipole): declaring the NLO dipoles costs +10 % of the six-gluon generation (38.5 against 35.0 s); at seven gluons 784 against 419 s, whose metric tables go to `native-cells`' next run. LC/NLC correlations of identical-flavour quark lines use the graded blocks of the colour row. **Independent oracles since M-FINAL-b** (MF-CORR, 821ae808): Recola's colour, spin and spin-colour correlations on four processes equal the one-loop `evaluate_correlated_many` and the tree `Runtime.evaluate_correlated_many` (Born ≤ 6.5×10^{-14}, Laurent coefficients ≤ 4.1×10^{-10} of the row scale, Table 6); a spin-correlated leg without a four-component vector source is refused at generation, as on the tree route (MF-SMALL2); correlations with an LC selection are refused (`selected-correlators`).
Per-order partials	supported	Every squared coupling order at tree level and at one loop, mixed Born orders, per-correction loop orders.
Poles at runtime	supported	`poles = "all" | "none" | "uv" | "ir" | "split"` (WP2.10, on loop since 24 October): finite-only mode with the finite part bitwise unchanged and the pole work skipped (natively per counterterm interaction since generation-scale-2's I5 record); `"uv"` and `"ir"` with the user's semantics of 23 October, from a lean stored companion of the certified-scaleless families (bitwise at $gg \to ggg$ and $u\bar{u} \to gggg$, equal to rounding at six and seven gluons) and in the SDK v4 too; the IR poles sum one Catani operator per correction, QED included (ew-poles, 8b52179b; a correction without a prediction gives none); since

Feature	Status	Boundary
		e6e7543a a mixed QCD×EW sector is predicted only where the requested corrections complete its loops, and joint requests predict it exactly; since MQ-EWJOIN a sector whose QCD loop selection omits IR-divergent loops (gluon exchange between quark lines joined by a W/Z in the loop, `separated-exchange`) is not predicted and checked against MadLoop per split order. Photon-only QED corrections on `loop_qcd_qed_sm` are refused (`veto-unlabelled-counterterms`: the veto reaches counterterms with empty or nominal loop labels; full EW on that model matches Catani; EWI-7). The UV residue comes from an interim local module until the NinjaRS re-pin; a `"uv"` estimate needs a rotation check. **EW pole split** (MS-POLES, `42b2794f`): on EW requests `"uv"` and `"split"` come from the model's pole regulators (`epsUV`, `epsIR`) under both mass schemes, at the cost of one more regulated pass of the tree program; the conventions record the regulators (= MadLoop on DY01 and TT01).
Stability estimates	supported	**Recommended checks** (MQ-STAB, `a2aa0d84`): `loops.v2.stability.RECOMMENDED_CHECKS`, one generic exact rotation and the rescaled reduction check, the O1 estimate of the campaign and G1 (calibration: binary64 misses 0 % / 0.05–0.10 % of points, DD none; true error ≤ 10 E at every campaign point; R3, DECISIONS 2026-10-29 ~18:25). Rotation checks (any number, 0 = estimate absent), reduction direction, exact rescaling, in any combination; no automatic escalation by default. **Rescue stack** (MF-RESCUE, `2234ee33`; user Q9: only in pyAmpliCol's Python native evaluate API, never in an SDK, the CLI or the config): `stability=VirtualRescuePolicy(...)` per call or as the `Runtime.load(..., stability=)` default, `DEFAULT_RESCUE` double → double-double → arbitrary precision (57 digits) to a target of 10^{-5}, each level with an Euler rotation (`euler:0.1,0.2,0.3`) and the rescaled check; per-lane `VirtualEvaluation.rescue` records and `rescue_counts`, typed refusals; the legacy `VirtualStabilityPolicy` is removed. A non-finite counterterm coupling fails only its own points (MF-SMALL, `isolate_scale_failures`). Generation's post-build validation re-checks a binary64 miss alone against a 32-digit reference: double-double must agree at the same tolerance, binary64 Borns at the tolerance and binary64 virtuals within the larger of the tolerance and their own estimate (one exact rotation and the reduction direction; an estimate above 10^{-9} or absent refuses), else generation refuses; an accepted point is logged (onshell-consistency, `4fc72e39`). The rotation and reduction-direction checks flag the fast binary64 lanes' known misses (precision row): use them for such kinematics or model units.
Precision carriers	partial	Binary64, DoubleFloat, arbitrary precision and exact rationals in the native executor; since G-int the native per-point inputs run in binary64, DoubleFloat and arbitrary precision (DESIGN §5.5). Since `7db32901` binary64 runs SymJIT plane applications and needs no Symbolica licence; DoubleFloat and arbitrary precision run under pyAmpliCol's library key. Since precision-ap (`a8409830`) a master-integral invariant of one external leg is that leg's squared mass at the point's parameters: DoubleFloat and arbitrary-precision virtuals were O(1) wrong at binary64-rounded massive momenta and now agree with a MadLoop quadruple-precision run within 10^{-15} (and DoubleFloat massive spinors along $-z$ are fixed); since generation-scale-2 derived masses (e.g. M_W of `loop_sm`) are parameter-program outputs, bound exactly (genericity M2). Since `onshell-threshold` (`6e123fdb`, G2-M3) binary64 lanes at the fast engine's absolute thresholds (light masses, invariants near a mass shell) go to the generic binary64 engine. **Known limitations of the fast binary64 lanes** (NinjaRS spinor package paused): Ninja's light-like spinors are off by up to ~10^{-5} for legs near the $-z$ axis, and models whose units are ~10^4 or more away from GeV can give silent percent-level errors with stability checks off; double-double is exact there and the recommended checks flag them. **Binary64 evaluation of a stored artifact never calls Symbolica** (user 2026-10-30), complex-mass artifacts included: a cut-function argument that becomes real when complex-mass widths are exactly zero takes its real part at run time (MS-MASTERS, `71e2041e`; binary64 = DD = AP(32) on the CM01 zero-width fuzz, = MadLoop where its double and quadruple precision agree).
Native executor, schedules	supported	Rust, 16 lanes (also for a one-point call, a G1 lever), threads; native tree closures in lanes of points with runtime AVX-512/AVX2 widths (lc-perf); in finite-only mode the counterterm interactions that only feed poles are skipped natively (generation-scale-2); G1 decision on the schedule-dependent stages (Table 18, provisional until compiled trees): M for pure-gluon LC from $n = 6$, B for every other class at $n \leq 6$; F as oracle. The NinjaRS reduction runs serially. Development report: stage times, modelled kernel work, master-integral time.
Plan v4 artifacts and API	partial	Published and loaded through Python, TOML and CLI; the artifact stores the native process (`point_inputs = "native-v2"`). Since `7db32901` (addendum G) its SymJIT -O2 applications are stored as IR and recompiled at load for the loading CPU (AVX and FMA3 required), no host string is embedded, and binary64 values are bitwise equal after relocation to another machine, user and path; since M-FINAL-b SymJIT 2.28.3 (MF-UP: siravan/symjit #26, #30, #31) makes the stored runtime members byte-reproducible between generations, and the parameter program's evaluator bytes no longer depend on how the model was obtained (MF-SMALL2: the runtime-bound scalar predicates in their own id block). **Permutation aliases and process-group members** (MF-MAP, `9ad0c77f`): a one-loop alias or a permuted member of a plan v4 process is served from the stored target at permuted momenta, colour and spin correlations composed, LC anchors relabelled; numerical process-group analysis stays refused (`process-group-analysis`) and helicity/colour-flow selection is refused for mapped processes. **Virtual C ABI version 2** (MF-CLEAN, `25d39f6b`): the legacy agreement check and the SDK policy argument are removed from the Rust SDK, the C ABI, the C/C++/Fortran wrappers and UMAMI. Since model-determinism (`4e9262b5`) the compiled model and the artifact's digest no longer depend on the process history; since native-lowering (`79f2a5bd`) the plan JSON is written by the Rust plan model (Python writer as oracle). Since generation-scale-2 (`11542fd5`) no call of a stored process regenerates (every pole selection with the stored companion, correlated, resolved, flavour and catalogue calls), a process the native export cannot store fails its generation with a typed refusal (genericity M1; the plan-only, regenerated-at-load artifacts are gone), and the Rust/C/C++/Fortran SDK v4 evaluates these artifacts with the `"uv"`/`"ir"` selections, bitwise

Feature	Status	Boundary
		the Python API. Since sdk-parity (f6619f34) one parameter rule holds in the Python API, V2Runtime, the Rust SDK and the C/C++/Fortran clients: ASCII decimal values checked before names, unknown, generation-fixed (naming who fixed them) and scale parameters refused with typed errors (G2-M2). The release natives need glibc ≥ 2.38, a lower glibc giving the typed NativePlatformError at load (glibc-floor, 5301e24a). Since M-SM external-photon and complex-mass artifacts are native-v2 too (ew-native, MS-CMS-N: complex pole masses through the native trees and executor, bitwise after relocation). Open: the plane check's outcome depends on the parameter order (G2-M4: a listed plane-mismatch refusal, never a wrong value).
Counterterms (model UV, R_2)	supported	Plain tree vertices of the recursion inserted exactly once (WP2.2, a18b7599; addendum J, b4d76cc3), admitted by coupling orders and the public particle_veto; no witness table or loop matching in production. **Stock loop_sm compatibility rule** (MQ-MODEL): its four-G^0 vertex V_1001 names the three-slot constant SSS1 where the four-slot SSSS1 (same exact value) is meant; model preparation moves such references by structure, only on vertices a request reaches, and records each repair (pyamplicol_source_repairs, source_compatibility_diagnostics in the dry run); any other slot mismatch is refused (lorentz-slot-arity). Vertices whose couplings vanish at the restriction leave the loop catalogue (fail-closed-2, loop-vertex-outside-tree-model otherwise). An unlowerable counterterm vertex that no tree of the request reaches is inert and recorded in the artifact; a reachable one is refused. Site closure is the test oracle. Per point they are formed natively (G-int), the Python stages kept as oracle; generation of the counterterm route has output-identical savings (ct-perf, 201bece5: compile memos, a content-addressed lowered-model cache), and since A4 (1b13005a) the native DAG builder is the production route for the loop blocks and the counterterm program (its counterterm emission byte-identical to the Python oracle; 11 listed features keep a recorded per-request Python fallback, any other native failure fails closed), at parity with the site oracle. In EW requests ('t Hooft–Feynman gauge, EWI-4b, 8b19dae8) the Goldstone–vector mixing counterterms are lowered as vertex insertions. The UFO's EW counterterms (ew-ct, 392d3f4a) are lowered through the model's own helper bodies with MadLoop's branch side for exactly real arguments: every Laurent piece of MadLoop's model library equals pyAmpliCol's to 3.7×10^{-15} in binary64 and 6.0×10^{-31} in double-double (against MadLoop's quadruple precision). Since EWI-3R (59d47cb3) the lowering runs at scale: native, parallel kernel-equivalence proofs (Python the oracle), one reference pair per Laurent coefficient, shared Horner schemes; about 26 s cold at the target instead of 259 s, 8.4 % of the G_μ $gg \to t\bar{t}$ (EW) generation (criterion < 20 %, user 2026-10-28); a non-polynomial coupling term or one below ε^{-2} is refused when reachable. Lorentz Epsilon is lowered (EWI-7b, 6da37352; sign = MadLoop to 10^{-12}), and the Goldstone–vector mixing sign, from the tree engine's momenta, matches MadLoop's counterterm selection. **UFO-complete R_2 insertion** on the EW UFOs (user Q7): every R_2 vertex a tree of the request reaches is inserted. Compatibility note: MadLoop never inserts R2_G0tt_b on the unpatched models (its label [W+, b] equals no loop's propagator set; mg5amcnlo#475, up to 8.4 % on $t\bar{t} \to zh$); on the shipped patched models MadLoop's exports call every R_2 coupling pyAmpliCol inserts (MS-VALIDATE).
Decays $1 \to n$	supported	**Since M-FINAL-b** (MF-DEC-1 eddb4b49, MF-DEC-2 28b24a2a): one initial leg is admitted generically ($1 \to n$, $n \geq 2$) at tree level and one loop, QCD and EW, through the same recursion; the decaying particle at its real mass with zero own width (the existing width guards; internal CMS widths allowed), MadLoop's convention (amputated loops, its absorptive self-energy dropped, δZ from the UFO counterterms); the rest-frame massive Dirac spinor fixed (was $2 \times \lvert M\rvert^2$ at $\lvert p\rvert = 0$). XX01–XX08 = MadLoop's quadruple precision (Table 4: 8 of 8 pass, DD ≤ 9.6×10⁻¹⁴), $Z \to u\bar{u}$ = the massless form factor, Ward on $t \to bW^+g$, LC/NLC by the N_c scan, public route and native inputs. Refused typed: loop-induced decays (loop-induced), chains and $ (madloop-syntax).
UFO genericity: HC_NLO_X0	partial	**Since M-FINAL-b** (MF-GM-HC, e2f0c78e): the Higgs characterisation UFO at NLO, HC01–HC04 and HC06 = MadLoop 3.7.0 to ≤ 2.4×10⁻¹³ (Table 5) through generic fixes (UFO namespace head, Feynman-gauge refusal only where massive vectors run in the loops, positional counterterm keys, a colour-singlet contact split through an adjoint auxiliary). Open: HC05 ($gg \to x_0 gg$) is refused typed (unlowered-counterterm): its five-point $ffff$ R_2 contact needs the multi-trace colour state (DESIGN §6.1).
Electroweak, complex mass	supported	**M-SM** (Section 4, Table 8): QCD, QED and joint [virt=QCD QED] corrections on the shipped loop_qcd_qed_sm and loop_qcd_qed_sm_Gmu reproduce MadLoop in every class of m-sm/PLAN.md §1.1 (69 of 71 rows pass, LL02 and SC02 partial: F7, accepted). EW native inputs (ew-native, EWI-N), the EW ladder (ew-validate, EWI-7: ghosts of vetoed vector bosons, Goldstone-side root mixing), **per-leg photon tagging** (MS-PHOTON: loop.external_photons = "final" by default, "all", "none" or a leg list, as MG5's !a!; $\alpha(0)$ and the conversion counterterm on the tagged legs only), the **complex-mass scheme** (MS-CMS: MG5's transform per (mass, width) pair, coloured widths included; MS-CMS-N: native complex pole masses; = MadLoop CMS on CM01–CM05 and CM07–CM10), the EW pole split (MS-POLES), the EW generation stages in Rust (MS-GEN: catalogue segment polynomials and the parameter program's renaming behind PyO3, the Python stages the oracles, outputs byte-identical, EW generation 2.6–3.2× faster) and Symbolica-free binary64 (MS-MASTERS); since M-FINAL-b the EW export's kernel-string reading and parameter-program renaming run in Rust too (MF-RUST, b36af289: Python stages the oracles, byte-identical; EW $2 \to 3$ export 62.3 → 32.4 s). Before M-SM on loop: the QED Catani operator, EW fermion lines, 't Hooft–Feynman gauge for trees, loops and counterterms, on-shell photons at $\alpha(0)$, the UFO's EW counterterms lowered at scale, Lorentz Epsilon, lepton lines through the loop (EWI-0 to EWI-7b). Refused, typed: the Missing box of Section 1; EW LC/NLC never.
Required set	supported	All eight generate and agree with their references in all coefficients with native inputs: seven in Table 21 (validated on the WP3.1 branch before its integration), the EW target in the M-SM campaign (TV07 against MadLoop, Recola and OpenLoops, Section 4); $gg \to ggggg$ against NJet's double-double values (its binary64 points 4 and 8 at $O(10^{-8})$ accepted by the user, within the estimate); the generator is on loop since G-int, and G1 times all seven in batches

Feature	Status	Boundary
		at full and leading colour (Section 6; the $2 \to 5$ rows since `measure-g1-7leg`; all re-timed at 71e2041e). The seven-gluon Ward check (addendum H): passed (MQ-WARD). The G1 accuracy check asserts binary64 where $E \le 10^{-9}$ and DD elsewhere (user E1 ruling, g1-accuracy); R3 for QCD is recorded (DECISIONS 2026-10-29 ~18:25: two providers per process within their agreement, $gg \to ggggg$ against NJet alone, addendum F). G1 times the EW target indicatively (Section 4). The provider reference trees no longer read the old working root (reference-relocate, 48aeb123; provider values bitwise, and the toolkit refuses provider files outside the configured reference root).
Performance gates	**partial**	G1 on loop (Section 6, native per-point inputs, 71e2041e) at $B = 128$: 10 of 10 gated processes pass; 3 below 5 ms per point are not gated (user, 2026-10-24); $gg \to ggggg$ without a matched provider: no gate; the gate is taken at $B = 128$ only (user 2026-10-28): $B = 1$ and the $2 \to 5$ rows' $B = 16$ are reported without a verdict (Table 12). The Python per-point inputs before G-int failed every gate they generated (history). The EW row $gg \to t\bar{t}e^-e^+g$ (EW) is timed indicatively: ×0.15 of Recola's time at $B = 128$, no verdict ([U5]). Missing: the binding EW speed gate and the provider protocol-mode columns (M-FINAL).
$2 \to 6$	**in progress**	M-FINAL item 1 (user ruling 2026-10-31): $gg \to 6g$ ($N_f = 5$, no top) at LC and full colour, $u\bar{u} \to 6g$ and $u\bar{u} \to u\bar{u}gggg$ at LC generate, store, load and evaluate on the engine of M-SM (**preliminary** rows of Table 1, records the N = 8 reduction adapter is tested, MF-8G-0 7cdd32e9); the internal checks are incomplete and the evaluation peaks large: the user stopped every 8-leg run on 2026-11-01 for optimisation rounds (RAM first), which run on the slice after this pin.

10 Architecture of the v2 engine

10.1 The computed quantity

For Born order a, loop order b and squared coupling order $s = a + b$,

$$V_{s,k} = \mathcal{N} \sum_{a+b=s} 2\,\mathrm{Re}\left[B_a^\dagger G_{BL} L_{b,k}\right], \quad k \in \{0,-1,-2\}, \quad V = C_\varepsilon \left(V_0 + \frac{V_{-1}}{\varepsilon} + \frac{V_{-2}}{\varepsilon^2}\right), \quad C_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)},$$

where G_{BL} is the colour metric between Born and loop colour structures and $\mathcal{N}$ holds the initial-state averages and identical-particle factors (MadLoop/BLHA convention; couplings included, nothing divided by the Born). Directed colour and spin correlations replace $2\,\mathrm{Re}(\ldots)$ by the two cross terms $B_A^\dagger G L_B + L_A^\dagger G B_B$.

10.2 Pipeline

For every denominator family f, sector s and observable column o, v2 builds the four-dimensional numerator $N_{f,s,o}(q) = \sum_\alpha c_\alpha q^\alpha$ that NinjaRS reduces: Born-weighted and helicity-summed, produced by three recursions that share all work (Figure 2).

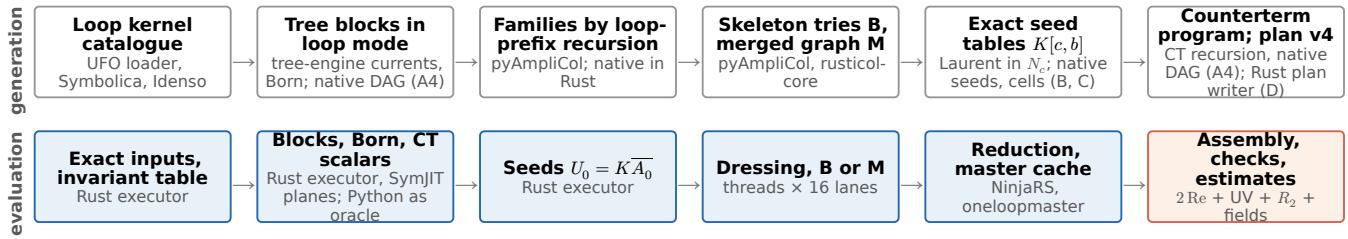


Figure 2: **v2 pipeline.** Upper lane: generation, once per process, into a plan v4 artifact. Lower lane: evaluation per batch of points. Blue: native batched numerics in the Rust executor (the per-point inputs since G-int); orange: Python per batch; white: generation. The owning code or dependency is named under each step.

Trees. The tree engine's colour-ordered Berends–Giele recursion computes, once per point, every off-shell current over every leg subset a loop vertex can absorb (the *blocks*), the Born colour components A_0 and, with WP2.2, every tree with one counterterm vertex. A loop-aware reachability test keeps exactly the currents some family position or root needs. Blocks with transverse sources use the exact transverse Yang–Mills kernels; loop segments, counterterm currents and Ward sources use the full ones.

Families and skeletons. A Recola-type recursion over loop prefixes (ordered leg sets, propagator species and masses) yields the denominator families, keyed by the canonical cyclic sequence of (leg mask, mass) and equal to Recola's tensor-integral identities. Per family it yields the colour-ordered *skeletons*: rings of blocks, each block summing all trees on its leg subset (Figure 3). Skeletons are never stored as a list; they are the leaves of the schedule tries. Only certified structural zeros are dropped (scaleless rings, external wave-function corrections, VEV tadpoles, colour zeros); nothing is restricted to 1PI.

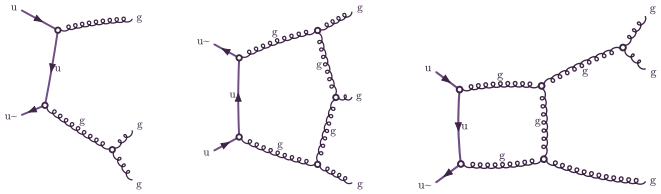


Figure 3: Three graphs of $u\bar{u} \to ggg$ drawn by FeynKit's renderer from its own generation (an oracle, not a production input). In v2 the Born (left) is a closure of blocks; the pentagon (middle) is a ring of five single-leg blocks; the box with a tree attachment (right) is a ring of four blocks, one of them the two-leg current, computed once per point and shared by every skeleton that contains that leg subset.

Loop numerators and schedules. An open-loop recursion dresses skeletons one segment at a time with polynomial-valued objects, $O' = O \cdot Y + \mathrm{raise}\big(O \cdot Z_\mu, q^\mu\big)$, in the NinjaRS monomial layout ($\binom{r+4}{4}$ coefficients at rank r, no rank cap). The seed is the colour-summed Born interference, and each block's helicities are summed as it is attached. One generator feeds three schedules (Table 25).

Table 25: **Three schedules on one generator** (DESIGN §2.2, §5.6).

	F: forward	B: suffix trie (default)	M: merged, primitive-seeded
Born weighting	at closure	at the leaves (one seed per skeleton class)	at the cut, one seed per colour ordering
Shared	prefixes across families	all prefixes with equal family and remainder	prefixes across families and compositions
Colour certificate	none needed	none needed	certified classes: adjoint rings, traces, quark rings
Role	oracle	default; generic fallback	pure-gluon LC from $n = 6$ (G1); planned default from $n \geq 8$

Colour. Colour lives only in exact seed tables $K[c, b]$, Laurent polynomials in N_c between seed classes c (the colour vector obtained by closing the cut of a skeleton's colour flow) and Born basis elements b; per point the seeds are $U_0 = \sum_b K[c, b]\overline{A_0[b]}$, with a symmetric-group FFT where certified. LC and NLC differ from full colour only in which classes are generated and how cells are truncated: at LC each colour ordering is seeded by its own Born amplitude and skeletons below the leading N_c degree are never generated. Certificates (adjoint rings, fundamental traces, quark rings) are proved at loop level, never inherited from the tree level.

Reduction and masters. Families close directly into NinjaRS `TensorNumerator` lanes: one reduction per (family, sector, column), families with equal (propagators, rank) batched as lanes of one call, scalar masters from oneloopmaster cached per event. A second reduction in a relabelled frame reuses the numerators and the cached masters as the cheap stability check.

Counterterms. The model's UV and R_2 vertices, two-point ones included, are ordinary vertices of the tree recursion, and each contribution contains exactly one of them; external field factors are separate sources. Admission is the requested coupling orders, the exactly-once constraint and a public `particle_veto`: a coupling term whose loop particles include a vetoed particle is dropped, and a vetoed particle is removed from every tree vertex, loop line and propagator of the generation, so loops and counterterms stay consistent. Nothing is matched to the loop (user, 23 October; addendum J). On `loop` the counterterms come from the WP2.2 recursion (one counterterm order in the tree recursion, the default route, whose $gg \to ggggg$ program leaves no rule to site closure) with this admission (`ct-simplify`, b4d76cc3): no witness table and no loop matching. Since A4 (`1b13005a`) the native DAG builder emits this program (byte-identical to the Python builder, which stays as oracle) and builds the loop blocks. The UFO's electroweak counterterms (`392d3f4a`) enter the same way, lowered through the model's own helper bodies with MadLoop's branch side for exactly real arguments, and in 't Hooft–Feynman gauge the Goldstone–vector mixing counterterms as vertex insertions (`8b19dae8`). A counterterm vertex that the model cannot lower and that no tree of the request reaches (the tree engine's own reachability, with the program's order limits and `particle_veto`) is inert and recorded in the artifact; a reachable one is refused. `particle_veto = [t]` replaces the internal top-free switch of Table 23; site closure and the witness machinery remain only as test oracles.

Precision, stability and assembly. Every kernel is generic over binary64, DoubleFloat, arbitrary-precision floats and exact rationals. A failed lane fails its point and never branches a kernel. The assembly takes $2\,\mathrm{Re}$ of $i/(16\pi^2)$ times the reduced family sum, plus local UV, R_2 and field terms, per sector and column, with optional rotation, reduction-direction and rescaling estimates and the finite-only mode. The public API (TOML, CLI or Python; v2 is the only one-loop engine since MQ-LEGACY-REMOVE) stores plan v4 as a relocatable PACBIN member (zlib JSON of exact kernel term lists, written by the Rust plan model since `79f2a5bd`, no compiled code, digest checked at load) and the

native process (point_inputs = "native-v2": the tree program with the Born, counterterm and field components, their saved exact evaluators and the binary64 SymJIT applications of every tree kernel and of the parameter program, stored as IR and recompiled at load) with the companion of the certified-scaleless families and the observables' tables, so no call of a stored process regenerates it (generation-scale-2), on another machine, user and path too; a process the native export cannot store is refused at generation. The Rust/C/C++/Fortran SDK v4 loads the same artifact.

Table 26: **Asymptotic classes per event** for n gluons, full colour (DESIGN §1.4): $F_n \approx (n-1)!\left(\frac{1}{\ln 2}\right)^n/2$ families, 2^n helicities. The required set ($n \le 7$) is far below the crossover of schedule B with the unitarity class ($n \approx 30$–50); schedule M is planned as the default from $n \ge 8$.

Stage or code	Class
Tree blocks	$\Theta(n \cdot n!)$ currents; $\Theta(n! \cdot 2^n)$ current-helicity rows
Dressing, schedule B	$\Theta((n-1)! \cdot 4^n)$ seed injection; trie dressing dominates for $n \le 8$
Dressing, schedule M	$\Theta(n! \cdot 2^n \cdot n^4)$
NinjaRS	$\Theta(F_n)$ reductions, $\Theta(F_n \cdot n^8)$ work, $\Theta(4^n)$ distinct masters after caching
MadLoop (per diagram, per helicity)	$\Theta(D_n \cdot 2^n \cdot n^4)$, D_n super-exponential ($gg \to gggg$: 26 975 loop diagrams)
Unitarity with exact helicity sums (NJet)	$\Theta((n-1)! \cdot 2^n \cdot n^{6\ldots9})$

Table 27: **Ownership of the mathematics.** pyAmpliCol reuses its dependencies for algebra, tensors, graphs and integrals and owns the recursions and the assembly.

Component	Responsibility
ufo_model_loader	Lossless UFO loop-model import: Laurent-valued parameters and couplings, UV/R_2 categories, loop groups
Symbolica	Expressions, coefficient extraction in q, evaluators; exact rationals and arbitrary precision
Spenso, Idenso (Community)	Lorentz and colour tensors behind the kernel catalogue and the colour metrics
FeynKit (Community)	Test oracles: graph and family counts, overall factors, drawings (Figure 3)
NinjaRS, oneloopmaster	Integrand reduction with Laurent expansion; scalar masters in all carriers
pyAmpliCol tree engine	Blocks, Born, coupling orders and colour flow (Python lane; Rust recurrence)
pyAmpliCol loop v2	Families, skeletons, schedules, seed tables, counterterm program, plan v4, native executor, API

11 Sources and reproducibility

- **Build:** typst compile benchmarks.typ benchmarks.pdf in docs/performance_reports/loop/ (Typst 0.15).
- **Status, in-flight work, decisions:** data/milestone.json, from the integration merges of f3d9e361 since 15d4ef5a (MF-CORR 821ae808, MF-RESCUE 2234ee33, MF-SMALL e1a1fea2, MF-GM-HC e2f0c78e, MF-UP dbc92efb, MF-CLEAN 25d39f6b, MF-MAP 9ad0c77f, MF-DEC-1 eddb4b49, MF-8G-0 7cdd32e9, MF-RUST b36af289, MF-DEC-2 28b24a2a, MF-SMALL2 617b59cd, mf-map-fix f3d9e361; commit subjects "Integrate work/v2-..."); workspace/reports/engine-v2/slice-status.md (integration blocks, DECISIONS 2026-10-23 to 29, the milestone pushes m1025a of 2026-10-25 ~13:45 UTC, m1025c of 2026-10-26 ~06:35 UTC (after the m1025b attempt at db13c260, not pushed) and m1026a of 2026-10-27 ~21:00 UTC (8335e9e8), and milestone m1028a (launched 2026-10-28 ~09:00 UTC; pin 28a16255, native loop-native-v2-slice-5cb5ea83; the user's rules of 2026-10-28: the speed gate at $B = 128$ only (~08:20 item 2, gate_decisions.gate_batch), the counterterm-lowering share below 20 % (~08:20 item 1) and the glibc floor (~07:40; 2.38 since glibc-floor); the known limitations from the coordinator's milestone brief and known_limitations in data/g1.json; the identical-flavour LC limitation of m1026a, DECISIONS 2026-10-26 ~21:10 item 1, is removed by identical-flavour-lc), milestone M-QCD (DECISIONS 2026-10-29 ~17:20: pin e18a024b, native loop-native-v2-slice-d0fef168; the user's milestone plan of 2026-10-28 ~12:50 and its rulings ~15:25, ~15:35/40 and ~16:10, the user's rulings of 2026-10-29 ~03:45 (E1) and ~10:20 (precision), the coordinator's ~11:25 (F7, F11) and R3 for QCD ~18:25; workspace/reports/engine-v2/m-qcd/PLAN.md §1, §5) and milestone M-SM (DECISIONS 2026-10-31 ~07:45: pin 71e2041e, native loop-native-v2-slice-71e2041e; the plan workspace/reports/engine-v2/m-sm/PLAN.md §1, §5 and its §6a user answers of 2026-10-30 ~08:10 (Q1 per-leg photons, Q2 multiparticle members, Q3 decays in M-FINAL, Q4 EW LC/NLC never, Q5 both EW UFOs, Q7 the UFO-complete R_2); the user's rules of 2026-10-30 ~08:40 (UFOmodeldb as ground truth), ~08:50 (M-FINAL order), ~12:45 (the issue-1 patch, with the coordinator's ~13:40) and ~17:35 (binary64 without Symbolica); the accepted F7 and the SC03 τ-mass relaxation; mg5amcnlo#475 and UFOmodeldb issue 1); the user's gate decision of 2026-10-24, 16:05 UTC, is recorded in data/milestone.json gate_decisions and no longer applies after the re-measure; the user's rules of 2026-10-24, 22:35 UTC — uncoloured generation, green-to-red run times, summary pages with only Table 1, Figure 1 and legends, and the gate rule "below 5 ms: not gated (user 2026-10-24)", recorded as gate_decisions.threshold_rule and applied by scripts/build_summary.py and this report); milestone M-FINAL-b (DECISIONS 2026-11-01 ~23:35 and 2026-11-02 ~00:20, ~01:05, ~03:08: pin f3d9e361 = 617b59cd + the test-only mf-map-fix, native loop-native-v2-slice-b36af289; the plan workspace/reports/engine-v2/m-final/PLAN.md §2, §3 and the user's rulings of 2026-10-31 ~11:00 (M-FINAL order, rescue only in the Python API, decays at the real mass with zero own width, EW LC/NLC never), ~14:20 (direct R_2 dropped), Q11 (one final retime) and 2026-11-01 ~23:15 (8-leg runs stopped for optimisation rounds); the G1 regression spot check at 617b59cd against 71e2041e: workspace/tmp/g1-m1101-mfb/logs/compare.txt); HANDOFF_24_10_2026.md §6; MASTER_PLAN 2026-10-23/24 entries; REQUIREMENTS_ADDENDUM A, D, F, G, H, I, J; DESIGN §5.5, §3.3, §5.7 (G1), §6.1, §6.2, §7.2, §7.5, §7.9, §12.1, §12.4, WP3.1, WP3.2; the 24 October pushes (59badc7a and b4d76cc3, gates in slice-status.md), the G-int push (79df367a, 25

October), the m1025a push (bae3b9ea, 25 October), the m1025c push (36f43250, 26 October) and the m1026a push (8335e9e8, 27 October). WP3.2 increments and dependencies: workspace/reports/engine-v2/wp32/PLAN.md; the EW counterterm comparison with MadLoop and measurement M1: wp32/M1-ew-ct.md and DESIGN §6.1; the genericity audits: workspace/reports/engine-v2/genericity-audit/REPORT.md and genericity-audit-2/REPORT.md (at 11542fd5, with the first audit's follow-up); precision-ap's MadLoop comparison: workspace/reports/engine-v2/precision-ap/; reference-relocate's inventory and evidence: workspace/reports/engine-v2/reference-relocate/; model-determinism: DESIGN WP2.5; native-lowering: DESIGN §12.1 and WP3.1 (Rust migration D); onshell-consistency: DESIGN §11.1 and docs/user/configuration.md; native-seeds and native-cells: DESIGN §7.2, §7.9 and workspace/reports/engine-v2/wp31/RUST-MIGRATION.md; fermion-lines: DESIGN §3.3 and the fixtures tests/fixtures/loop/required/uux_{ddxssx,uuxddx,uuxuux,uuxepem}.json; generation-scale-2: DESIGN WP3.1 (its status) and §12.1; EWI-4b and EWI-6b: DESIGN §12.4 and wp32/PLAN.md; the packages of m1028a: their integration blocks in slice-status.md (ew-ct EWI-3R 59d47cb3, ew-epsilon EWI-7b 6da37352, fail-closed fb54246c, sdk-parity f6619f34, ew-leptons EWI-5 b82d8280 with its R2_G0tt_b evidence (DECISIONS 2026-10-28 ~05:20 item 2 and ~08:10 item 1), identical-flavour-lc 5cb5ea83, gate-b128 28a16255), DESIGN §5.7 ("Gate batch"), §12.4 and the listed generation refusals, docs/user/configuration.md and docs/user/ufo-model-coverage.md; the packages of M-QCD: their integration blocks in slice-status.md (Section 1), docs/user/loop-sm-and-madloop.md (the loop_sm repair rule) and the listed refusals of src/pyamplicol/loops/v2/refusals.py and processes/refusals.py.

- **Section 1 (M-SM), Section 4, Table 7, Table 8, Table 29:** data/msm.json (scripts/build_mqcd.py with --manifest tests/fixtures/loop/msm/manifest.json, from the campaign's records, rebuilt byte-identically), data/msm-findings.json (MS-VALIDATE's triage, unchanged at the pin), data/msm-public.json (the public-route test file's JUnit outcomes and cases), data/msm-selftest.json (wheel self-test), combined with the manifest by scripts/build_msm_matrix.py into data/msm-matrix.json; the G1 EW row data/msm-g1.json and data/msm-g1-values.json (python -m tools.loop_benchmark gate over the EW records; Table 1 through data/summary.json); the campaign's records, logs and scripts in workspace/reports/engine-v2/m-sm/records/m1031-msm/ (a copy of workspace/tmp/campaign-m1031-msm: results/, the JUnit and logs of the 32 jobs, wheel/, STATE.md). Re-run at M-FINAL-b (617b59cd): data/msm.json, msm-public.json and msm-selftest.json from workspace/reports/engine-v2/m-sm/records/m1101-mfb/ (a copy of workspace/tmp/campaign-m1101-mfb: results/, the JUnit and logs of its 33 jobs, wheel/out/, compare.py, msm-table.md, STATE.md). The M-QCD re-run at M-FINAL-b: data/mqcd.json, records in workspace/reports/engine-v2/m-qcd/records/m1101-mfb-mqcd/ (c1/, compare.json against the M-SM re-run's file, which came from records/m1031-msm-mqcd/).
- **Section 2, Table 4, Table 28:** data/dec.json (scripts/build_mqcd.py with --manifest tests/fixtures/loop/dec/manifest.json and the decay checks' JUnit, re-run at 617b59cd; dec-compare.json against MF-DEC-2′s file: no status change), combined with the manifest by scripts/build_dec_matrix.py into data/dec-matrix.json; records in workspace/reports/engine-v2/m-final/records/m1101-mfb-new/ (results/, junit/, logs/, scripts/, STATE.md).
- **Table 5, Table 6:** data/hc.json and data/corr.json, written by scripts/build_hc.py and scripts/build_corr.py from the whole-file jobs of tests/unit/test_loop_v2_hc.py and test_loop_v2_corr_recola.py (their artifacts reloaded, JUnit) in the same records; the rescue and mapping counts of Section 1: that campaign's junit/rescue.xml and mf-map-fix's run at f3d9e361 (slice-status UPDATE 2026-11-02 ~01:05).
- **Table 1 (2 → 6 rows), Table 2, Table 3:** data/2to6.json, written by scripts/build_2to6.py from the 58 records listed in workspace/reports/engine-v2/m-final/records/m1101-mfb-2to6/args.txt (the benchmark's workspace/tmp/bench-2to6/ runs, logs and dry runs, workspace/artifacts/2to6/*/generation.json, the 8g0-report); reproduced byte for byte; Table 1 through data/summary.json preliminary.
- **Section 1 (M-QCD), Section 5, Table 10, Table 30:** data/mqcd.json (scripts/build_mqcd.py from the campaign's records), data/mqcd-findings.json (MQ-VALIDATE's triage, re-checked at the pin), data/mqcd-public.json (the public-route representatives), data/mqcd-fft-cost.json and data/mqcd-fft-worst.json (FFT cost and worst points), data/mqcd-selftest.json (wheel self-test), combined by scripts/build_mqcd_matrix.py into data/mqcd-matrix.json; the campaign's records, logs and scripts in workspace/reports/engine-v2/m-qcd/records/m1029-mqcd/ (a copy of workspace/tmp/campaign-m1029-mqcd: c1/ results, JUnit and logs, x/ the FFT and stock-loop_sm test logs, public/, fft/, wheel/, STATE.md).
- **Table 1, Figure 1:** data/summary.json, written by scripts/build_summary.py from data/g1.json, the EW row of data/msm-g1.json, data/measured.json, data/milestone.json, tools/loop_benchmark/processes.json and the preliminary rows of data/2to6.json by its main-configuration rule (the most recent measured configuration of each cell; a newer check that only repeats part of an older comparison does not replace it; revision ancestry from data/milestone.json); every entry names its configuration and source (README).
- **Table 20:** HANDOFF §0.1 (addendum F); the provider generation records of Table 1 (data/summary.json): data/g1.json for the G1 rows and the reference runs' records (workspace/reference/runs/required/<process>/, collected into data/measured.json) for the 2 → 5 rows; MadLoop's production-build times of the 2 → 5 rows: the same runs' madloop-production-result.json. **Table 21, Section 7, Section 8:** the native validations on work/v2-generation-scale, records workspace/tmp/generation-scale/gint2/after/, gint/val1/ and i7/val/ (the earlier seven-gluon points: gint/val1/), collected into data/measured.json; the earlier $gg \to t\bar{t}e^-e^+g$ record: DESIGN WP3.1 fourth increment. **Conventions:** MadLoop standalone, Recola, OpenLoops 2.
- **Section 6 (Table 11 to Table 19):** data/g1.json and the per-point values data/g1-values.json, written by python -m tools.loop_benchmark gate (README and docstrings of tools/loop_benchmark); streams tools/loop_benchmark/streams/; run records archived in workspace/reports/engine-v2/measure-g1/records/ (first campaign runs/, fix round runs2/, the re-measure at G-int (4e9b5a9e) gint/ with its paired binary64-input runs confirm/<process>/ab.json, milestone m1025a's at 1d27426b m1025a/ (with the frozen-heap launcher gc_launcher.py and its protocol experiment gc/), milestone m1026a's at 11542fd5 m1026a/ (a copy of workspace/tmp/g1-m1026a/, with the cold/warm generation A/B ab/ and in killed/ the first full-colour seven-gluon run, no gate input), milestone m1028a's at 28a16255 m1028a/ (a copy of the campaign workspace/tmp/g1-m1028a/: runs/ with every row's timing, generation, accuracy check and stability rows, round 2 of the 2 → 2 to 2 → 4 rows (m1028a-r2-*, deciding) and the seven-gluon stores' records, logs/ with /usr/bin/time -v and host-load samples (hostload.txt, hostload-

r2.txt), the public-path confirmation confirm/, the same-core old/new A/B and bisect runs ab/ with logs/ab-summary.txt, the gate inputs out/inputs-final.txt and labels out/meta-final.json, the campaign's scripts scripts/), milestone M-QCD's at e18a024b m1029-mqcd/ (a copy of workspace/tmp/g1-m1029-mqcd: runs/, logs/, bin/, the gate inputs inputs-final.txt, labels meta-final.json, compare.json, gate.log, STATE.md; the seven-gluon stores are not copied) and milestone M-SM's at 71e2041e m1031-msm/ (a copy of workspace/tmp/g1-m1031-msm: the same files, the EW gate's msm-g1.json, msm-g1-values.json, msm-gate.log and the public-path check confirm/); provider records reused from the first campaign and the fix round, not re-run); the earlier $2 \to 5$ records and the provider runs of those rows in workspace/reports/engine-v2/measure-g1-7leg/records/ (runs-f2/, runs-f4/), MadLoop exports of the G1-only processes in workspace/reference/runs/g1/; recalibration tools/loop_v2_costmodel.py --g1 data/g1.json. Builds: loop-native-v2-slice-71e2041e (loop, 71e2041e); history: loop-native-measure-g1-genscale-39824f9b (work/v2-generation-scale 889af02c plus the timers of scratch commit 39824f9b, tag g1-native-inputs-39824f9b: earlier runs, one lane, schedule M, stability) and loop-native-measure-g1-033edf15 (Python inputs); the earlier runs of M-QCD: loop-native-v2-slice-d0fef168, of m1028a: loop-native-v2-slice-5cb5ea83, of m1026a: loop-native-v2-slice-11542fd5, of m1025a: loop-native-v2-slice-7db32901, of G-int: loop-native-v2-slice-4e9b5a9e, release with one codegen unit and thin LTO; the harness is the committed tools/loop_benchmark of 71e2041e for every row (the heap freeze in the harness; m1025a's launcher and measure-g1-7leg's run worktree 2265a9f7 are history). Schedule M at seven legs: DESIGN §5.6.

- **Table 22** and NJet prose: workspace/reports/engine-v2/njet-benchmark.md §2, §5, §6 (raw workspace/reference/runs/njet/timing/; the seven-gluon single evaluations are read from those records into data/measured.json); NJet, arXiv:1209.0100 Table 5.
- **Table 23:** data/measured.json (scripts/collect_milestone.py <workspace>); top-free values at lower multiplicity: DESIGN WP3.1 and test_loop_v2_topfree.py; the seven-gluon generation history: DESIGN WP3.1 eighth and ninth increments. Memory in GiB = 2^{30} bytes.
- **Table 24, Section 10:** docs/development/loop_v2/DESIGN.md §1–§3, §5.5, §5.6, §6, §10, §12, REQUIREMENTS_ADDENDUM J (counterterm rule) and the integration records above.
- **History:** the legacy per-diagram engine's snapshot of 2026-10-18 (data/measurements.json, provider-cold.json, legacy-labels.json, colour-measurements.json) stays for the DESIGN citations and is no longer shown; its table is in the earlier report revisions (README).

Appendix: the decay campaign rows

Table 28: **Rows of the decay campaign** (data/dec.json, written by scripts/build_mqcd.py with the decay manifest tests/fixtures/loop/dec/manifest.json from the campaign's records at 617b59cd, native loop-native-v2-slice-b36af289 through data/dec-matrix.json rows). Per row: gauge and the Catani route (python: the tools' operator; engine: the runtime's poles = "ir"), tiers (d default, n nightly) with points · checks · failures, binary64 against the MadLoop reference per coefficient (Born, finite, ε^{-1}, ε^{-2}; information: rule 2 asserts binary64 against DD), the coefficients whose binary64 estimate is out of scope (DD asserted there), the extras and the reference fixture.

Row	Process	Gauge · Catani	Tiers	Born	Finite	ε^{-1}	ε^{-2}	Out of scope	Extras	Reference
XX01	$t \to bw^+$	unitary · python	d 2 · 42 · 0; n 2 · 52 · 0	7.2e−16	4.5e−15	1.4e−14	5.8e−16	—	uv, ap	t_bwp_qcd.json
XX02	$t \to bw^+$	feynman · engine	n 4 · 134 · 0	9.0e−16	5.1e−15	2.0e−13	2.2e−15	—		t_bwp_ew.json
XX03	$z \to e^-e^+$	feynman · python	n 4 · 82 · 0	1.0e−15	3.6e−15	2.3e−15	6.1e−16	—		z_emep_ew.json
XX04	$z \to u\bar{u}$	feynman · python	n 4 · 134 · 0	1.1e−15	7.6e−15	3.9e−13	1.3e−15	—		z_uux_ew.json
XX05	$w^+ \to e^+$ ve	feynman · python	n 4 · 82 · 0	1.3e−15	1.9e−15	5.3e−14	1.3e−15	—		wp_epve_ew.json
XX06	$h \to b\bar{b}$	unitary · python	n 4 · 84 · 0	7.7e−16	2.9e−15	1.2e−15	—	4 × double pole (QCD2 QED2)	uv	h_bbx_4fs.json
XX07	$t \to bw^+g$	unitary · python	n 4 · 92 · 0	6.5e−15	1.4e−13	6.2e−14	5.9e−15	—	uv, ward	t_bwpg_qcd.json
XX08	$z \to u\bar{u}g$	feynman · python	n 4 · 134 · 0	1.2e−13	3.6e−12	5.2e−11	4.9e−14	—		z_uuxg_ew.json

Appendix: the M-SM campaign rows

Table 29: **Rows of the M-SM validation campaign** (data/msm.json, written by scripts/build_mqcd.py with its M-SM manifest from the campaign's records at 617b59cd, native loop-native-v2-slice-b36af289 through data/msm-matrix.json rows). Per row: class, process, model and restriction, tiers (d default, n nightly, c campaign) with their points, per-point inputs (n native), the reference (ML MadLoop in quadruple precision, "ML d" its double-precision controls, Rc Recola, OL OpenLoops), the largest deviations over the row and its gating variants (DD from the reference, rule 1; binary64 from DD, rule 2; poles from Catani in DD, rule 4, "≤" a relaxed tolerance; — not formed or no predicted sector), the extras and gating variants, the findings that name the row (data/msm-findings.json) and the status.

Row	Class	Process	Model	Points	Reference	DD	b64/DD	Catani	Extras · variants	Findings	Status
LL01	W-lep	$e^-e^+ \to m^-m^+$	G_μ	n 3	ML d	1.9e−14	5.0e−14	2.0e−30	ap, uv		pass
LL02	W-lep	$e^-e^+ \to m^-m^+$	G_μ, m_μ	n 3	ML d	1.1e−14	6.1e−15	—		F7	partial
LL03	W-lep	$e^-e^+ \to e^-e^+$	qcd_qed_sm	n 3	ML	4.0e−15	6.9e−15	4.5e−31			pass
LL05	W-lep	$e^-e^+ \to w^+w^-$	G_μ	n 4	ML	1.6e−13	3.7e−13	5.7e−29			pass
LL08	W-lep	$e^-e^+ \to m^-m^+a$	G_μ a0	n 4	ML	4.6e−13	2.3e−12	3.5e−28	a0		pass
LL09	W-lep	$e^-e^+ \to t\bar{t}$	G_μ	n 4	ML	2.8e−15	7.9e−15	5.1e−31	uv		pass
LL10	W-lep	$e^-e^+ \to t\bar{t}h$	G_μ	n 4	ML, Rc	2.0e−14	1.7e−12	4.9e−29			pass
DY01	W-DY	$u\bar{u} \to e^-e^+$	G_μ	n 4	ML	1.2e−15	1.9e−14	2.7e−31	uv, hel, sp · mwmz		pass
DY02	W-DY	$u\bar{d} \to e^+\ \mathrm{ve}$	G_μ	n 4	ML	6.9e−15	4.7e−12	1.8e−29			pass
DY03	W-DY	$u\bar{u} \to e^-e^+$	qcd_qed_sm	n 4	ML	1.9e−15	3.1e−14	5.0e−31			pass
DY04	W-DY	$e^-u \to e^-u$	G_μ	n 4	ML	2.3e−15	2.2e−14	8.0e−31			pass
DY05	W-DY	$u\bar{u} \to e^-e^+g$	G_μ	n 3	ML	9.1e−13	4.8e−13	5.8e−30	ward, joint, ap	F4	pass
DY07	W-DY	$u\bar{u} \to e^-e^+m^-m^+$	G_μ	n 4	ML	8.1e−15	4.2e−12	1.1e−28			pass
DY08	W-DY	$u\bar{u} \to e^-e^+gg$	G_μ	n 4	ML	2.5e−14	1.5e−12	1.8e−29			pass
VJ01	W-V	$u\bar{u} \to zg$	G_μ	n 4	ML	3.6e−15	1.2e−13	3.3e−29	ward		pass
VJ02	W-V	$u\bar{d} \to w^+g$	G_μ	d 2 · n 2	ML	4.0e−15	7.8e−15	2.1e−30	uv		pass
VJ03	W-V	$u\bar{u} \to ag$	G_μ a0	n 3	ML	2.8e−14	9.5e−13	3.3e−30	a0, ward	F4	pass
VV01	W-V	$u\bar{u} \to w^+w^-$	G_μ	n 4	ML	6.7e−13	5.3e−13	6.3e−28	sp · 4tev		pass
VV02	W-V	$u\bar{u} \to zz$	G_μ	n 4	ML	2.0e−14	2.0e−12	5.8e−30	hel	F1	pass
VV03	W-V	$u\bar{d} \to w^+z$	G_μ	n 4	ML	2.4e−13	5.9e−12	4.5e−29			pass
VV05	W-V	$u\bar{d} \to w^+a$	G_μ a0	n 4	ML	1.9e−13	2.5e−11	2.3e−28	a0, ward, sp · zero	F3	pass
VV06	W-V	$u\bar{u} \to aa$	G_μ a0	n 3	ML	8.8e−13	1.3e−12	2.9e−30	a0	F4	pass
VV07	W-V	$u\bar{u} \to w^+w^-g$	G_μ	n 4	ML	5.3e−11	9.2e−11	1.1e−26	ward		pass
V301	W-V	$u\bar{u} \to w^+w^-z$	G_μ	n 4	ML	5.9e−14	4.8e−14	2.6e−28		F1	pass
VH01	W-V	$u\bar{u} \to zh$	G_μ	n 4	ML	9.7e−15	2.0e−13	1.6e−30		F1	pass
VH02	W-V	$u\bar{d} \to w^+h$	G_μ	n 4	ML	1.7e−14	1.8e−12	5.8e−29		F1	pass
VH03	W-V	$ud \to udh$	G_μ	n 4	ML	4.7e−13	8.4e−12	2.3e−28		F1	pass
TT01	W-top	$gg \to t\bar{t}$	G_μ	d 2 · n 1	ML, Rc, OL	5.9e−15	8.2e−15	6.6e−31	joint, ap, uv		pass
TT02	W-top	$u\bar{u} \to t\bar{t}$	G_μ	n 4	ML	2.7e−14	5.2e−14	1.0e−29	so, uv		pass
TT03	W-top	$gg \to t\bar{t}g$	G_μ	n 4	ML	1.4e−11	3.4e−11	2.1e−26	ward	F3	pass
TT04	W-top	$t\bar{t} \to t\bar{t}$	G_μ	n 4 · n	ML, Rc	1.4e−14	2.8e−12	1.8e−30			pass
TT05	W-top	$gt \to tt\bar{t}$	G_μ	n 4	ML, Rc	3.9e−12	1.0e−11	1.1e−29 ≤ 1×10⁻¹⁴	uv		pass
TT06	W-top	$gb \to t\bar{t}b$	G_μ with_b_mass	n 4	ML, Rc	1.2e−11	4.2e−16	4.2e−30 ≤ 1×10⁻¹³		F10	pass
TV01	W-top	$gg \to t\bar{t}h$	G_μ	n 4	ML	3.6e−13	7.3e−12	7.4e−29			pass
TV02	W-top	$gg \to t\bar{t}z$	G_μ	n 4	ML	2.1e−13	3.2e−12	6.6e−29			pass
TV03	W-top	$u\bar{d} \to t\bar{t}w^+$	G_μ	n 4	ML	3.8e−14	1.5e−11	8.7e−29	so	F1	pass
TV06	W-top	$gg \to t\bar{t}e^-e^+$	G_μ	n 3	ML, Rc, OL	1.2e−12	1.9e−10	1.8e−28			pass
TV07	W-top	$gg \to t\bar{t}e^-e^+g$	G_μ	c 3 · n	ML, Rc, OL	1.4e−12	5.1e−11	6.4e−26	ward, joint · qcd	F3	pass
ST01	W-singletop	$ub \to dt$	G_μ	n 4	ML	3.4e−14	1.9e−14	7.4e−31	uv		pass
ST02	W-singletop	$u\bar{d} \to t\bar{b}$	G_μ	n 3	ML, Rc	4.2e−13	4.7e−14	1.9e−30			pass
ST03	W-singletop	$gb \to tw^-$	G_μ	n 4	ML	1.7e−15	2.0e−14	1.5e−29	ward	F2, F3	pass
ST04	W-singletop	$u\bar{d} \to t\bar{b}$	G_μ with_b_mass	n 4	ML, Rc	3.0e−12	2.3e−11	4.8e−27		F10	pass
JJ01	W-jets	$u\bar{u} \to d\bar{d}$	G_μ	n 4	ML	6.7e−15	8.4e−14	9.6e−31	so, uv		pass
JJ02	W-jets	$u\bar{u} \to u\bar{u}$	G_μ	n 4	ML	2.8e−14	6.4e−13	7.8e−29			pass
JJ03	W-jets	$u\bar{u} \to gg$	G_μ	n 4	ML	6.7e−15	6.5e−13	1.8e−30			pass
JJ04	W-jets	$u\bar{u} \to d\bar{d}g$	G_μ	n 4	ML	6.9e−13	7.6e−12	2.0e−28	so		pass
PI01	W-photon	$aa \to e^-e^+$	G_μ	n 4	ML	9.3e−15	1.7e−13	1.9e−30	a0 · all		pass
PI02	W-photon	$aa \to w^+w^-$	G_μ	n 4	ML	3.2e−15	4.9e−14	3.3e−30			pass
PI03	W-photon	$ag \to u\bar{u}$	G_μ	n 4	ML	1.9e−15	2.9e−13	1.7e−31	uv		pass
PI04	W-photon	$au \to au$	G_μ a0	n 4	ML	9.7e−15	2.7e−13	1.0e−29	a0		pass
CM01	W-CMS	$e^-e^+ \to m^-m^+$	G_μ	n 4	ML d	1.2e−13	1.0e−13	2.0e−30	ap, sp · zero, small, source		pass
CM02	W-CMS	$u\bar{u} \to e^-e^+$	G_μ	n 4	ML	8.8e−15	1.5e−13	2.2e−29	sp		pass
CM03	W-CMS	$u\bar{d} \to e^+\ \mathrm{ve}$	G_μ	n 4	ML	2.2e−14	3.9e−12	3.1e−30	sp		pass
CM04	W-CMS	$gg \to t\bar{t}$	G_μ	n 4	ML	1.3e−15	1.5e−14	9.1e−31	uv	F8	pass
CM05	W-CMS	$u\bar{u} \to e^-e^+g$	G_μ	n 4	ML	1.1e−13	5.1e−13	4.8e−30	ward		pass
CM07	W-CMS	$gg \to t\bar{t}e^-e^+g$	G_μ	c 3 · n	ML	6.9e−13	5.6e−11	7.2e−26	ward	F3, F8	pass
CM08	W-CMS	$u\bar{d} \to e^+\ \mathrm{ve}\ b\bar{b}$	G_μ	n 4	ML	1.3e−14	1.8e−13	—			pass
CM09	W-CMS	$u\bar{d} \to e^+\ \mathrm{ve}\ a$	G_μ a0	n 4	ML	2.7e−14	1.6e−12	2.4e−29	a0, ward		pass
CM10	W-CMS	$u\bar{d} \to w^+b\bar{b}$	loop_sm no_b_mass	n 4	ML	1.6e−13	1.8e−10	—			pass
MP01	W-multi	$pp \to e^-e^+$	G_μ	variants	ML	4.2e−14	1.5e−12	2.9e−30	P2, P3, P4, P5	F2	pass
MP02	W-multi	$pp \to t\bar{t}$	G_μ	variants	ML	2.7e−14	1.6e−13	1.0e−29	P0, P2, P3, P4, P5, P6	F2	pass

Row	Class	Process	Model	Points	Reference	DD	b64/DD	Catani	Extras · variants	Findings	Status
SC01	W-scheme	$e^-e^+ \to u\bar{u}$	G_μ	n 4	ML d	6.8e−15	7.9e−15	1.9e−31			pass
SC02	W-scheme	$u\bar{u} \to m^-m^+$	G_μ, m_μ	n 4	ML	4.9e−9	1.6e−14	—		F5, F7	partial
SC03	W-scheme	$u\bar{d} \to e^+\ \mathrm{ve}$	G_μ ckm	n 4	ML	3.0e−12	5.4e−12	1.1e−6 ≤ 1×10^{-5}	uv	F6	pass
SC04	W-scheme	$b\bar{b} \to w^+w^-$	G_μ with_b_mass	n 4	ML	2.3e−10	3.1e−14	4.8e−29		F1, F9, F10	pass
SC05	W-scheme	$u\bar{u} \to e^-e^+gg$	loop_sm	n 3	fixture	8.0e−13	2.8e−12	4.1e−14 ≤ 1×10^{-12}			pass
SC06	W-scheme	$w^+w^- \to w^+w^-$	G_μ	n 4 · n	ML	6.3e−10	1.1e−10	1.1e−26		F1	pass
SC07	W-scheme	$hh \to hh$	G_μ	n 4 · n	ML	8.2e−16	2.6e−12	1.6e−32		F1	pass
SC08	W-scheme	$t\bar{t} \to w^+w^-$	G_μ	n 4 · n	ML	6.3e−13	1.2e−12	5.0e−29		F2	pass
SC09	W-scheme	$t\bar{t} \to zh$	G_μ	n 4 · n	ML, Rc	3.0e−15	2.0e−11	1.3e−30			pass
SC10	W-scheme	$zz \to hh$	G_μ	n 4	ML, Rc	4.3e−13	7.0e−13	3.7e−32			pass

Appendix: the M-QCD campaign rows

Table 30: **Rows of the M-QCD validation campaign** (data/mqcd.json, written by scripts/build_mqcd.py from the campaign's records; 617b59cd (M-FINAL-b pin, native loop-native-v2-slice-b36af289). Per row: class, process and loop_sm restriction, tiers (d default, n nightly) with their points, the reference providers of the row and its gating variants (ML MadLoop, Rc Recola, OL OpenLoops, NJet; MadLoop in quadruple precision from MQ-REF), the largest deviations over the row and its gating variants (DD from the reference, rule 1; binary64 from DD where asserted, rule 2; poles from Catani in DD, rule 4; — none formed), the extras and gating variants (status where not pass), and the row's status; the failures and their findings are in data/mqcd.json and data/mqcd-findings.json.

Row	Class	Process	Restriction	Points	Reference	DD	b64/DD	Catani	Extras · variants	Status
A01	Q-massless	$gg \to gg$	no_b_mass	d 2 · n 1	ML, Rc, OL	1.1e−15	2.9e−14	1.2e−31	nc, ap · sectors	pass
A02	Q-massless	$u\bar{u} \to gg$	no_b_mass	n 3	ML	1.2e−14	4.2e−13	1.4e−31	madloop	pass
A03	Q-massless	$gg \to u\bar{u}$	no_b_mass	n 4	ML	6.6e−15	4.2e−14	1.6e−31		pass
A04	Q-massless	$ug \to ug$	no_b_mass	d 2 · n 2	ML	1.0e−15	7.3e−15	2.2e−30		pass
A05	Q-massless	$u\bar{u} \to d\bar{d}$	no_b_mass	n 4	ML	1.1e−15	6.8e−15	1.9e−31	uv	pass
A06	Q-massless	$u\bar{u} \to u\bar{u}$	no_b_mass	n 3	ML, Rc, OL	2.8e−14	5.8e−13	1.1e−28	sectors	pass
A07	Q-massless	$uu \to uu$	no_b_mass	n 4	ML	2.6e−15	2.9e−14	1.6e−30		pass
A08	Q-massless	$gg \to ggg$	no_b_mass	n 4	ML	3.4e−14	4.5e−13	6.1e−30	nc, ward, hel	pass
A09	Q-massless	$u\bar{u} \to ggg$	no_b_mass	d 2 · n 2	ML, Rc, OL	7.8e−13	4.5e−11	1.2e−22	nc, sp · soft, collinear fail	fail
A12	Q-massless	$u\bar{u} \to d\bar{d}g$	no_b_mass	d 2 · n 1	ML, Rc, OL	5.0e−14	4.5e−13	7.7e−30	nc · sectors	pass
A13	Q-massless	$u\bar{u} \to u\bar{u}g$	no_b_mass	n 3	ML, Rc, OL	6.3e−14	7.9e−13	5.3e−30	nc, ap · sectors	pass
A14	Q-massless	$gg \to gggg$	no_b_mass	n 3	ML, Rc, OL	3.3e−14	4.6e−11	1.4e−23	ward	pass
A15	Q-massless	$u\bar{u} \to gggg$	no_b_mass	n 4	ML	3.4e−14	4.9e−12	1.7e−29	nc	pass
A18	Q-massless	$gg \to u\bar{u}u\bar{u}$	no_b_mass	n 4	ML, Rc, OL	1.2e−11	1.1e−10	1.4e−27	nc	pass
A19	Q-massless	$u\bar{u} \to d\bar{d}s\bar{s}$	no_b_mass	n 3	ML, Rc, OL	8.4e−16	2.0e−13	8.1e−30	sectors	pass
A20	Q-massless	$u\bar{u} \to u\bar{u}d\bar{d}$	no_b_mass	n 3	ML, Rc, OL	1.9e−14	2.4e−12	2.5e−30	sectors	pass
A21	Q-massless	$u\bar{u} \to u\bar{u}u\bar{u}$	no_b_mass	n 3	ML, Rc, OL	3.7e−14	4.1e−12	2.3e−29	nc · sectors	pass
A22	Q-massless	$u\bar{u} \to d\bar{d}s\bar{s}g$	no_b_mass	n 3	ML, Rc	7.0e−13	4.8e−13	1.3e−28		pass
A23	Q-massless	$u\bar{u} \to ggggg$	no_b_mass	n 3	ML, Rc	3.2e−12	1.3e−12	2.7e−29		pass
A24	Q-massless	$gg \to ggggg$	no_b_mass	n 2	NJet	8.0e−17	3.2e−12	—	uv	partial
A25	Q-massless	$ud \to ud$	no_b_mass	n 4	ML	2.1e−15	5.8e−14	9.2e−30		pass
A26	Q-massless	$\bar{u}\bar{u} \to \bar{u}\bar{u}$	no_b_mass	n 4	ML	2.6e−15	8.4e−14	1.6e−30		pass
B01	Q-top	$gg \to t\bar{t}$	no_b_mass	d 2 · n 1	ML, Rc, OL	1.7e−15	1.2e−10	3.0e−26	ap, uv, hel, sp · threshold, mt170, sectors	pass
B02	Q-top	$u\bar{u} \to t\bar{t}$	no_b_mass	n 3	ML	3.9e−14	8.7e−14	4.0e−30	madloop	pass
B03	Q-top	$gg \to t\bar{t}g$	no_b_mass	n 4	ML	3.4e−13	3.4e−11	7.5e−27	nc, ward	pass
B05	Q-top	$gg \to t\bar{t}gg$	no_b_mass	n 3	ML, Rc, OL	1.3e−13	3.2e−11	1.4e−27	sectors	pass
B06	Q-top	$u\bar{u} \to t\bar{t}d\bar{d}$	no_b_mass	n 4	ML	1.5e−14	1.5e−12	1.8e−28		pass
B07	Q-top	$gg \to t\bar{t}t\bar{t}$	no_b_mass	n 4	ML	2.5e−14	1.4e−13	1.4e−29		pass
B08	Q-heavy	$gg \to t\bar{t}$	default	n 4	ML	1.9e−15	3.1e−14	1.5e−30		pass
C01	Q-heavy	$gg \to b\bar{b}$	default	d 2 · n 2	ML, Rc	9.3e−16	6.2e−15	8.1e−29	nc, ap, ward, sp · params	pass
C02	Q-heavy	$gg \to gg$	default	n 4	ML	5.0e−14	9.0e−12	1.7e−29	uv	pass
C03	Q-heavy	$b\bar{b} \to h$	default	n 4	ML	3.9e−16	4.3e−16	9.9e−31		pass
C04	Q-heavy	$b\bar{b} \to hg$	default	n 4	ML	5.9e−16	1.9e−15	3.2e−30		pass
C05	Q-heavy	$gg \to b\bar{b}h$	default	n 4	ML	5.0e−15	2.3e−10	1.6e−29		pass
C06	Q-heavy	$gg \to t\bar{t}b\bar{b}$	default	n 4	ML	1.9e−15	1.5e−13	1.9e−29		pass
C07	Q-singletop	$ug \to dt\bar{b}$	default	n 4	ML	8.0e−15	1.6e−14	—		pass
C08	Q-heavy	$gg \to c\bar{c}$	c_mass	n 4	ML, Rc	1.7e−15	4.5e−15	4.9e−29		pass
C09	Q-heavy	$gs \to w^- c$	c_mass	n 4	ML, Rc	4.6e−16	1.5e−14	1.1e−28		pass
D01	Q-EWBorn	$u\bar{u} \to e^- e^+$	no_b_mass	d 2 · n 2	ML	5.3e−16	4.9e−15	1.0e−31	ap, ff	pass
D02	Q-EWBorn	$u\bar{u} \to e^- e^+$	no_b_mass	n 4	ML	7.1e−16	4.3e−15	2.1e−31		pass
D05	Q-EWBorn	$u\bar{u} \to e^- e^+ g$	no_b_mass	d 2 · n 2	ML	2.3e−14	2.5e−14	1.6e−31	nc, ward, uv, hel	pass
D06	Q-EWBorn	$u\bar{u} \to e^- e^+ gg$	no_b_mass	n 3	ML, Rc	8.6e−14	1.1e−12	2.7e−31	madloop	pass
D07	Q-EWBorn	$u\bar{u} \to e^- e^+ ggg$	no_b_mass	n 3	ML, Rc, OL	1.3e−14	5.2e−13	2.4e−30	sectors	pass
D08	Q-EWBorn	$u\bar{u} \to e^- e^+ d\bar{d}g$	no_b_mass	n 3	ML, Rc, OL	1.1e−14	8.9e−13	1.4e−30	sectors	pass
D09	Q-EWBorn	$u\bar{u} \to e^- e^+ m^- m^+$	no_b_mass	n 3	ML	8.9e−16	2.7e−14	5.3e−31	sectors	pass
D10	Q-EWBorn	$u\bar{u} \to u\bar{u}e^+ e^-$	no_b_mass	n 3	ML, Rc	3.1e−15	1.7e−12	3.0e−30	sectors	pass
D11	Q-EWBorn	$u\bar{u} \to zg$	no_b_mass	n 4	ML	7.1e−15	3.4e−14	2.5e−30		pass
D12	Q-EWBorn	$u\bar{d} \to e^+ \text{ ve } g$	no_b_mass	n 4	ML	1.3e−15	1.2e−14	1.4e−31		pass
D13	Q-EWBorn	$ug \to au$	no_b_mass	n 4	ML	1.2e−15	1.2e−14	4.9e−31		pass
D15	Q-EWBorn	$u\bar{u} \to zz$	no_b_mass	n 4	ML	3.2e−15	2.8e−13	1.5e−30		pass
D16	Q-EWBorn	$u\bar{u} \to w^+ w^-$	no_b_mass	n 4	ML	1.2e−13	1.4e−12	3.2e−28		pass
D17	Q-EWBorn	$b\bar{b} \to w^+ w^-$	default	n 4	ML	4.6e−14	6.6e−13	6.1e−29	5F	pass
D18	Q-EWBorn	$u\bar{u} \to zh$	no_b_mass	n 4	ML	8.5e−16	2.0e−12	1.1e−29		pass
D19	Q-EWBorn	$ud \to udh$	no_b_mass	n 4	ML	5.9e−16	3.3e−15	—		pass
D20	Q-EWBorn	$e^+ e^- \to u\bar{u}g$	no_b_mass	n 4	ML	7.3e−14	2.0e−12	7.0e−30		pass
D21	Q-EWBorn	$e^+ e^- \to t\bar{t}$	no_b_mass	n 3	ML	1.3e−14	1.8e−14	1.2e−31		pass
D23	Q-EWBorn	$ag \to u\bar{u}$	no_b_mass	n 4	ML	4.3e−16	3.1e−13	3.2e−31		pass
D24	Q-EWBorn	$e^- u \to e^- u$	no_b_mass	n 4	ML	4.8e−16	4.1e−15	8.0e−32		pass
D25	Q-EWBorn	$u\bar{d} \to w^+ a$	no_b_mass	n 4	ML	4.8e−15	4.4e−13	1.3e−28	sp · zero	pass
D26	Q-heavy	$u\bar{u} \to e^- e^+ g$	default	n 4	ML	2.3e−14	8.5e−14	1.5e−31		pass
D27	Q-EWBorn	$gu \to e^+ \text{ ve } d$	no_b_mass	n 4	ML	1.3e−15	4.3e−14	6.5e−31		pass
D28	Q-EWBorn	$u\bar{u} \to w^+ w^- g$	default	n 4	ML	1.8e−12	1.4e−11	1.6e−28	ward	pass

Row	Class	Process	Restriction	Points	Reference	DD	b64/DD	Catani	Extras · variants	Status
D29	Q-EWBorn	$b\bar{b} \to w^+w^-g$	default	n 4	ML	1.7e−13	8.8e−12	7.3e−28	ward · 5F	pass
E01	Q-ttX	$gg \to t\bar{t}h$	no_b_mass	n 4	ML	3.6e−13	7.4e−12	6.7e−30	nc, ward, uv	pass
E02	Q-ttX	$gg \to t\bar{t}z$	no_b_mass	n 4	ML	7.2e−14	3.2e−12	5.8e−30		pass
E05	Q-ttX	$e^+e^- \to t\bar{t}h$	no_b_mass	n 3	ML	5.8e−15	6.6e−15	2.4e−31	ap	pass
E06	Q-ttX	$gg \to t\bar{t}e^-e^+g$	no_b_mass	n 3	ML, Rc, OL	6.7e−14	4.4e−12	5.7e−27	sectors	pass
F01	Q-singletop	$ub \to dt$	default	n 4	ML	1.2e−15	8.7e−16	1.3e−31	nc, uv, ap · 5F	pass
F02	Q-singletop	$bu \to dt$	default	n 4	ML	1.2e−15	1.8e−15	1.1e−31	5F	pass
F03	Q-singletop	$u\bar{d} \to t\bar{b}$	no_b_mass	d 2 · n 2	ML	1.7e−15	4.1e−16	8.6e−32		pass
F04	Q-singletop	$gb \to tw^-$	default	n 4	ML	6.2e−16	3.2e−15	3.6e−30	ward · 5F	pass
G01	Q-mixed	$u\bar{u} \to d\bar{d}$	none	d 2 · n 1	ML	4.6e−15	2.4e−15	5.0e−32	uv, ap	pass
G02	Q-mixed	$u\bar{u} \to t\bar{t}$	no_b_mass	n 4	ML	2.0e−15	3.1e−14	3.7e−30		pass
H01	Q-CKM	$u\bar{s} \to w^+g$	zeromass_ckm	n 4	ML	1.4e−15	7.2e−14	4.7e−29	uv, ap	pass
H02	Q-CKM	$gs \to tw^-$	zeromass_ckm	n 4	ML	1.2e−15	7.6e−15	3.2e−30		pass
H03	Q-CKM	$gd \to tw^-$	ckm	n 4	ML	1.2e−15	5.6e−15	4.6e−30		pass
M01	Q-multi	$pp \to t\bar{t}$	no_b_mass		ML	2.0e−15	4.1e−14	4.0e−30	P0, public-P0, P2, public-P2	pass
R01	Q-EWBorn	$u\bar{u} \to m^-m^+$	lepton_masses	n 4	ML	3.1e−16	1.2e−15	3.9e−32		pass